



US012398317B2

(12) **United States Patent**
Han et al.

(10) **Patent No.:** **US 12,398,317 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **ORGANOMETALLIC COMPOUND AND ORGANIC LIGHT-EMITTING DEVICE INCLUDING THE SAME**

(71) Applicant: **SAMSUNG DISPLAY CO., LTD.**,
Yongin-si (KR)

(72) Inventors: **Junghoon Han**, Yongin-si (KR);
Soobyung Ko, Yongin-si (KR);
Sungbum Kim, Yongin-si (KR); **Sujin Shin**, Yongin-si (KR); **Eunsoo Ahn**,
Yongin-si (KR); **Jaesung Lee**,
Yongin-si (KR); **Mina Jeon**, Yongin-si
(KR); **Youngkook Kim**, Yongin-si (KR)

(73) Assignee: **Samsung Display Co., Ltd.**, Yongin-si
(KR)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 1027 days.

(21) Appl. No.: **16/713,406**

(22) Filed: **Dec. 13, 2019**

(65) **Prior Publication Data**

US 2020/0194694 A1 Jun. 18, 2020

(30) **Foreign Application Priority Data**

Dec. 17, 2018 (KR) 10-2018-0163306

(51) **Int. Cl.**

C09K 11/06 (2006.01)
C07F 15/00 (2006.01)
H10K 50/11 (2023.01)
H10K 85/30 (2023.01)
H10K 85/40 (2023.01)
H10K 85/60 (2023.01)
H10K 101/10 (2023.01)

(52) **U.S. Cl.**

CPC **C09K 11/06** (2013.01); **C07F 15/0086**
(2013.01); **H10K 85/322** (2023.02); **H10K**
85/346 (2023.02); **H10K 85/361** (2023.02);
H10K 85/40 (2023.02); **C09K 2211/1044**
(2013.01); **C09K 2211/185** (2013.01); **H10K**
50/11 (2023.02); **H10K 85/658** (2023.02);
H10K 2101/10 (2023.02)

(58) **Field of Classification Search**

CPC C07F 15/0086; C07F 1/08; C07F 1/10;
C07F 1/12; C07F 7/28; C07F 15/002;
C07F 15/0033; C07F 15/0046; C07F
15/006; C07F 15/0073; H01L 51/0087;
H10K 85/346; H10K 85/30; H10K
85/341; H10K 85/342; H10K 85/344;
H10K 85/348; H10K 85/351; H10K
85/371; C09K 11/06; C09K 2211/185;
C09K 2211/183; C09K 2211/186; C09K
2211/187; C09K 2211/188

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,059,412 B2	6/2015	Zeng et al.	
10,033,000 B2	7/2018	Xia et al.	
2002/0121860 A1 *	9/2002	Seo	H10K 50/18 313/506
2005/0064238 A1 *	3/2005	Lee	H05B 33/14 548/440
2005/0260444 A1	11/2005	Forrest et al.	
2005/0260445 A1 *	11/2005	Walters	H05B 33/14 313/506
2006/0263635 A1	11/2006	Ise	
2010/0108995 A1 *	5/2010	Yagi	C07D 209/08 257/E51.026

(Continued)

FOREIGN PATENT DOCUMENTS

CN	105315308 A	2/2016
CN	105367605 A	3/2016

(Continued)

OTHER PUBLICATIONS

Office action issued Apr. 22, 2024 in corresponding CN Patent
Application No. 201911290297.9, 8pp.

(Continued)

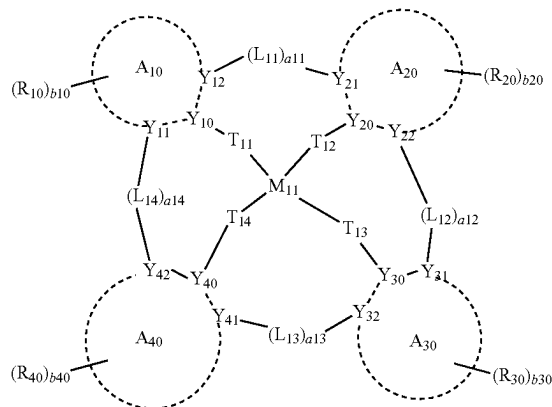
Primary Examiner — Dylan C Kershner

(74) *Attorney, Agent, or Firm* — Womble Bond Dickinson
(US) LLP

(57) **ABSTRACT**

Provided are an organometallic compound and an organic
light-emitting device including the same. An organic layer of
the organic light-emitting device includes an organometallic
compound represented by Formula 1:

<Formula 1>



(56)

References Cited

U.S. PATENT DOCUMENTS

2015/0295197 A1 10/2015 Adamovich et al.
2015/0380666 A1 * 12/2015 Szigethy C09K 11/06
546/4
2016/0028028 A1 1/2016 Li et al.
2017/0294587 A1 10/2017 Kim et al.

FOREIGN PATENT DOCUMENTS

CN 107163082 A 9/2017
CN 107267138 A 10/2017
JP 2004-352655 A 12/2004
KR 10-0522697 B1 10/2005
KR 10-0929504 B1 12/2009
KR 10-2013-0043459 A 4/2013
KR 10-2015-0056467 A 5/2015
KR 10-2016-0001679 A 1/2016
WO WO-2008117889 A1 * 10/2008 C09K 11/06

OTHER PUBLICATIONS

Notice of Allowance issued in corresponding KR Patent Application
No. 10-2018-0163306, dated Jul. 29, 2024, 3pp.

* cited by examiner

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150
110

ORGANOMETALLIC COMPOUND AND ORGANIC LIGHT-EMITTING DEVICE INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

Korean Patent Application No. 10-2018-0163306, filed on Dec. 17, 2018, in the Korean Intellectual Property Office, and entitled: "Organometallic Compound and Organic Light-Emitting Device Including the Same," is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

Embodiments relate to an organometallic compound and an organic light-emitting device including the same.

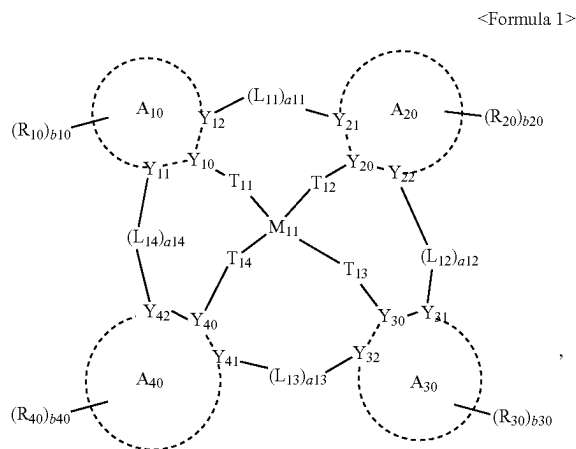
2. Description of the Related Art

Organic light-emitting devices are self-emission devices that produce full-color images, and also have wide viewing angles, high contrast ratios, short response times, and excellent characteristics in terms of brightness, driving voltage, and response speed, compared to devices in the art.

An example of such organic light-emitting devices may include a first electrode disposed on a substrate, and a hole transport region, an emission layer, an electron transport region, and a second electrode, which are sequentially disposed on the first electrode. Holes provided from the first electrode may move toward the emission layer through the hole transport region, and electrons provided from the second electrode may move toward the emission layer through the electron transport region. Carriers, such as holes and electrons, recombine in the emission layer to produce excitons. These excitons transit from an excited state to a ground state, thereby generating light.

SUMMARY

Embodiments are directed to an organic light-emitting device, including a first electrode, a second electrode, and an organic layer between the first electrode and the second electrode and including an emission layer, the organic layer including an organometallic compound represented by Formula 1:



wherein, in Formula 1,

M₁₁ may be platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), or thulium (Tm),

Y₁₀, Y₂₀, Y₃₀, and Y₄₀ may each independently be N or C,

Y₁₁, Y₁₂, Y₂₁, Y₂₂, Y₃₁, Y₃₂, Y₄₁, and Y₄₂ may each independently be N or C,

A₁₀, A₂₀, A₃₀, and A₄₀ may each independently be a C₅-C₆₀ carbocyclic group or a C₁-C₆₀ heterocyclic group,

T₁₁ to T₁₄ may each independently be a single bond, *—O—*, or *—S—*,

L₁₁ may be a substituted or unsubstituted C₃-C₁₀ cycloalkylene group, *—O—*, *—S—*, *—C(R₁)(R₂)—*, *—C(R₁)=*, *—C(R₁)—*, *—C(R₁)=C(R₂)—*, *—C(=S)—*, *—B(R₁)—*, *—N(R₁)—*, *—P(R₁)—*, *—Si(R₁)(R₂)—*, *—P(R₁)(R₂)—*, or *—Ge(R₁)(R₂)—*,

a₁₁ may be 1, 2, 3, 4, or 5,

at least one of L₁₁(s) in the number of a₁₁ may be a substituted or unsubstituted C₃-C₁₀ cycloalkylene group,

L₁₂ to L₁₄ may each independently be a single bond, *—O—*, *—S—*, *—C(R₁)(R₂)—*, *—C(R₁)=*, *—C(R₁)—*, *—C(R₁)=C(R₂)—*, *—C(=O)—*, *—C(=S)—*, *—C≡C—*, *—B(R₁)—*, *—N(R₁)—*, *—P(R₁)—*, *—Si(R₁)(R₂)—*, *—P(R₁)(R₂)—*, or *—Ge(R₁)(R₂)—*,

a₁₂, a₁₃, and a₁₄ may each independently be 0, 1, 2, 3, 4, or 5,

R₁, R₂, R₁₀, R₂₀, R₃₀, and R₄₀ may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted C₁-C₆₀ heteroaryloxy group, a substituted or unsubstituted C₁-C₆₀ heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁)(Q₂)(Q₃), —B(Q₁)(Q₂), —N(Q₁)(Q₂), —P(Q₁)(Q₂), —C(=O)(Q₁), —S(=O)(Q₁), —S(=O)₂(Q₁), —P(=O)(Q₁)(Q₂), or —P(=S)(Q₁)(Q₂),

R₁ and R₂ may be linked to form a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group,

R₁ and R₁₀; R₁ and R₂₀; R₁ and R₃₀; or R₁ and R₄₀ may be linked to form a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group,

b₁₀, b₂₀, b₃₀, and b₄₀ may each independently be 1, 2, 3, 4, 5, 6, 7, or 8,

* and *¹ each indicate a binding site to a neighboring atom,

at least one substituent of the substituted C₅-C₆₀ carbocyclic group, the substituted C₁-C₆₀ heterocyclic group, the substituted C₃-C₁₀ cycloalkylene group, the substituted C₁-C₆₀ alkyl group, the substituted C₂-C₆₀ alkenyl group, the substituted C₂-C₆₀ alkynyl group, the substituted C₁-C₆₀ alkoxy group, the substituted C₃-C₁₀ cycloalkyl group, the substituted C₁-C₁₀ heterocycloalkyl group, the substituted C₃-C₁₀ cycloalkenyl group, the substituted C₁-C₁₀ heterocycloalkenyl group, the substituted C₆-C₆₀ aryl group, the substituted C₆-C₆₀ aryloxy group, the substituted C₆-C₆₀ arylthio group, the substituted C₁-C₆₀ heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may be:

deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, or a C₁-C₆₀ alkoxy group;

a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, or a C₁-C₆₀ alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁₁)(Q₁₂)(Q₁₃), —N(Q₁₁)(Q₁₂), —B(Q₁₁)(Q₁₂), —C(=O)(Q₁₁), —S(=O)₂(Q₁₁), or —P(=O)(Q₁₁)(Q₁₂);

a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group;

a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₂₁)(Q₂₂)(Q₂₃), —N(Q₂₁)(Q₂₂), —B(Q₂₁)(Q₂₂), —C(=O)(Q₂₁), —S(=O)₂(Q₂₁), or —P(=O)(Q₂₁)(Q₂₂); or —Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), or —P(=O)(Q₃₁)(Q₃₂), and

Q₁ to Q₃, Q₁₁ to Q₁₃, Q₂₁ to Q₂₃, and Q₃₁ to Q₃₃ may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a C₁-C₆₀ alkyl group substituted with deuterium, —F, or a cyano group, a C₆-C₆₀ aryl group substituted with deuterium, —F, or a cyano group, a biphenyl group, or a terphenyl group.

Embodiments are also directed to an organometallic compound represented by Formula 1.

BRIEF DESCRIPTION OF THE DRAWING

These and/or other aspects will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with FIGURE which is a schematic view of an organic light-emitting device according to an example embodiment.

DETAILED DESCRIPTION

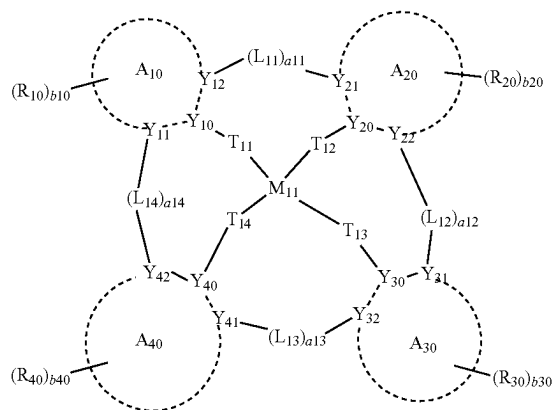
Example embodiments will now be described more fully hereinafter with reference to the accompanying drawing; however, they may be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey exemplary implementations to those skilled in the art.

In the drawing figure, the dimensions of layers and regions may be exaggerated for clarity of illustration. Like reference numerals refer to like elements throughout.

As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

An organometallic compound according to an example embodiment is represented by Formula 1:

<Formula 1>



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In Formula 1, M_{11} may be platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), or thulium (Tm).

In an example embodiment, M_{11} may be Pt, Pd, Cu, Ag, or Au.

For example, M_{11} may be Pt.

In Formula 1, Y_{10} , Y_{20} , Y_{30} , and Y_{40} may each independently be N or C.

In an example embodiment, in Formula 1,

Y_{10} , Y_{20} , and Y_{30} may each be C, and Y_{40} may be N, Y_{10} , Y_{20} , and Y_{40} may each be C, and Y_{30} may be N, Y_{10} , Y_{30} , and Y_{40} may each be C, and Y_{20} may be N, Y_{20} , Y_{30} , and Y_{40} may each be C, and Y_{10} may be N, Y_{10} and Y_{40} may each be C, and Y_{20} and Y_{30} may each be N,

Y_{10} and Y_{40} may each be N, and Y_{20} and Y_{30} may each be C,

Y_{10} and Y_{20} may each be C, and Y_{30} and Y_{40} may each be N,

Y_{10} and Y_{20} may each be N, and Y_{30} and Y_{40} may each be C,

Y_{10} and Y_{30} may each be C, and Y_{20} and Y_{40} may each be N, or

Y_{10} and Y_{30} may each be N, and Y_{20} and Y_{40} may each be C.

In an example embodiment, at least one of Y_{10} and Y_{20} may be C,

(i) when Y_{10} is C, a bond between Y_{10} and T_{11} or a bond between Y_{10} and M_{11} may be a coordinate bond, and

(ii) when Y_{20} is C, a bond between Y_{20} and T_{12} or a bond between Y_{20} and M_{11} may be a coordinate bond.

In an example embodiment, Y_{30} and Y_{40} may each be C.

In Formula 1, A_{10} , A_{20} , A_{30} , and A_{40} may each independently be a C_5 - C_{60} carbocyclic group or a C_1 - C_{60} heterocyclic group.

In an example embodiment, A_{10} to A_{40} may each independently be a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a 1,2,3,4-tetrahydronaphthalene group, a furan group, a thiophene group, a silole group, an indene group, a fluorene group, an indole group, a carbazole group, a benzofuran group, a dibenzofuran group, a benzothiophene group, a dibenzothiophene group, a benzosilole group, a dibenzosilole group, an indeno pyridine group, an indolopyridine group, a benzofuopyridine group, a benzothienopyridine group, a benzosilolopyridine group, an indeno pyrimidine group, an indolopyrimidine group, a benzofuopyrimidine group, a benzothienopyrimidine group, a benzosilolopyrimidine group, a dihydropyridine group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a 2,3-dihydroimidazole group, a triazole group, a 2,3-dihydrotriazole group, an oxazole group, an isooxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a 2,3-dihydrobenzimidazole group, an imidazopyridine group, a 2,3-dihydroimidazopyridine group, an imidazopyrimidine group, a 2,3-dihydroimidazopyrimidine group, an imidazopyrazine group, a 2,3-dihydroimidazopyrazine group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothi-

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adiazole group, a 5,6,7,8-tetrahydroisoquinoline group, or a 5,6,7,8-tetrahydroquinoline group.

In an example embodiment, at least one of A_{10} and A_{20} may be an N-containing C_1 - C_{60} heterocyclic group.

In an example embodiment, A_{10} and A_{20} may each independently be an N-containing C_1 - C_{60} heterocyclic group.

In an example embodiment, A_{10} and A_{20} may each independently be a pyrrole group, a pyrazole group, an imidazole group, a 2,3-dihydroimidazole group, a triazole group, a 2,3-dihydrotriazole group, an oxazole group, an isooxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a 2,3-dihydrobenzimidazole group, an imidazopyridine group, a 2,3-dihydroimidazopyridine group, an imidazopyrimidine group, a 2,3-dihydroimidazopyrimidine group, an imidazopyrazine group, a 2,3-dihydroimidazopyrazine group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a 5,6,7,8-tetrahydroisoquinoline group, or a 5,6,7,8-tetrahydroquinoline group.

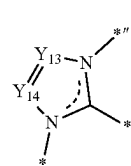
In an example embodiment, at least one of A_{10} and A_{20} may be an N-containing 5-membered heterocyclic group or an N-containing 7-membered heterocyclic group, or

at least one of A_{10} and A_{20} may be condensed with a 6-membered carbocyclic group or a 6-membered heterocyclic group.

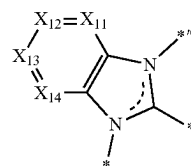
In an example embodiment, A_{10} and A_{20} may each independently be an N-containing 5-membered heterocyclic group or an N-containing 7-membered heterocyclic group, and

A_{10} and A_{20} may be condensed with a 6-membered carbocyclic group or a 6-membered heterocyclic group.

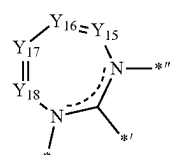
In an example embodiment, A_{10} and A_{20} may each independently be a group represented by one of Formulae 10-1 to 10-3:



10-1



10-2



10-3

In Formulae 10-1 to 10-3,

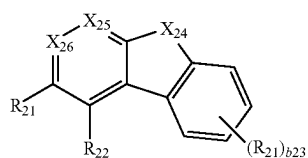
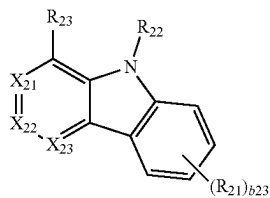
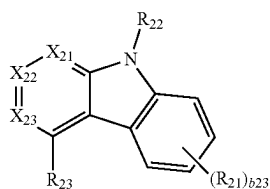
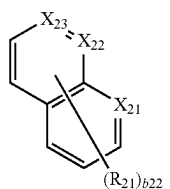
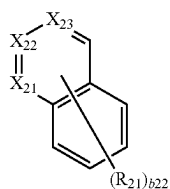
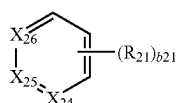
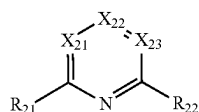
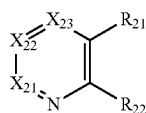
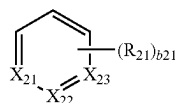
Y_{13} to Y_{18} and X_{11} to X_{14} may each independently be N or C(R_{10}),

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R₁₀ is the same as described above, and

*, **, and *** each indicate a binding site to a neighboring atom.

In an example embodiment, A₃₀ and A₄₀ may each independently be a group represented by one of Formulae 2-1 to 2-43:



2-1 10

2-2 15

2-3 20

2-4 25

2-5 30

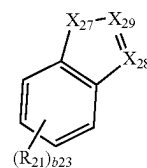
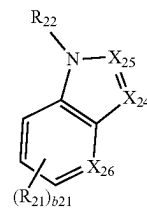
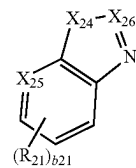
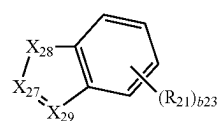
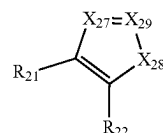
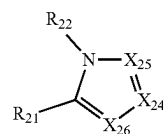
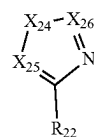
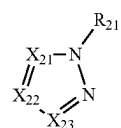
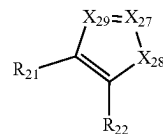
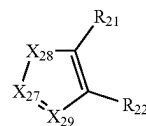
2-6

2-7 45

2-8

2-9 60

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8

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2-10

2-11

2-12

2-13

2-14

2-15

2-16

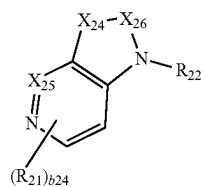
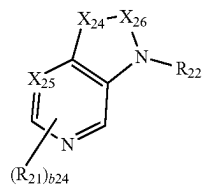
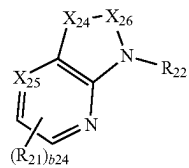
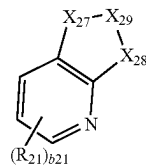
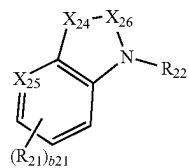
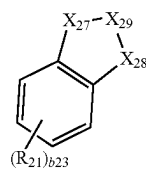
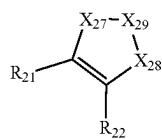
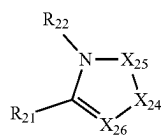
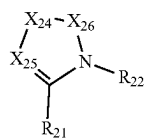
2-17

2-18

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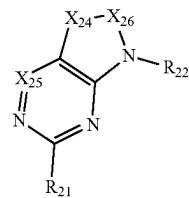


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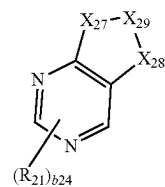
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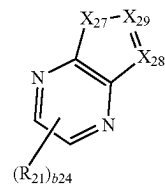
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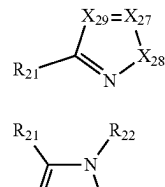
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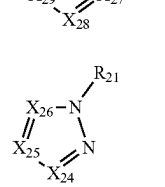
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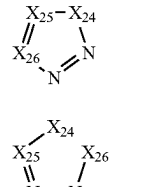
2-24

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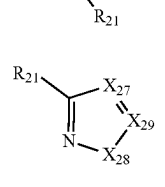
2-25

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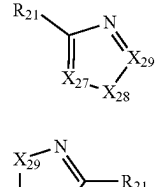
2-26

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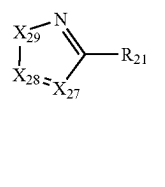
2-27

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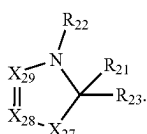
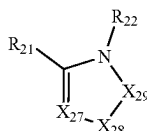
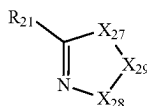
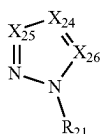
2-37

2-38

2-39

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-continued



In Formulae 2-1 to 2-43,

X₂₁ to X₂₃ may each independently be C(R₂₄) or C—*, and at least two of X₂₁ to X₂₃ may be C—*,

X₂₄ may be N—*.

X₂₅ and X₂₆ may each independently be C(R₂₄) or C—*, and at least one of X₂₅ and X₂₆ may be C—*,

X₂₇ and X₂₈ may each independently be N, N(R₂₅), or N—*, and X₂₉ may be C(R₂₄) or C—*, wherein i) at least one of X₂₇ and X₂₈ may be N—*, and X₂₉ may be C—*, or ii) X₂₇ and X₂₈ may each be N—*, and X₂₉ may be C(R₂₄),

R₂₁ to R₂₅ may each independently be the same as described in connection with R₁₀,

b21 may be 1, 2, or 3,

b22 may be 1, 2, 3, 4, or 5,

b23 may be 1, 2, 3, or 4,

b24 may be 1 or 2, and

* indicates a binding site to a neighboring atom.

In an example embodiment, in Formula 1, A₃₀ and A₄₀ may each independently be a group represented by one of Formulae 2-1 to 2-9.

In an example embodiment, in Formula 1, A₃₀ and A₄₀ may each independently be a benzene group, a pyridine group, a pyrimidine group, or a triazine group.

In Formula 1, T₁₁ to T₁₄ may each independently be a single bond, *—O—*, or *—S—*.

In an example embodiment, in Formula 1, each of T₁₁ to T₁₄ may be a single bond;

T₁₁ may be O or S, and T₁₂ to T₁₄ may each be a single bond;

T₁₂ may be O or S, and T₁₁, T₁₃, and T₁₄ may each be a single bond;

T₁₃ may be O or S, and T₁₁, T₁₂, and T₁₄ may each be a single bond; or

T₁₄ may be O or S, and T₁₁, T₁₂, and T₁₃ may each be a single bond.

For example, in Formula 1, each of T₁₁ to T₁₄ may each be a single bond.

In an example embodiment, a bond between Y₁₀ and T₁₁ or a bond between Y₁₀ and M₁₁ may be a coordinate bond.

In an example embodiment, a bond between Y₂₀ and T₁₂ or a bond between Y₂₀ and M₁₁ may be a coordinate bond.

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In an example embodiment, a bond between Y₃₀ and T₁₃ or a bond between Y₃₀ and M₁₁ may be a covalent bond.

In an example embodiment, a bond between Y₄₀ and T₁₄ or a bond between Y₄₀ and M₁₁ may be a covalent bond.

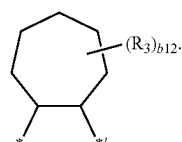
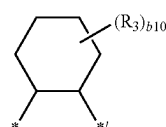
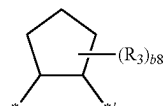
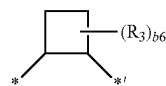
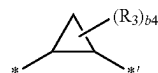
In Formula 1, L₁₁ may be a substituted or unsubstituted C₃-C₁₀ cycloalkylene group, *—O—*, *—S—*, *—C(R₁)(R₂)—*, *—C(R₁)=*, *—C(R₁)—*, *—C(R₁)=C(R₂)—*, *—C(=O)—*, *—C(=S)—*, *—B(R₁)—*, *—N(R₁)—*, *—P(R₁)—*, *—Si(R₁)(R₂)—*, *—P(R₁)(R₂)—*, or *—Ge(R₁)(R₂)—*, a11 may be 1, 2, 3, 4, or 5, and at least one of L₁₁(s) in the number of a11 may be a substituted or unsubstituted C₃-C₁₀ cycloalkylene group.

In an example embodiment, at least one of L₁₁(s) in the number of a11 may be:

a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, or a cyclodecanylene group; or

a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, or a cyclodecanylene group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₂-C₂₀ alkenyl group, a C₂-C₂₀ alkynyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, a triazinyl group, a dibenzofuranyl group, or a dibenzothiophenyl group.

In an example embodiment, at least one of L₁₁(s) in the number of a11 may be a group represented by one of Formulae 3-1 to 3-5:



In Formulae 3-1 to 3-5,

R₃ may be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an

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amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₂-C₂₀ alkenyl group, a C₂-C₂₀ alkynyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, a triazinyl group, a dibenzofuranyl group, or a dibenzothiophenyl group,

b4 may be an integer from 1 to 4,

b6 may be an integer from 1 to 6,

b8 may be an integer from 1 to 8,

b10 may be an integer from 1 to 10,

b12 may be an integer from 1 to 12, and

* and *' each indicate a binding site to a neighboring atom.

In an example embodiment, a11 may be 1, and

L₁₁ may be:

a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, or a cyclodecanylene group; or

a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, or a cyclodecanylene group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkenyl group, a C₁-C₂₀ alkynyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, a triazinyl group, a dibenzofuranyl group, or a dibenzothiophenyl group.

In an example embodiment, a11 may be 1, and L₁₁ may be a group represented by one of Formulae 3-1 to 3-5.

In Formula 1, L₁₂ to L₁₄ may each independently be a single bond, *—O—*, *—S—*, *—C(R₁)(R₂)—*, *—C(R₁)=*, *—C(R₁)=C(R₂)—*, *—C(=O)—*, *—C(=S)—*, *—C≡C—*, *—B(R₁)—*, *—N(R₁)—*, *—P(R₁)—*, *—Si(R₁)(R₂)—*, *—P(R₁)(R₂)—*, or *—Ge(R₁)(R₂)—*.

In an example embodiment, L₁₂ to L₁₄ may each independently be a single bond, *—O—*, *—S—*, *—N(R₁)—*, *—C(R₁)(R₂)—*, *—Si(R₁)(R₂)—*, or *—B(R₁)—*.

In an example embodiment, in Formula 1, L₁₂ may be *—C(R₁)(R₂)—* or *—Si(R₁)(R₂)—*, and R₁ and R₂ may be linked to form a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group,

L₁₂ may be *—C(R₁)(R₂)—* for *—Si(R₁)(R₂)—*, and R₁ and R₂ may be linked to form a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group, or

L₁₄ may be *—N(R₁)—*, *—C(R₁)(R₂)—*, or *—Si(R₁)(R₂)—*, and R₁ and R₂ may be linked to form a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group.

In an example embodiment, in Formula 1, L₁₂ may be *—N(R₁)—*, *—C(R₁)(R₂)—*, or *—Si(R₁)(R₂)—*, and R₁ and R₂₀ may be linked to form a substituted or

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unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group,

L₁₂ may be *—N(R₁)—*, *—C(R₁)(R₂)—*, or *—Si(R₁)(R₂)—*, and R₁ and R₂₀; or R₁ and R₃₀ may be linked to form a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group, or

L₁₄ may be *—N(R₁)—*, *—(R₁)(R₂)—*, or *—Si(R₁)(R₂)—*, and R₁ and R₄₀ may be linked to form a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group.

In an example embodiment, in Formula 1, L₁₂ and L₁₄ may each be a single bond, and

L₁₃ may be *—O—*, *—S—*, *—N(R₁)—*, *—C(R₁)(R₂)—*, *—Si(R₁)(R₂)—*, or *—B(R₁)—*.

In Formula 1, a12, a13, and a14 may each independently be 0, 1, 2, 3, 4, or 5. When a12 is 0, A₂₀ and A₃₀ may not be linked, when a13 is 0, A₃₀ and A₄₀ may not be linked, and when a14 is 0, A₃₀ and A₄₀ may not be linked.

In an example embodiment, a12, a13, and a14 may each independently be 1, 2, or 3

In Formula 1, R₁, R₂, R₁₀, R₂₀, R₃₀, and R₄₀ may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted C₁-C₆₀ heteroaryloxy group, a substituted or unsubstituted C₁-C₆₀ heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁)(Q₂)(Q₃), —B(Q₁)(Q₂), —N(Q₁)(Q₂), —P(Q₁)(Q₂), —C(=O)(Q₁), —S(=O)(Q₁), —S(=O)₂(Q₁), —P(=O)(Q₁)(Q₂), or —P(=S)(Q₁)(Q₂),

R₁ and R₁₀; R₁ and R₂₀; R₁ and R₃₀; or R₁ and R₄₀ may be linked to form a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group,

b10, b20, b30, and b40 may each independently be 1, 2, 3, 4, 5, 6, 7, or 8, and

Q₁ to Q₃ may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, or a terphenyl group.

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In an example embodiment, R₁, R₂, R₁₀, R₂₀, R₃₀, and R₄₀ may each independently be:

- hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C₁-C₂₀ alkyl group, or a C₁-C₂₀ alkoxy group;
- a C₁-C₂₀ alkyl group or a C₁-C₂₀ alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, or a biphenyl group;
- a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentacenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoindolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a benzoisoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, a thiadiazolyl group, an oxadiazolyl group, a triazinyl group, a carbazolyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a benzocarbazolyl group, a naphthobenzofuranyl group, a naphthobenzothiophenyl group, a naphthobenzosilolyl group, a dibenzocarbazolyl group, a dinaphthofuranyl group, a dinaphthothiophenyl group, a dinaphthosilolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an oxazolopyridinyl group, a thiazolopyridinyl group, a benzonaphthyridinyl group, an azafuorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azadibenzosilolyl group, an indenopyrrolyl group, an indolopyrrolyl group, an indeno carbazolyl group, or an indolocarbazolyl group;
- a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentacenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoin-

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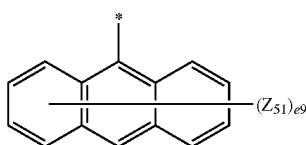
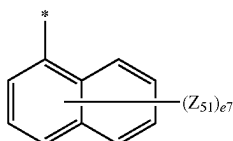
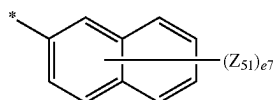
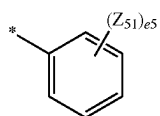
dolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a benzoisoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, a thiadiazolyl group, an oxadiazolyl group, a triazinyl group, a carbazolyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a benzocarbazolyl group, a naphthobenzofuranyl group, a naphthobenzothiophenyl group, a naphthobenzosilolyl group, a dibenzocarbazolyl group, a dinaphthofuranyl group, a dinaphthothiophenyl group, a dinaphthosilolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an oxazolopyridinyl group, a thiazolopyridinyl group, a benzonaphthyridinyl group, an azafuorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azadibenzosilolyl group, an indenopyrrolyl group, an indolopyrrolyl group, an indeno carbazolyl group, or an indolocarbazolyl group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentacenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, a silolyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an indolyl group, an isoindolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a benzoisoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzosilolyl group, a benzothiazolyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, a thiadiazolyl group, an oxadiazolyl group, a triazinyl group, a carbazolyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a benzocarbazolyl group, a naphthobenzofuranyl group, a naphthobenzothiophenyl group, a naphthobenzosilolyl group, a dibenzocarbazolyl group, a dinaphthofuranyl group, a dinaphthothiophenyl group, a dinaphthosilolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an

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oxazopyridinyl group, a thiazopyridinyl group, a benzonaphthyridinyl group, an azafluorenyl group, an azaspiro-bifluorenyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, an azadibenzosilolyl group, an indenopyrrolyl group, an indolopyrrolyl group, an indeno carbazolyl group, an indolocarbazolyl group, $-\text{Si}(\text{Q}_{30})(\text{Q}_{32})(\text{Q}_{33})$, $-\text{N}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{B}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{C}(=\text{O})(\text{Q}_{31})$, $-\text{S}(=\text{O})(\text{Q}_{31})$, $-\text{S}(=\text{O})_2(\text{Q}_{31})$, $-\text{P}(=\text{O})(\text{Q}_{31})(\text{Q}_{32})$, or $-\text{P}(=\text{S})(\text{Q}_{31})(\text{Q}_{32})$; or $-\text{Si}(\text{Q}_1)(\text{Q}_2)(\text{Q}_3)$, $-\text{B}(\text{Q}_1)(\text{Q}_2)$, $-\text{N}(\text{Q}_1)(\text{Q}_2)$, $-\text{P}(\text{Q}_1)(\text{Q}_2)$, $-\text{C}(=\text{O})(\text{Q}_1)$, $-\text{S}(=\text{O})(\text{Q}_1)$, $-\text{S}(=\text{O})_2(\text{Q}_1)$, $-\text{P}(=\text{O})(\text{Q}_1)(\text{Q}_2)$, or $-\text{P}(=\text{S})(\text{Q}_1)(\text{Q}_2)$, and Q_1 to Q_3 and Q_{31} to Q_{33} may each independently be hydrogen, deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, a C_1 - C_{60} alkoxy group, a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl group, a C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} heterocycloalkenyl group, a C_6 - C_{60} aryl group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryl group, a C_1 - C_{60} heteroaryloxy group, a C_1 - C_{60} heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a C_1 - C_{60} alkyl group substituted with at least deuterium, $-\text{F}$, or a cyano group, a C_6 - C_{60} aryl group substituted with deuterium, $-\text{F}$, or a cyano group, a biphenyl group, or a terphenyl group.

In an example embodiment, R_1 , R_2 , R_{10} , R_{20} , R_{30} , and R_{40} may each independently be:

hydrogen, deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a cyano group, a C_1 - C_{20} alkyl group, or a C_1 - C_{20} alkoxy group; a C_1 - C_{20} alkyl group or a C_1 - C_{20} alkoxy group, each substituted with deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a cyano group, a phenyl group, or a biphenyl group; a group represented by one of Formulae 5-1 to 5-25; or a group represented by one of Formulae 6-1 to 6-55, and R_1 and R_2 may be linked to form a cyclopentane group, a cyclohexane group, a benzene group, a naphthalene group, or a fluorene group,



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5-2 50

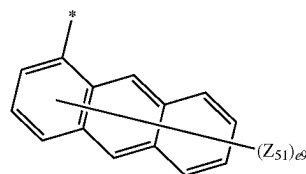
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5-4 60

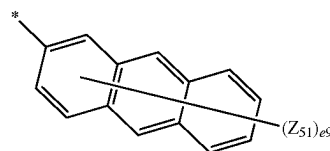
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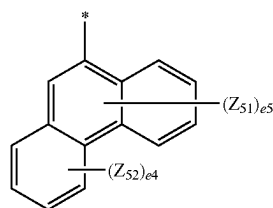
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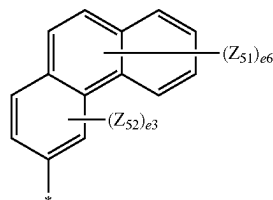
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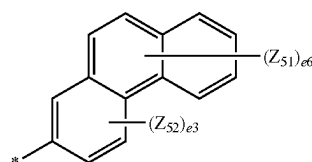
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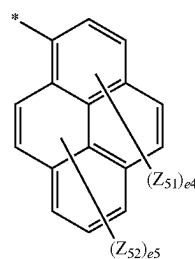
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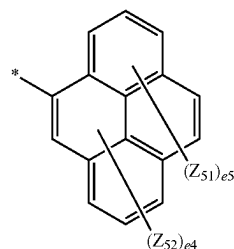
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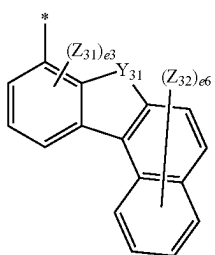
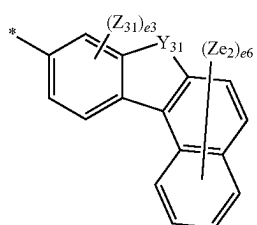
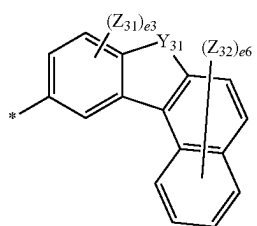
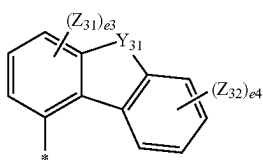
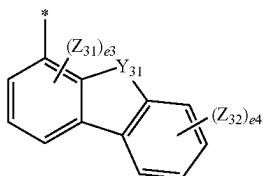
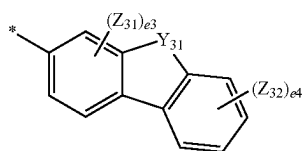
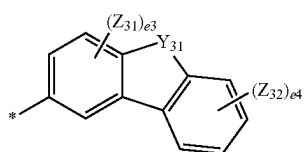
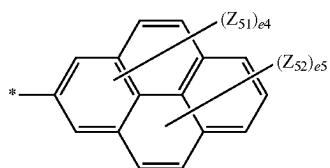
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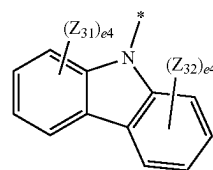
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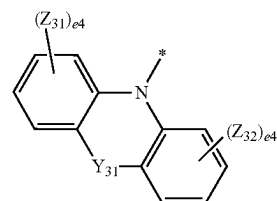
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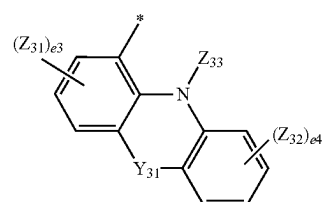
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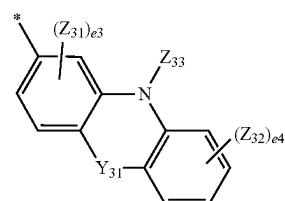
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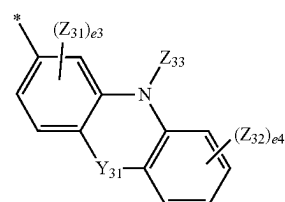
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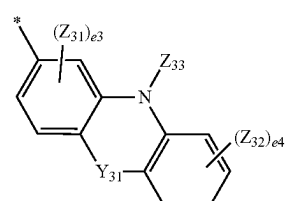
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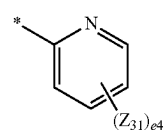
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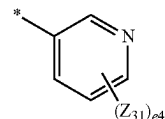
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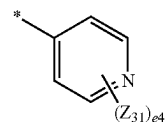
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5-20

5-21

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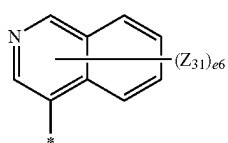
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5-25

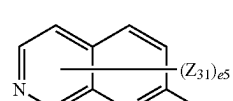
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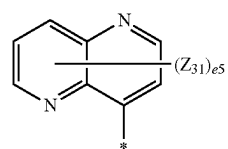


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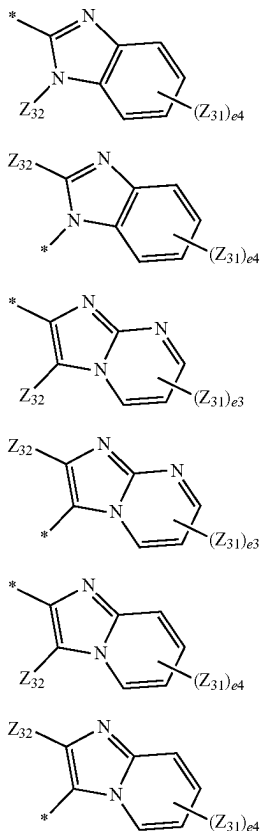
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In Formulae 5-1 to 5-25 and 6-1 to 6-55,

Y_{31} may be O, S, $C(Z_{34})(Z_{35})$, $N(Z_{34})$, Or $Si(Z_{34})(Z_{35})$,

Z_{31} to Z_{35} may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_2 - C_{20} alkenyl group, a C_2 - C_{20} alkynyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, or a triazinyl group,

e_2 may be 1 or 2,

e_3 may be an integer from 1 to 3,

e_4 may be an integer from 1 to 4,

e_5 may be an integer from 1 to 5,

e_6 may be an integer from 1 to 6,

e_7 may be an integer from 1 to 7,

e_9 may be an integer from 1 to 9, and

* indicates a binding site to a neighboring atom.

In an example embodiment, R_1 , R_2 , R_{10} , R_{20} , R_{30} , and R_{40} may each independently be:

hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C_1 - C_{20} alkyl group, or a C_1 - C_{20} alkoxy group;

a C_1 - C_{20} alkyl group or a C_1 - C_{20} alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, or a biphenyl group;

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group,

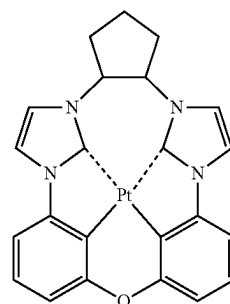
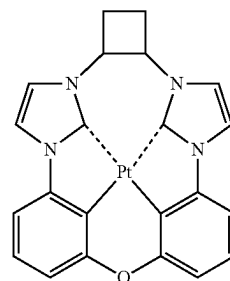
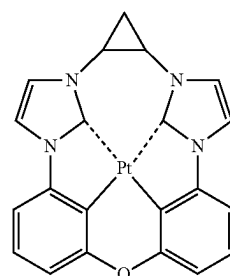
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a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, or a dibenzocarbazolyl group; or

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, or a dibenzocarbazolyl group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, or a biphenyl group, and

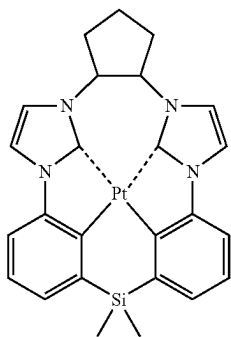
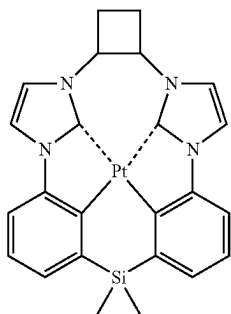
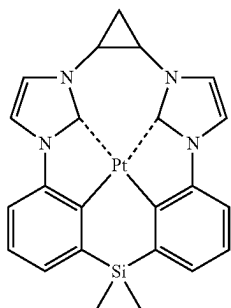
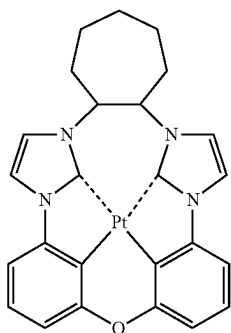
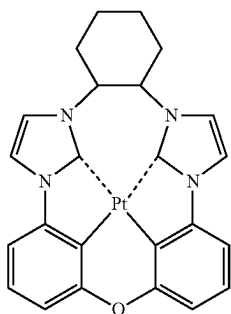
R_1 and R_2 may be linked to form a cyclopentane group, a cyclohexane group, or a fluorene group.

In an example embodiment, the organometallic compound may be one of Compounds 1 to 40:



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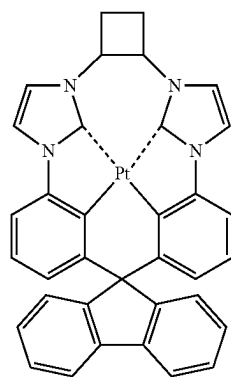
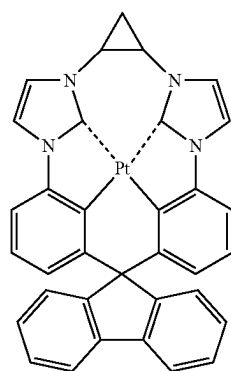
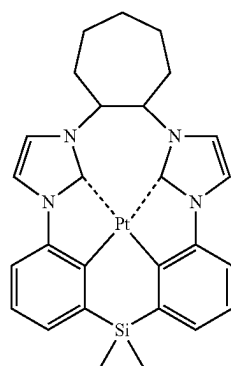
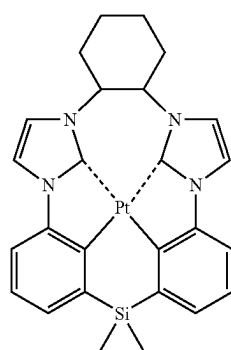
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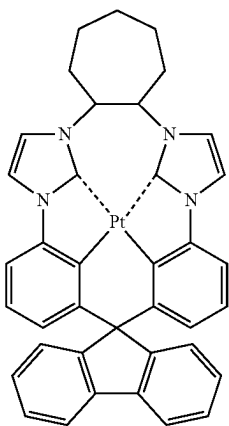
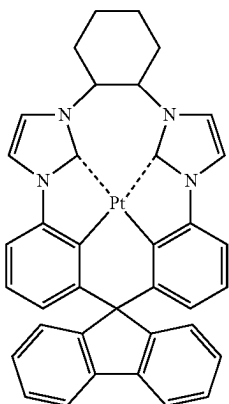
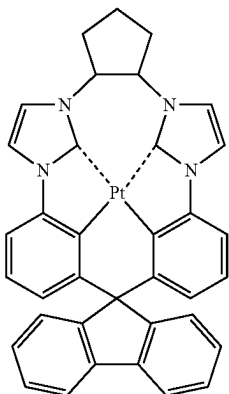
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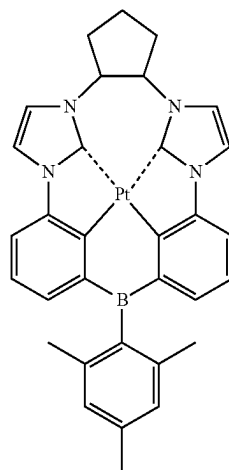
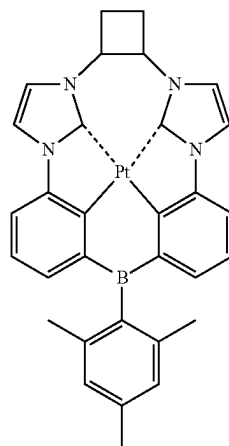
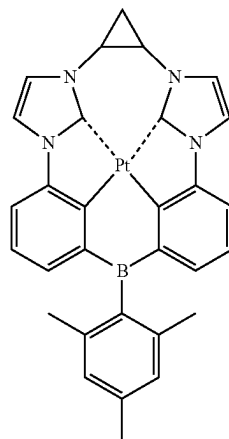
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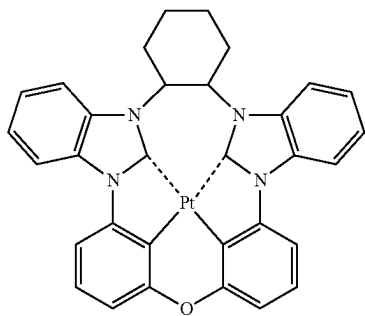
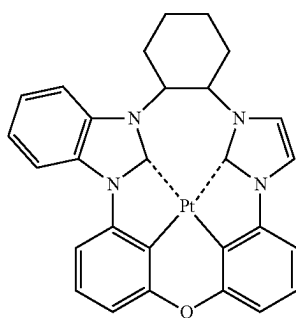
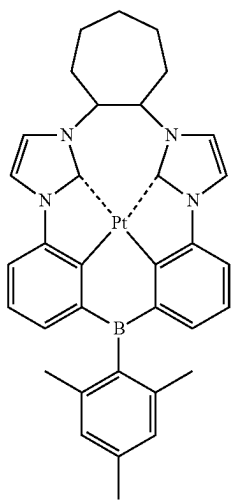
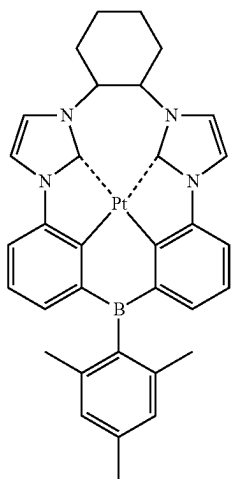
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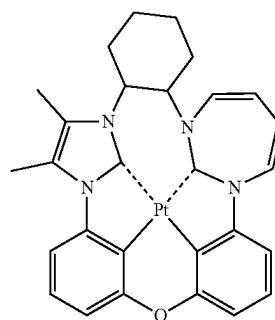
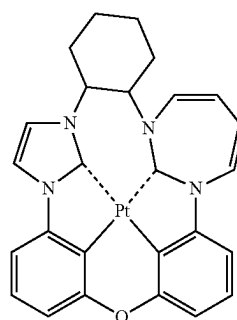
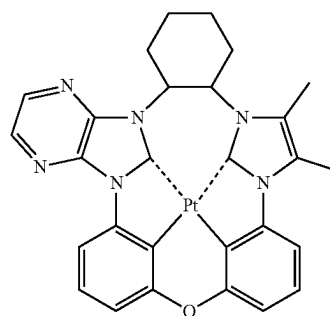
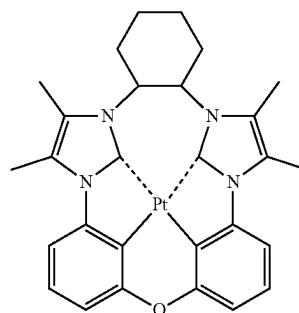
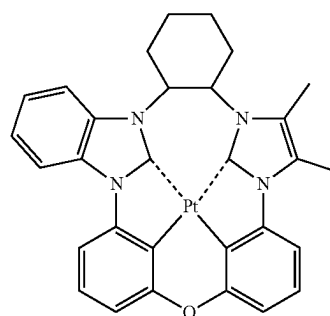
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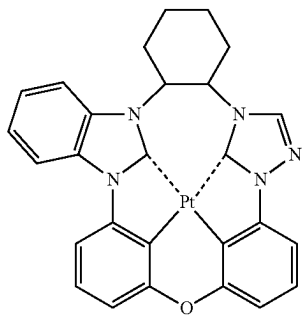
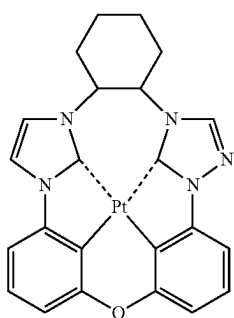
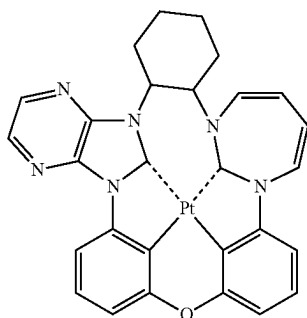
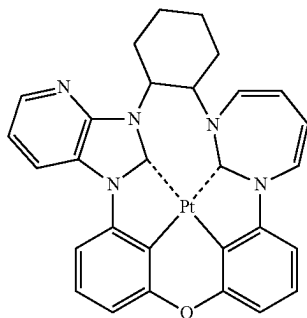
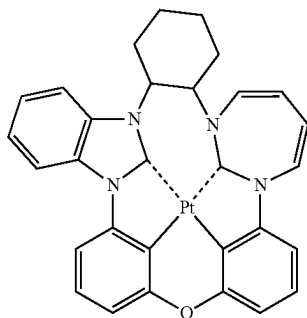
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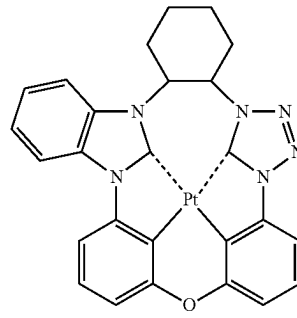
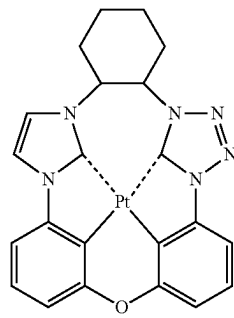
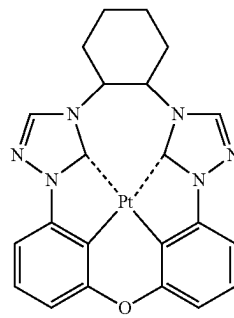
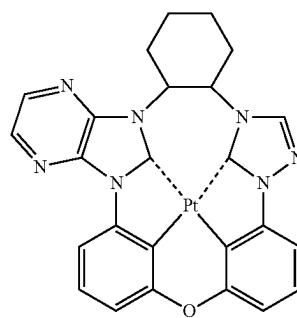
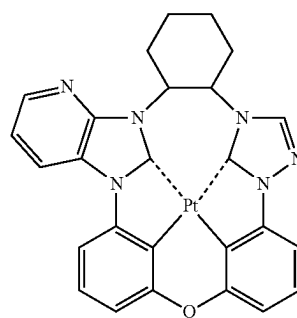
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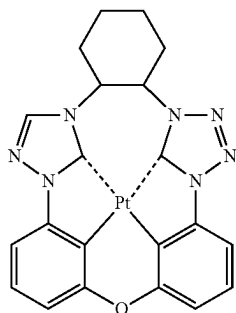
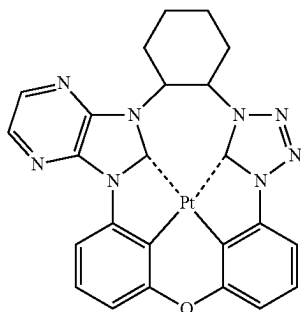
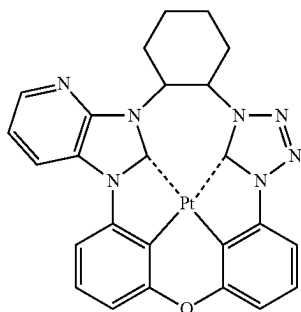
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In an example embodiment, the organometallic compound represented by Formula 1 has a structure in which at least one of $L_{11}(s)$ in the number of a_{11} is a substituted or unsubstituted C_3 - C_{10} cycloalkylene group. Without being bound by theory, it is believed that, since the cycloalkylene group is positioned outside a plane (in which a metal M_{11} and rings A_{11} to A_{14} are included) to form a three-dimensional structure, interaction with other molecules may be suppressed, and the formation of excimer and exciplex may be inhibited. Therefore, when the organometallic compound represented by Formula 1 according to the present example embodiment is applied to the organic light-emitting device, it may be advantageous to energy transfer, luminance efficiency may be high, high color purity may be realized, and lifespan characteristics may be improved.

In addition, when ring A_{10} in the organometallic compound represented by Formula 1 is carbene, high color purity and excellent lifespan characteristics may be provided due to strong σ -donation and π -backdonation effects.

In an example embodiment, the organometallic compound represented by Formula 1 includes a moiety providing a rigid structure. Thus, an organic light-emitting device including the organometallic compound may have high durability and a long lifespan.

The organometallic compound may emit blue light. For example, the organometallic compound may emit blue light

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(bottom emission $CIE_{x,y}$, color coordinates 0.135, 0.15 to 0.30) having a wavelength of maximum emission of about 440 nm to about 500 nm, for example, a wavelength of maximum emission of about 460 nm to about 495 nm. Therefore, the organometallic compound represented by Formula 1 may be usefully used to manufacture an organic light-emitting device that emits blue light.

A synthesis method for the organometallic compound represented by Formula 1 will be apparent to those of skill in the art by referring to the examples below.

At least one of the organometallic compound of Formula 1 may be used between a pair of electrodes of an organic light-emitting device. For example, the organometallic compound may be included in an emission layer. The organometallic compound may act as a dopant in the emission layer.

According to an example embodiment, an organic light-emitting device includes a first electrode, a second electrode facing the first electrode, and an organic layer between the first electrode and the second electrode. The organic layer may include an emission layer. The organic layer may include at least one organometallic compound represented by Formula 1.

The expression “(an organic layer) includes at least one organometallic compound” as used herein may include a case in which “(an organic layer) includes identical compounds represented by Formula 1” and a case in which “(an organic layer) includes two or more different organometallic compounds represented by Formula 1.”

For example, the organic layer may include, as the organometallic compound, only Compound 1. In this regard, Compound 1 may exist in an emission layer of the organic light-emitting device. In an example embodiment, the organic layer may include, as the organometallic compound, Compound 1 and Compound 2. In this regard, Compound 1 and Compound 2 may exist in an identical layer (for example, Compound 1 and Compound 2 may all exist in an emission layer), or different layers (for example, Compound 1 may exist in an emission layer and Compound 2 may exist in an electron transport region).

In an example embodiment,

the first electrode of the organic light-emitting device may be an anode,

the second electrode of the organic light-emitting device may be a cathode,

the organic layer may further include a hole transport region between the first electrode and the emission layer and an electron transport region between the emission layer and the second electrode,

the hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or a combination thereof, and

the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or a combination thereof.

The term “organic layer” as used herein refers to a single layer and/or a plurality of layers disposed between the first electrode and the second electrode of the organic light-emitting device. A material included in the “organic layer” is not limited to an organic material.

In an example embodiment, the emission layer may include an organometallic compound represented by Formula 1, and may further include a host, wherein an amount of the host in the emission layer may be greater than that of the organometallic compound in the emission layer.

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In an example embodiment, the emission layer may further include a host, and the host may include a silyl-containing compound, a phosphine oxide-containing compound, or a combination thereof.

In an example embodiment, the hole transport region may include an electron blocking layer, and the electron blocking layer may include the organometallic compound; or

the electron transport region may include a hole blocking layer, and the hole blocking layer may include the organometallic compound.

In an example embodiment, the hole transport region may include a p-dopant having a lowest unoccupied molecular orbital (LUMO) energy level of about -3.5 eV or less.

In an example embodiment, the electron transport region may include a hole blocking layer, and the hole blocking layer may directly contact the emission layer, and the hole blocking layer may include a phosphine oxide-containing compound or a silyl-containing compound.

DESCRIPTION OF THE FIGURE

The FIGURE is a schematic view of an organic light-emitting device **10** according to an example embodiment. The organic light-emitting device **10** includes a first electrode **110**, an organic layer **150**, and a second electrode **190**.

Hereinafter, the structure of the organic light-emitting device **10** according to an example embodiment and a method of manufacturing the organic light-emitting device **10** will be described in connection with the FIGURE.

[First Electrode **110**]

In the FIGURE, a substrate may be additionally disposed under the first electrode **110** or above the second electrode **190**. The substrate may be a glass substrate or a plastic substrate, each having excellent mechanical strength, thermal stability, transparency, surface smoothness, ease of handling, and water resistance.

The first electrode **110** may be formed by depositing or sputtering a material for forming the first electrode **110** on the substrate. When the first electrode **110** is an anode, the material for a first electrode may be selected from materials with a high work function to facilitate hole injection.

The first electrode **110** may be a reflective electrode, a semi-transmissive electrode, or a transmissive electrode. When the first electrode **110** is a transmissive electrode, a material for forming a first electrode may be selected from indium tin oxide (ITO), indium zinc oxide (ILO), tin oxide (SnO_2), zinc oxide (ZnO), and a combinations thereof. In an example embodiment, when the first electrode **110** is a semi-transmissive electrode or a reflective electrode, a material for forming a first electrode may be selected from magnesium (Mg), silver (Ag), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), and a combinations thereof.

The first electrode **110** may have a single-layered structure, or a multi-layered structure including two or more layers. For example, the first electrode **110** may have a three-layered structure of ITO/Ag/ITO.

[Organic Layer **150**]

The organic layer **150** is disposed on the first electrode **110**. The organic layer **150** may include an emission layer.

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The organic layer **150** may further include a hole transport region between the first electrode **110** and the emission layer, and an electron transport region between the emission layer and the second electrode **190**.

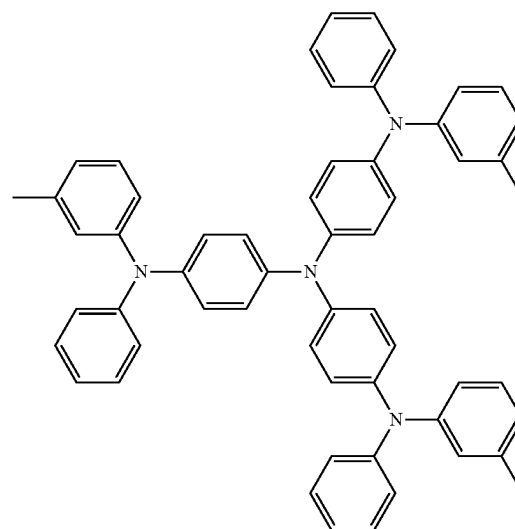
[Hole Transport Region in Organic Layer **150**]

The hole transport region may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

The hole transport region may include at least one layer selected from a hole injection layer, a hole transport layer, an emission auxiliary layer, and an electron blocking layer.

For example, the hole transport region may have a single-layered structure including a single layer including a plurality of different materials, or a multi-layered structure having a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein for each structure, constituting layers may be sequentially stacked from the first electrode **110** in this stated order.

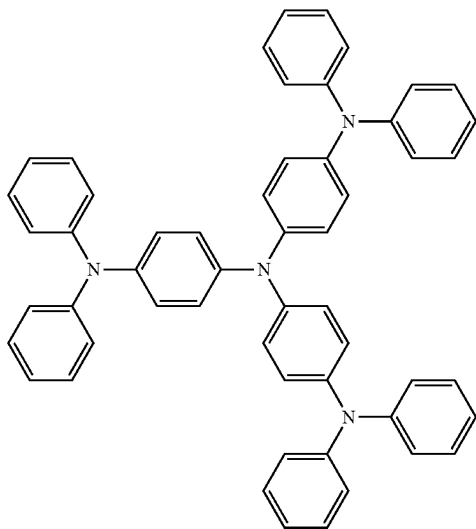
The hole transport region may include at least one selected from m-MTDATA, TDATA, 2-TNATA, NPB (NPB), β -NPB, TPD, Spiro-TPD, Spiro-NPB, methylated-NPB, TAPC, HMTDP, 4,4',4''-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), a compound represented by Formula 201, and a compound represented by Formula 202:



m-MTDATA

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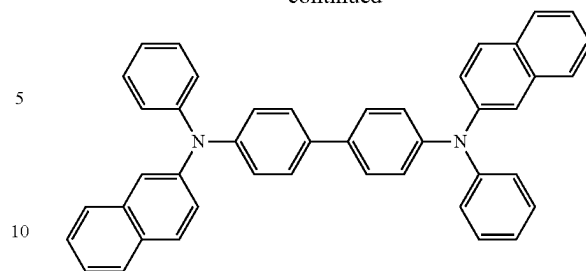
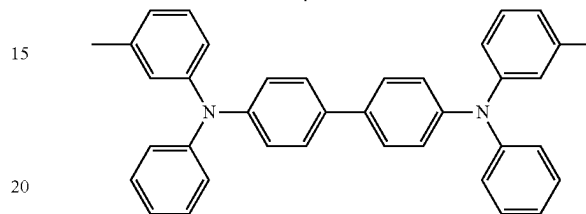
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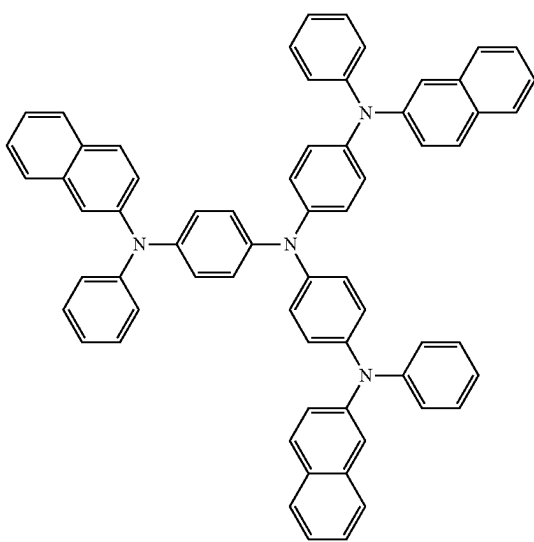
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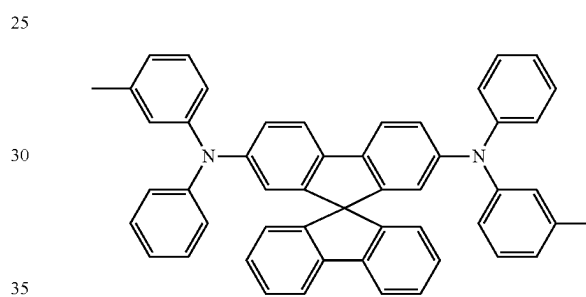
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 β -NPB

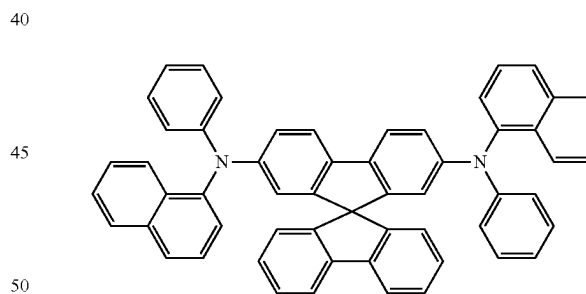
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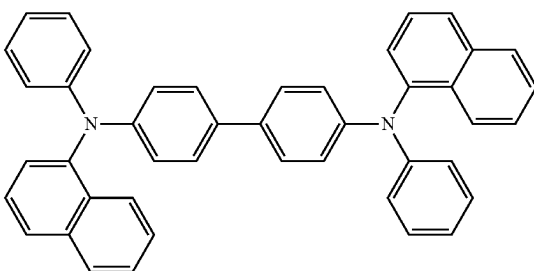
2-TNATA



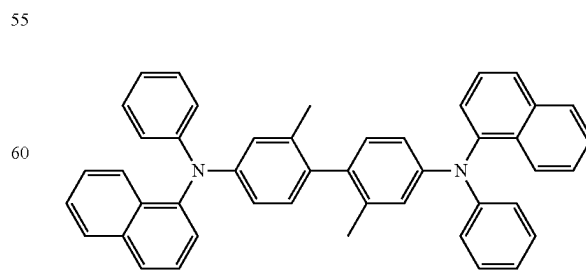
Spiro-TPD



Spiro-NPB



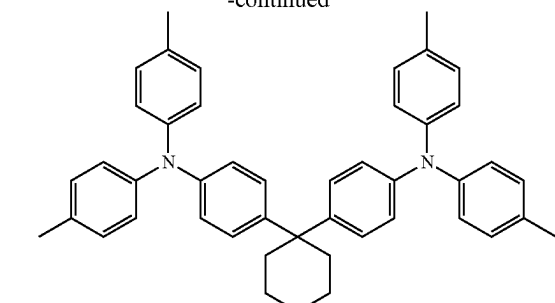
NPB



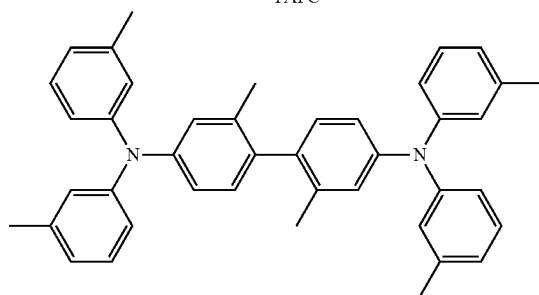
methylated NPB

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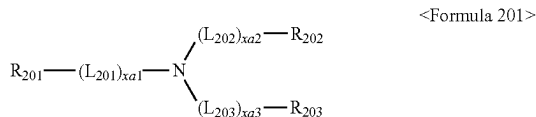
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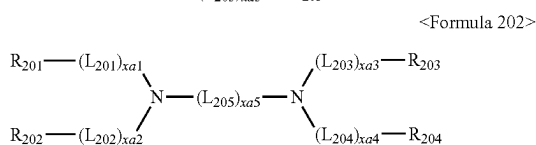
TAPC



HMTDP



<Formula 201>



<Formula 202>

In Formulae 201 and 202,

L_{201} to L_{204} may each independently be selected from a substituted or unsubstituted C_3 - C_{10} cycloalkylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkylene group, a substituted or unsubstituted C_3 - C_{10} cycloalkenylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenylene group, a substituted or unsubstituted C_6 - C_{60} arylene group, a substituted or unsubstituted C_1 - C_{60} heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

L_{205} may be selected from $^*\text{—O—}^*$, $^*\text{—S—}^*$, $^*\text{—N}(\text{Q}_{201})\text{—}^*$, a substituted or unsubstituted C_1 - C_{20} alkylene group, a substituted or unsubstituted C_2 - C_{20} alkenylene group, a substituted or unsubstituted C_3 - C_{10} cycloalkylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkylene group, a substituted or unsubstituted C_3 - C_{10} cycloalkenylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenylene group, a substituted or unsubstituted C_6 - C_{60} arylene group, a substituted or unsubstituted C_1 - C_{60} heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

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$\text{xa}1$ to $\text{xa}4$ may each independently be an integer of 0 to 3,

$\text{xa}5$ may be an integer of 1 to 10, and

R_{201} to R_{204} and Q_{201} may each independently be selected from a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{60} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

For example, in Formula 202, R_{201} and R_{202} may be linked via a single bond, a dimethyl-methylene group, or a diphenyl-methylene group, and R_{203} and R_{204} may be linked via a single bond, a dimethyl-methylene group, or a diphenyl-methylene group.

In an example embodiment, in Formulae 201 and 202,

L_{201} to L_{205} may each independently be selected from: a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylenylene group, a heptalenylene group, an indacenylene group, an acenaphthenylene group, a fluorenylene group, a spirobifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysénylene group, a naphthacenylenylene group, a picenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronénylene group, an ovalenylene group, a thiophenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylene group, a dibenzofuranylenylene group, a dibenzothiophenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylene group, and a pyridinylene group; and

a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylenylene group, a heptalenylene group, an indacenylene group, an acenaphthenylene group, a fluorenylene group, a spirobifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysénylene group, a naphthacenylenylene group, a picenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronénylene group, an ovalenylene group, a thiophenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylene group, a dibenzofuranylenylene group, a dibenzothiophenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylene group, and a pyridinylene group, each substituted with at least one selected from deuterium, —F , —Cl , —Br , —I , a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl

group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃), and —N(Q₃₁)(Q₃₂), and

Q₃₁ to Q₃₃ may each independently be selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

In an example embodiment, xa1 to xa4 may each independently be 0, 1, or 2.

In an example embodiment, xa5 may be 1, 2, 3, or 4.

In an example embodiment, R₂₀₁ to R₂₀₄ and Q₂₀₁ may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group; and

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br,

—I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃), and —N(Q₃₁)(Q₃₂), and

Q₃₁ to Q₃₃ are the same as described above.

In an example embodiment, in Formula 201, at least one selected from R₂₀₁ to R₂₀₃ may each independently be selected from:

a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; and

a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group.

In an example embodiment, in Formula 202, i) R₂₀₁ and R₂₀₂ may be linked via a single bond, and/or ii) R₂₀₃ and R₂₀₄ may be linked via a single bond.

In an example embodiment, in Formula 202, at least one selected from R₂₀₁ to R₂₀₄ may each independently be selected from:

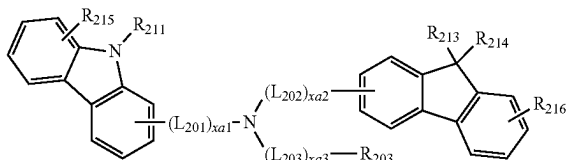
a carbazolyl group; and

a carbazolyl group substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group.

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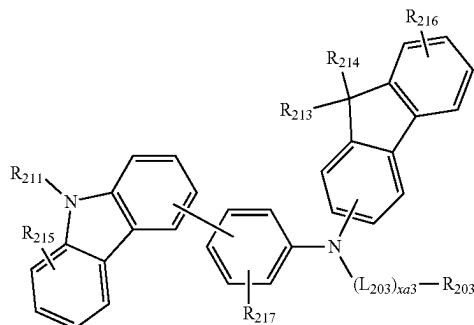
The compound represented by Formula 201 may be represented by Formula 201A:

<Formula 201A> 5



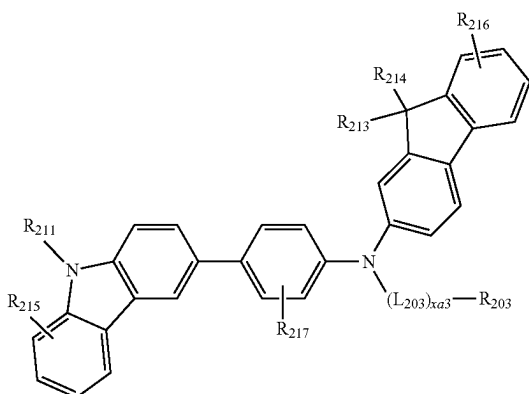
For example, the compound represented by Formula 201 may be represented by Formula 201A(1):

<Formula 201A(1)> 25



For example, the compound represented by Formula 201 may be represented by Formula 201A-1:

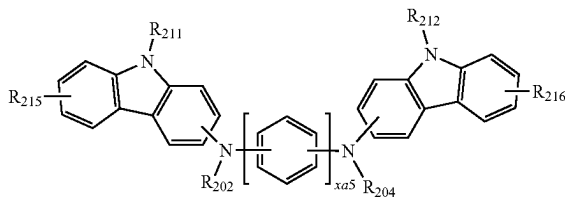
<Formula 210A-1> 50



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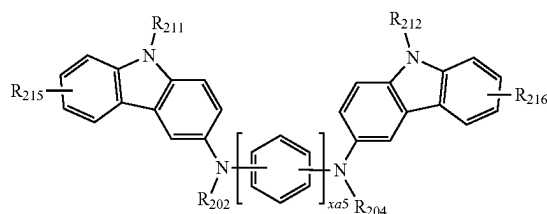
For example, the compound represented by Formula 202 may be represented by Formula 202A:

<Formula 202A>



For example, the compound represented by Formula 202 may be represented by Formula 202A-1:

<Formula 202A-1>



In Formulae 201A, 201A(1), 201A-1, 202A, and 202A-1, L_{201} to L_{203} , $xa1$ to $xa3$, $xa5$ and R_{202} , to R_{204} are the same as described above,

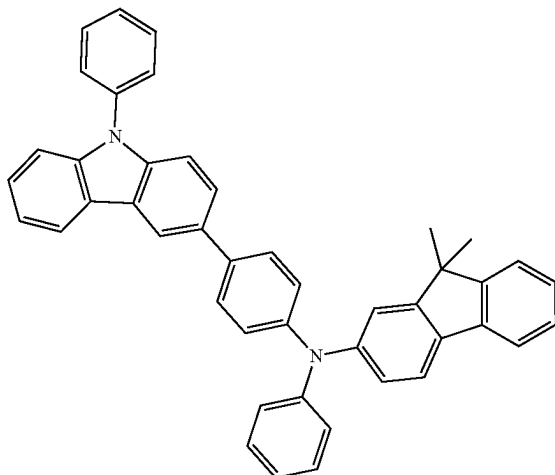
R_{211} and R_{212} may each independently be the same as defined in connection with R_{203} , and

R_{213} to R_{217} may each independently be selected from hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C_1 - C_{10} alkyl group, a phenyl group substituted with $-F$, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthenyl group, a fluorenyl group, a Spiro-bifluorenyl group, a benzo-fluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzo-furanyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

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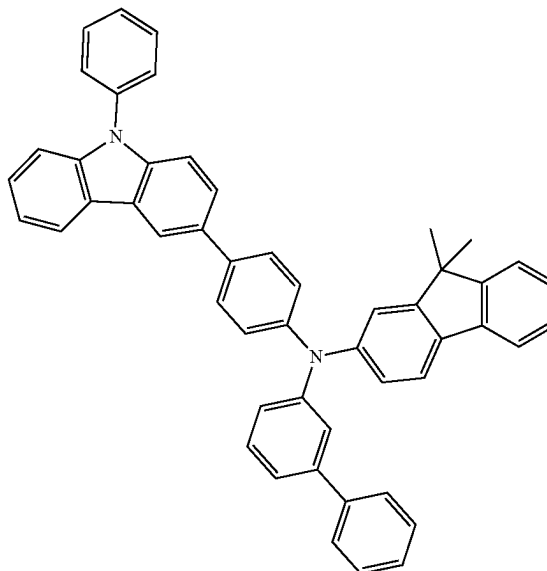
The hole transport region may include at least one compound selected from Compounds HT1 to HT39:

HT1

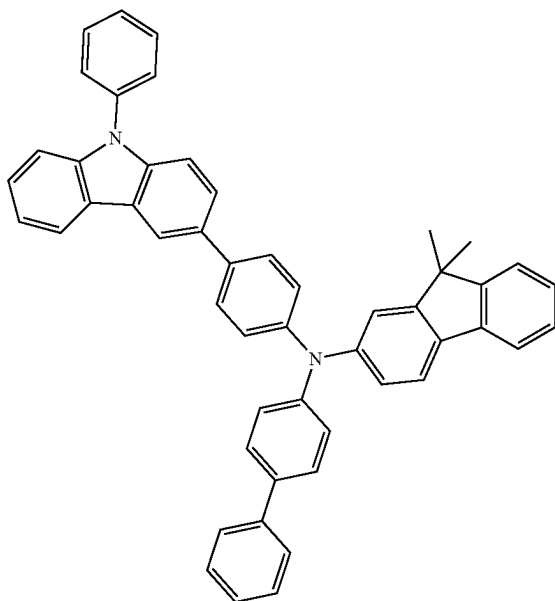


48

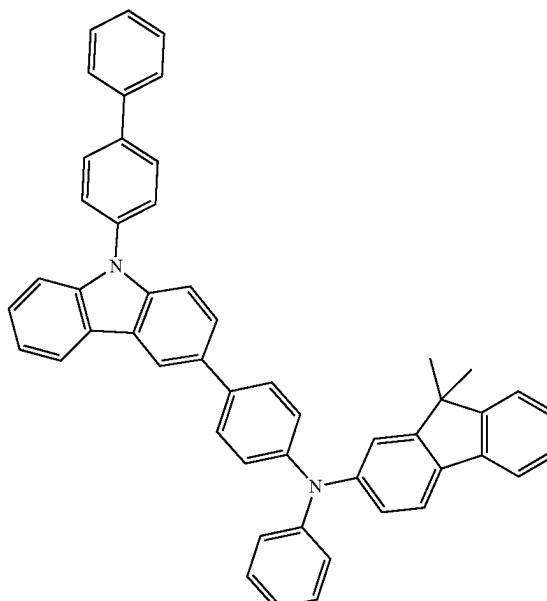
HT2



HT3

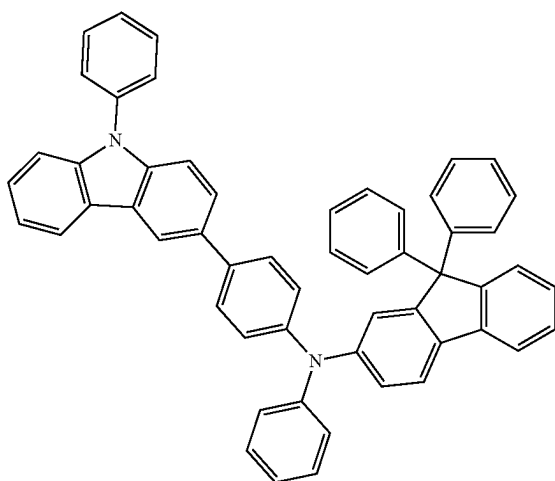


HT4



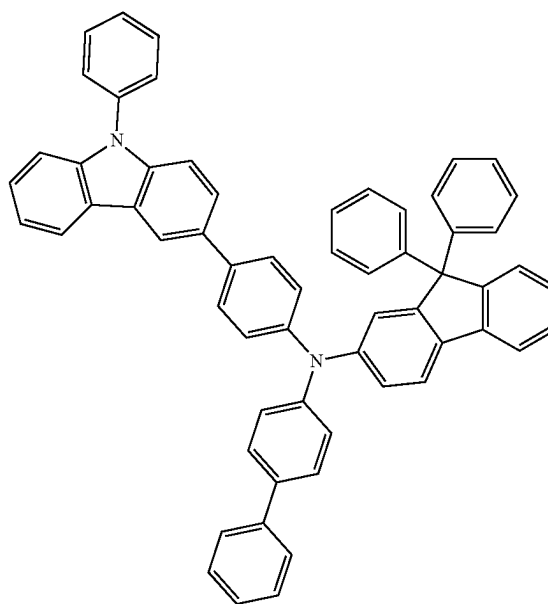
49

-continued
HT5

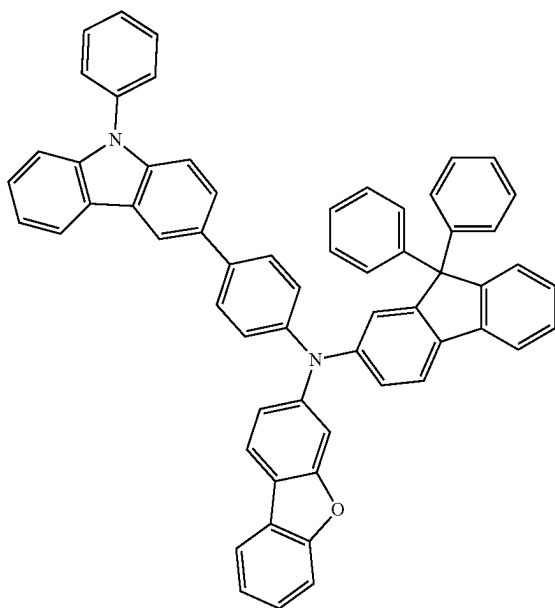


50

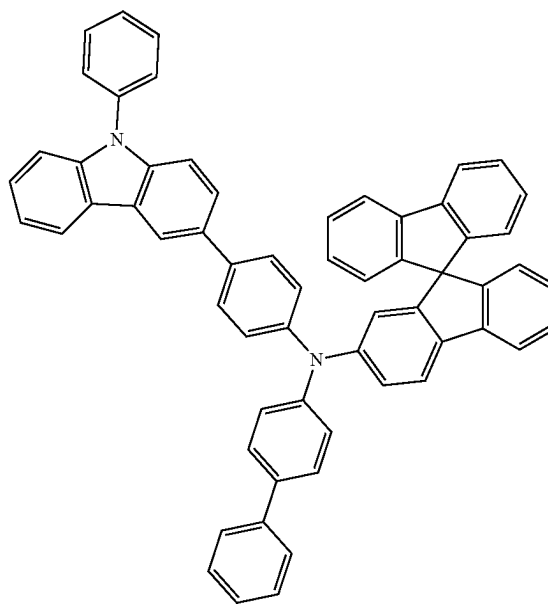
HT6



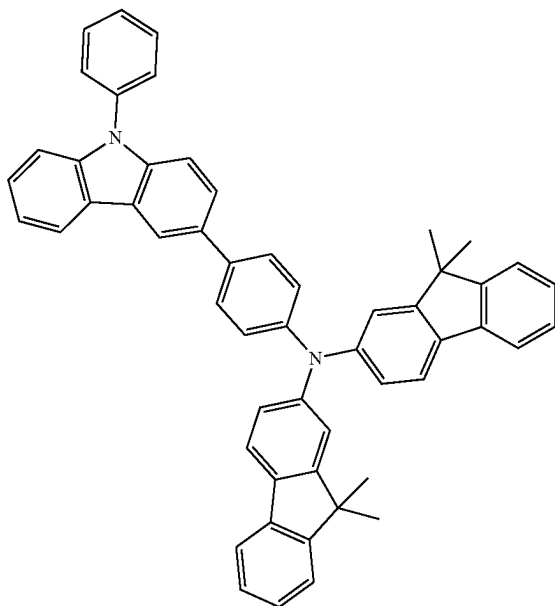
HT7



HT8

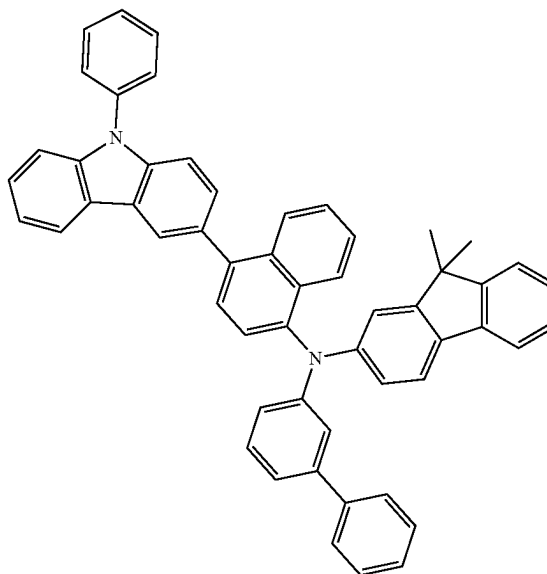


51

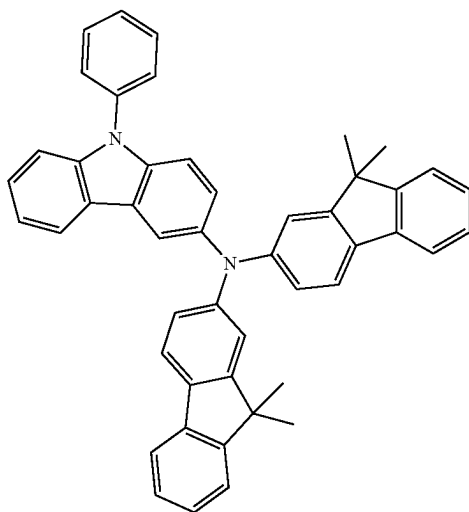
-continued
HT9

52

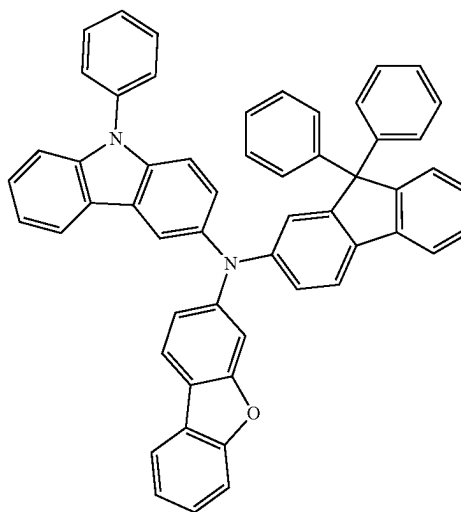
HT10



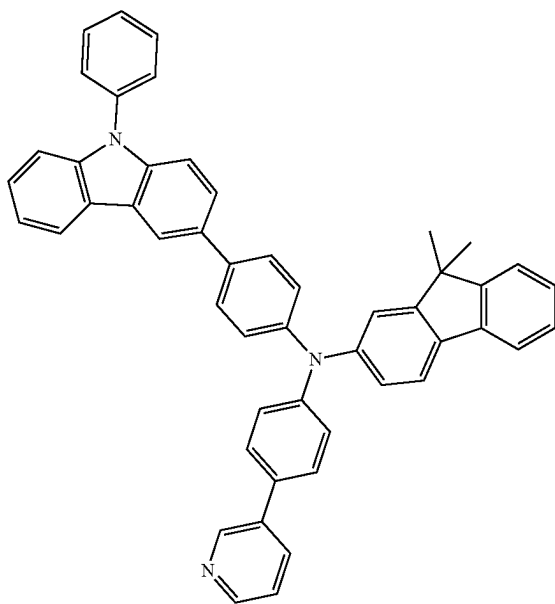
HT11



HT12

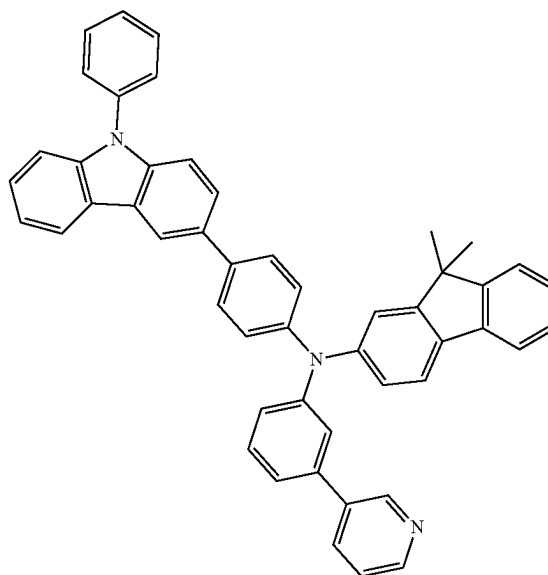


53

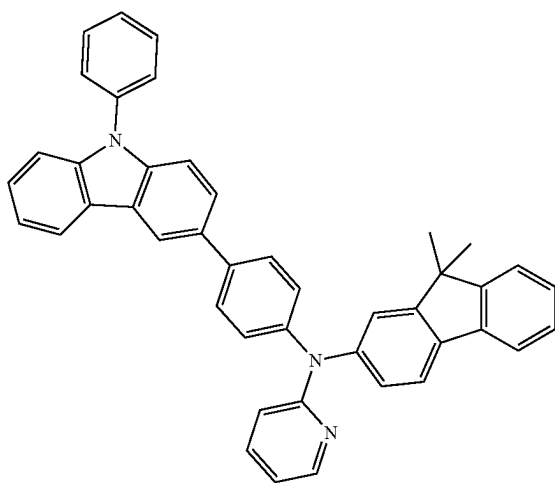
-continued
HT13

54

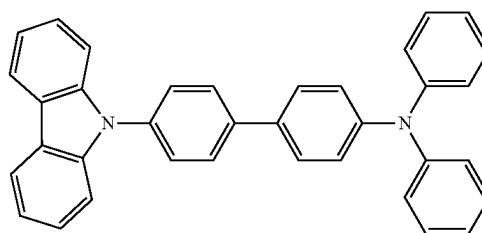
HT14



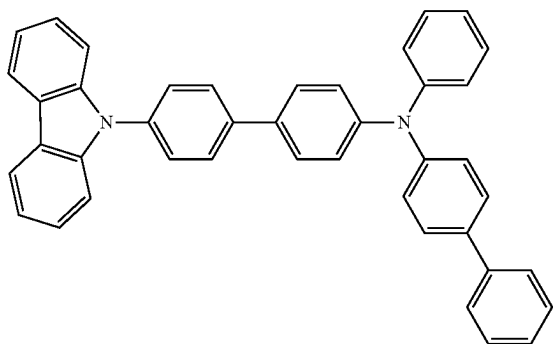
HT15



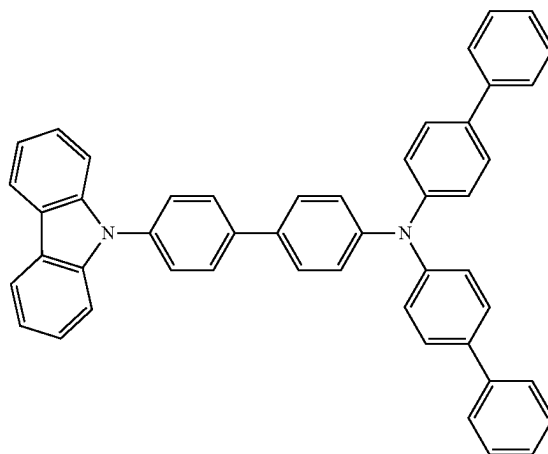
HT16



HT17



HT18

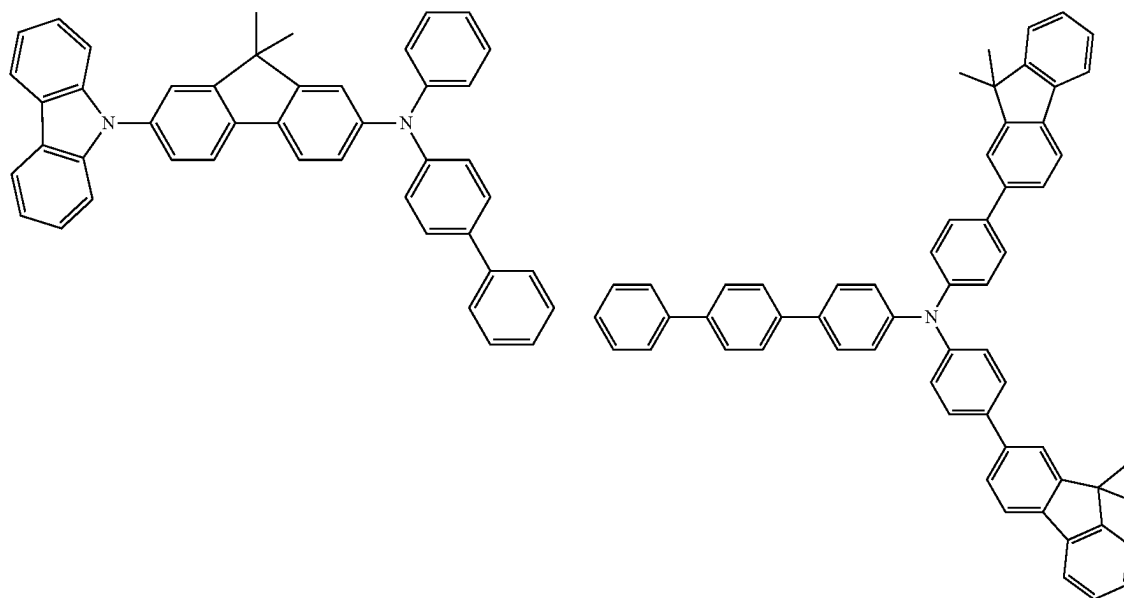


55

56

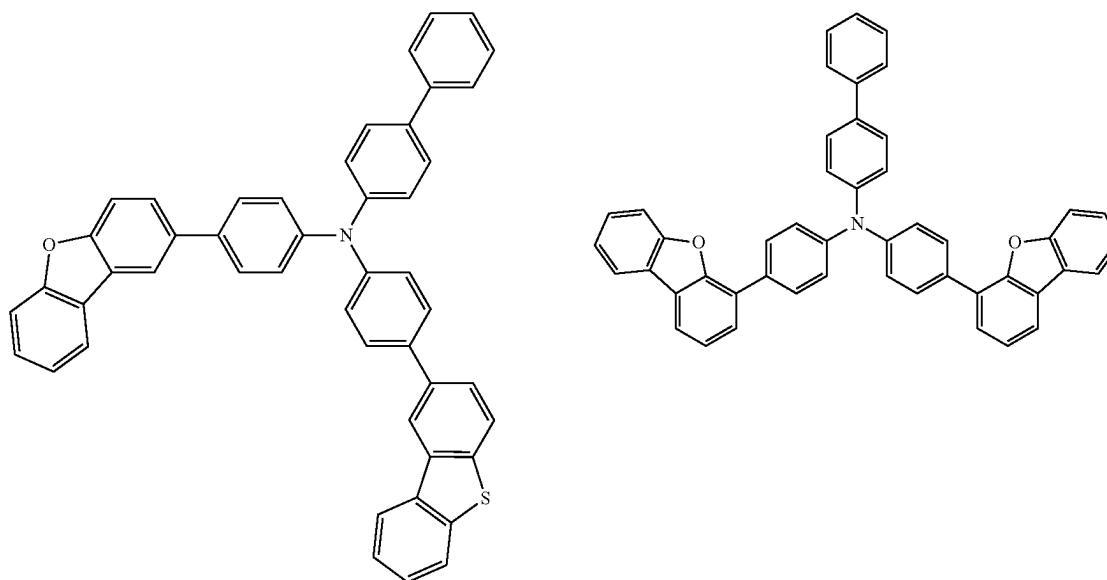
-continued
HT19

HT20



HT21

HT22

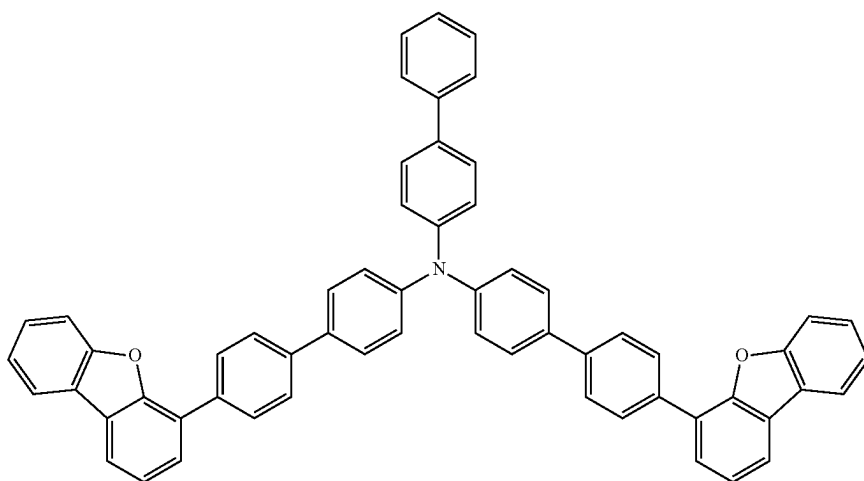


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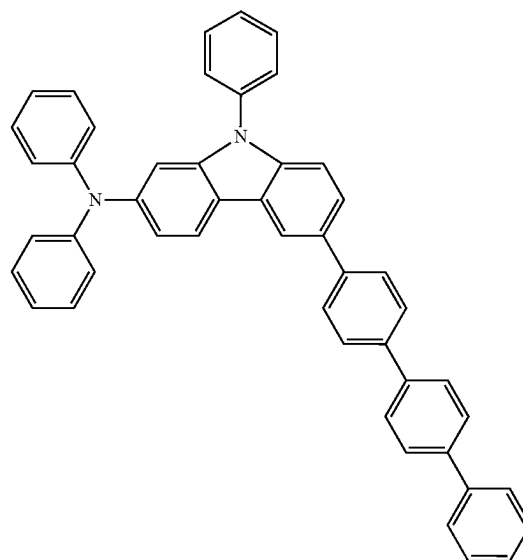
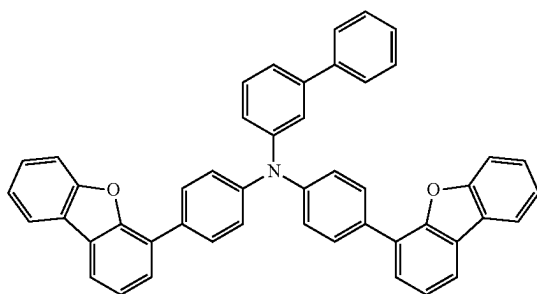
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HT23



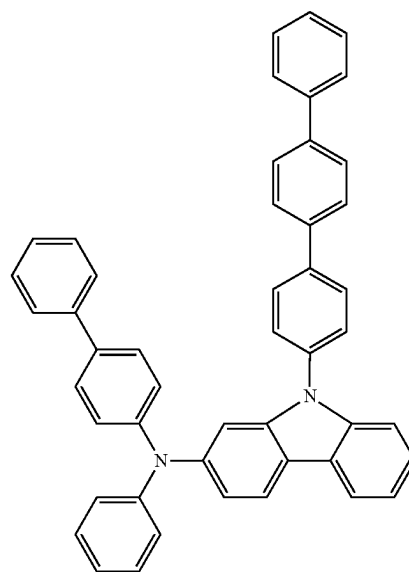
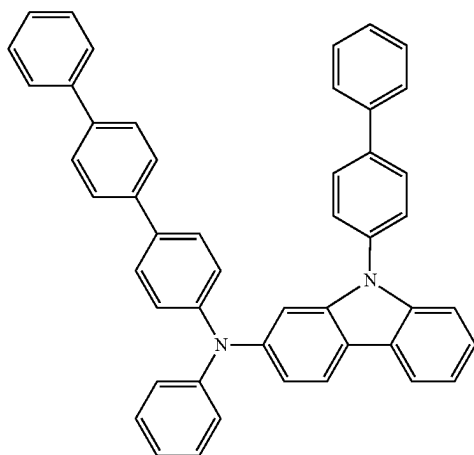
HT24

HT25



HT26

HT27

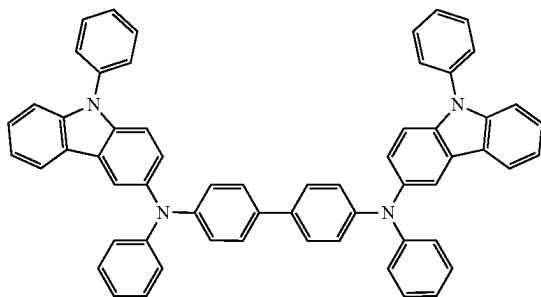


59

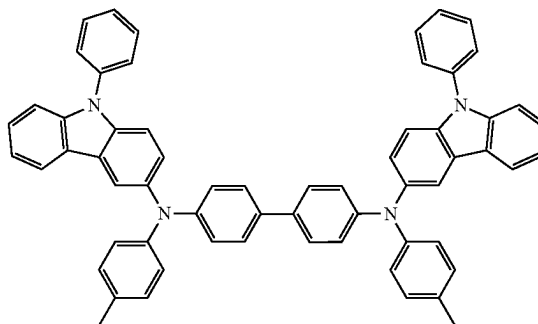
60

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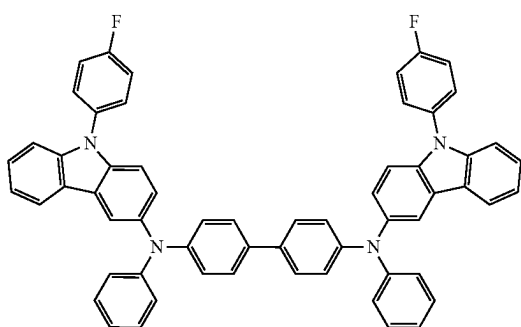
HT28



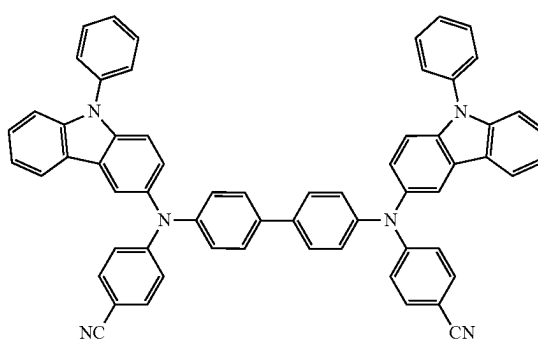
HT29



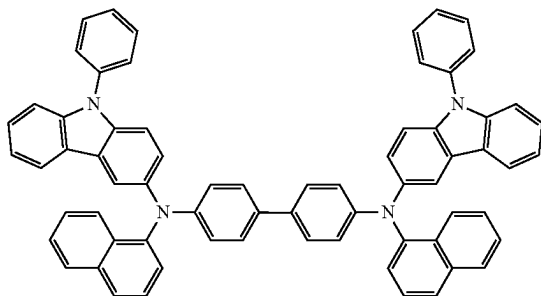
HT30



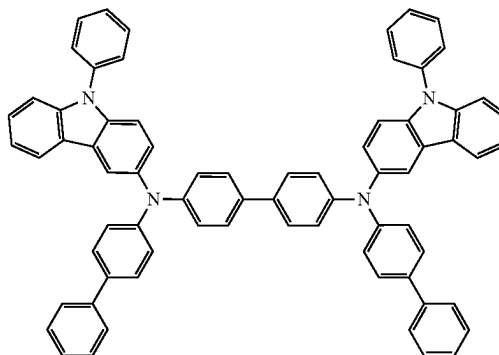
HT31



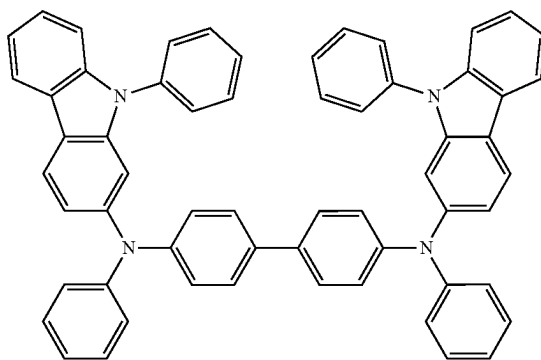
HT32



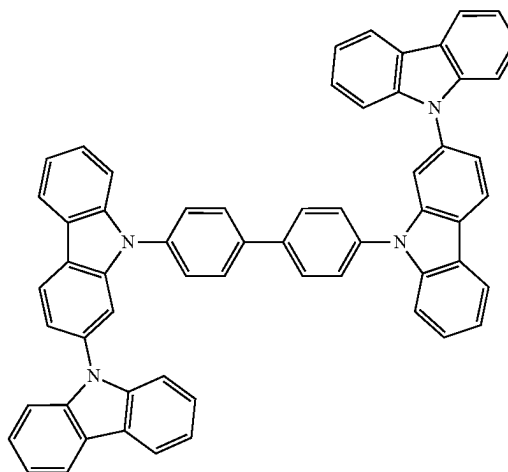
HT33



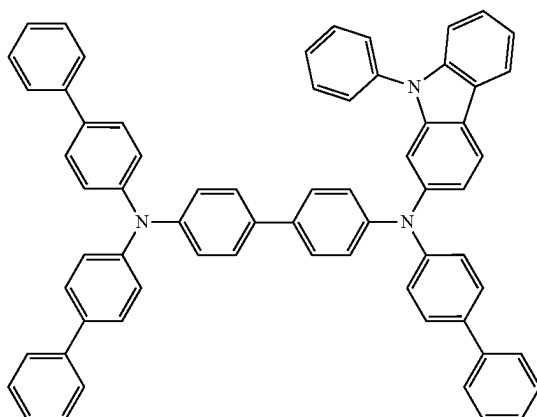
HT34



HT35

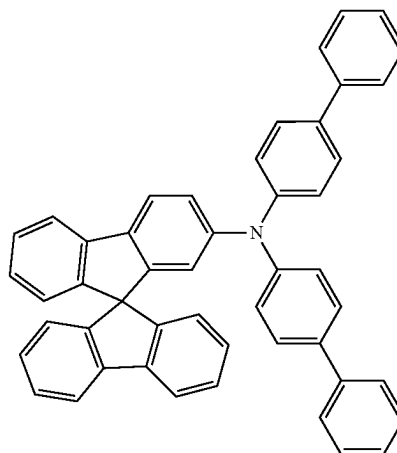


61

-continued
HT36

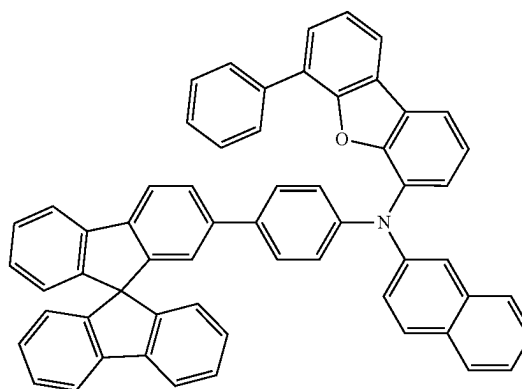
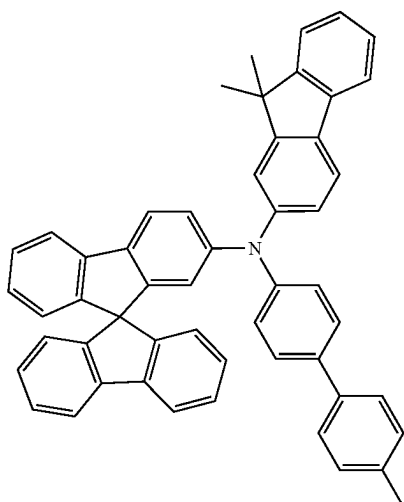
62

HT37



HT38

HT39



A thickness of the hole transport region may be in a range of about 100 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å. When the hole transport region includes at least one of a hole injection layer and a hole transport layer, a thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, for example, about 100 Å to about 1,000 Å, and a thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å, for example about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer, and the hole transport layer are within these ranges, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

The emission auxiliary layer may increase light-emission efficiency by compensating for an optical resonance distance according to the wavelength of light emitted by an emission layer, and the electron blocking layer may block the flow of electrons from an electron transport region. The emission auxiliary layer and the electron blocking layer may include the materials as described above.

[p-Dopant]

The hole transport region may further include, in addition to these materials, a charge-generation material for the improvement of conductive properties. The charge-generation material may be homogeneously or non-homogeneously dispersed in the hole transport region.

The charge-generation material may be, for example, a p-dopant.

In an example embodiment, the p-dopant may have a LUMO energy level of about -3.5 eV or less.

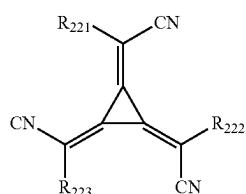
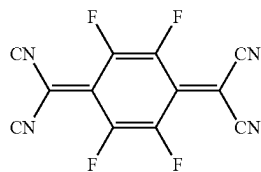
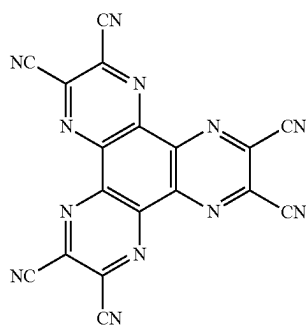
The p-dopant may include at least one selected from a quinone derivative, a metal oxide, and a cyano group-containing compound.

For example, the p-dopant may include at least one selected from:

- a quinone derivative, such as tetracyanoquinodimethane (TCNQ) or 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4-TCNQ);
- a metal oxide, such as tungsten oxide or molybdenum oxide;
- 1,4,5,8,9,12-hexaazatriphenylene-hexacarbonitrile (HAT-CN); and

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a compound represented by Formula 221:



In Formula 221,

R₂₂₁ to R₂₂₃ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, wherein at least one selected from R₂₂₁ to R₂₂₃ may have at least one substituent selected from a cyano group, —F, —Cl, —Br, —I, a C₁-C₂₀ alkyl group substituted with —F, a C₁-C₂₀ alkyl group substituted with —Cl, a C₁-C₂₀ alkyl group substituted with —Br, and a C₁-C₂₀ alkyl group substituted with —I.

[Emission Layer in Organic Layer 150]

When the organic light-emitting device 10 is a full-color organic light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, or a blue emission layer, according to a sub-pixel. In an example embodiment, the emission layer may have a stacked structure of two or more layers selected from a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers contact each other or are separated from each other. In an example embodiment, the emission layer may include two or more materials selected from a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials are mixed with each other in a single layer to emit white light.

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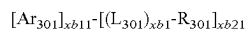
The emission layer may include a host and a dopant. The dopant may include at least one selected from a phosphorescent dopant and a fluorescent dopant. The phosphorescent dopant may include the organometallic compound represented by Formula 1:

In the emission layer, an amount of the dopant may be in a range of about 0.01 parts by weight to about 15 parts by weight based on 100 parts by weight of the host.

A thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å, for example, about 200 Å to about 600 Å. When the thickness of the emission layer is within this range, excellent light-emission characteristics may be obtained without a substantial increase in driving voltage.

[Host in Emission Layer]

In an example embodiment, the host may include a compound represented by Formula 301:



<Formula 301>

In Formula 301,

Ar₃₀₁ may be a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group,

xb11 may be 1, 2, or 3,

L₃₀₁ may be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkylene group, a substituted or unsubstituted C₃-C₁₀ cycloalkenylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenylene group, a substituted or unsubstituted C₆-C₆₀ arylene group, a substituted or unsubstituted C₁-C₆₀ heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

xb1 may be an integer of 0 to 5,

R₃₀₁ may be selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₃₀₁)(Q₃₀₂)(Q₃₀₃), —N(Q₃₀₁)(Q₃₀₂), —B(Q₃₀₁)(Q₃₀₂), —C(=O)(Q₃₀₁), —S(=O)₂(Q₃₀₁), and —P(=O)(Q₃₀₁)(Q₃₀₂),

xb21 may be an integer of 1 to 5,

Q₃₀₁ to Q₃₀₃ may each independently be selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

In an example embodiment, Ar₃₀₁ in Formula 301 may be selected from:

a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthra-

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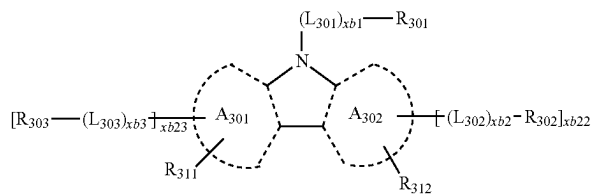
cene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group; and

a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)(Q₃₂), and

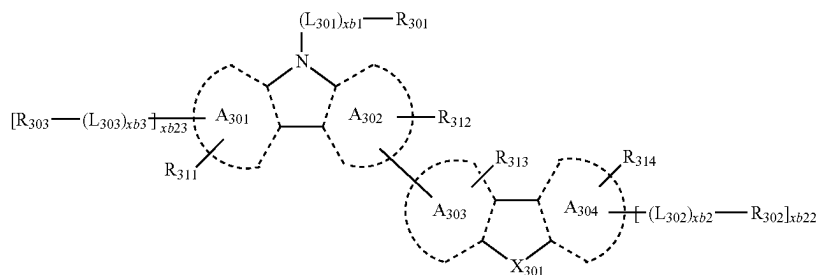
Q₃₁ to Q₃₃ may each independently be selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

In Formula 301, when xb11 is two or more, two or more Ar₃₀₁(s) may be linked via a single bond.

In an example embodiment, the compound represented by Formula 301 may be represented by Formula 301-1 or 301-2:



<Formula 301-1>



<Formula 301-2>

In Formulae 301-1 to 301-2,

A₃₀₁ to A₃₀₄ may each independently be selected from a benzene group, a naphthalene group, a phenanthrene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a pyridine group, a pyrimidine group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, an indole group, a carbazole group, a benzocarbazole group, a dibenzocarbazole group, a furan group, a benzofuran group, a dibenzofuran group, a naphthofuran group, a benzonaphthofuran group, a dinaphthofuran group, a thiophene group, a benzothiophene group, a dibenzothiophene group, a naphtho-

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thiophene group, a benzonaphthothiophene group, and a dinaphthothiophene group,

X₃₀₁ may be O, S, or N-[(L-304)xb4-R₃₀₄],

R₃₁₁ to R₃₁₄ may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group —Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)(Q₃₂),

xb22 and xb23 may each independently be 0, 1, or 2,

L₃₀₁, xb1, R₃₀₁, and Q₃₁ to Q₃₃ are the same as described above,

L₃₀₂ to L₃₀₄ may each independently be the same as defined in connection with L₃₀₁,

xb2 to xb4 may each independently be the same as defined in connection with xb1, and

R₃₀₂ to R₃₀₄ may each independently be the same as defined in connection with R₃₀₁.

For example, in Formulae 301, 301-1, and 301-2, L₃₁ to L₃₀₄ may each independently be selected from:

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylene group, a quinolinylene group, an isoquinolinylene group, a benzoquinolinylene group, a phtha-

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lazinylene group, a naphthyridinylene group, a quinoxalinylene group, a quinazolininylene group, a cinnolininylene group, a phenanthridinylene group, an acridinylene group, a phenanthrolinylene group, a phenazinylene group, a benzimidazolylene group, an isobenzothiazolylene group, a benzoxazolylene group, an isobenzoxazolylene group, a triazolylene group, a tetrazolylene group, an imidazopyridinylene group, an imidazopyrimidininylene group, and an azacarbazolylene group; and

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylene group, a thiophenylenylene group, a furanylene group, a carbazolylene group, an indolylene group, an isoindolylene group, a benzofuranylene group, a benzothiophenylenylene group, a dibenzofuranylene group, a dibenzothiophenylenylene group, a benzocarbazolylene group, a dibenzocarbazolylene group, a dibenzosilolylene group, a pyridinylene group, an imidazolylene group, a pyrazolylene group, a thiazolylene group, an isothiazolylene group, an oxazolylene group, an isoxazolylene group, a thiadiazolylene group, an oxadiazolylene group, a pyrazinylene group, a pyrimidinylene group, a pyridazinylene group, a triazinylene group, a quinolinylene group, an isoquinolinylene group, a benzoquinolinylene group, a phthalazinylene group, a naphthyridinylene group, a quinoxalinylene group, a quinazolinylene group, a cinnolinylene group, a phenanthridinylene group, an acridinylene group, a phenanthrolinylene group, a phenazinylene group, a benzimidazolylene group, an isobenzothiazolylene group, a benzoxazolylene group, an isobenzoxazolylene group, a triazolylene group, a tetrazolylene group, an imidazopyridinylene group, an imidazopyrimidininylene group, and an azacarbazolylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazoyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, an imidazopyrimidinyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazoyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an

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dazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazoyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)(Q₃₂), and

Q₃₁ to Q₃₃ are the same as described above.

In an example embodiment, in Formulae 301, 301-1, and 301-2, R₃₀₁ to R₃₀₄ may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazoyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazoyl group; and

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazoyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazoyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an

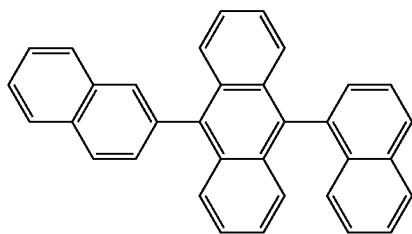
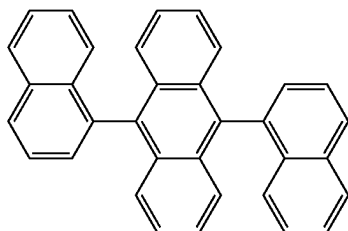
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amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoanthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuran group, a benzothiophenyl group, a benzofuran group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, $-Si(Q_{31})(Q_{32})(Q_{33})$, $-N(Q_{31})(Q_{32})$, $-B(Q_{31})(Q_{32})$, $-C(=O)(Q_{31})$, $-S(=O)_2(Q_{31})$, and $-P(=O)(Q_{31})(Q_{32})$, and

Q_{31} to Q_{33} are the same as described above.

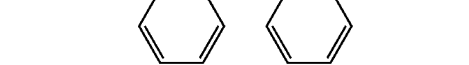
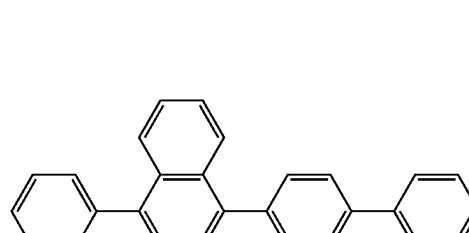
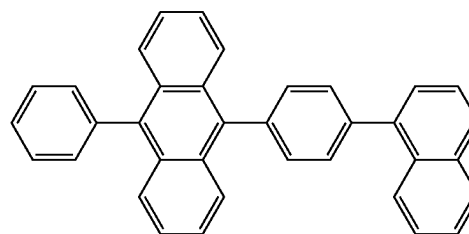
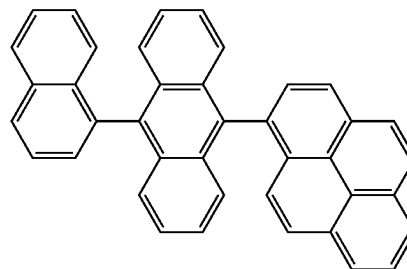
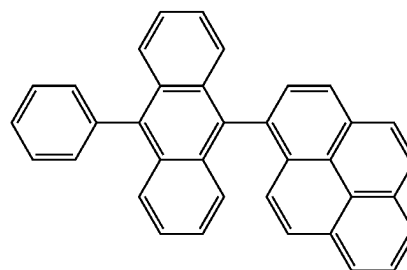
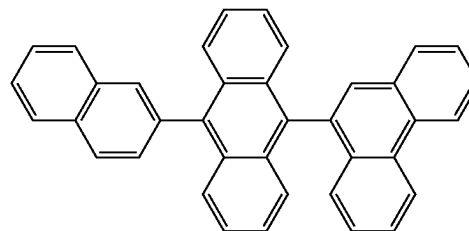
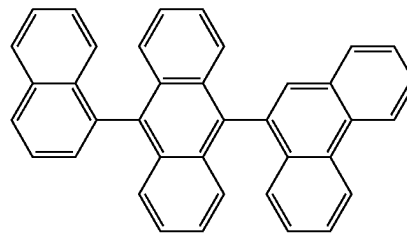
In an example embodiment, the host may include an alkaline earth metal complex. For example, the host may be selected from a Be complex (for example, Compound H55), a Mg complex, and a Zn complex.

The host may include at least one selected from 9,10-di(2-naphthyl)anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl)anthracene (MADN), 9,10-di(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 1,3-di-9-carbazolylbenzene (mCP), 1,3,5-tri(carbazol-9-yl)benzene (TCP), and Compounds H1 to H55:



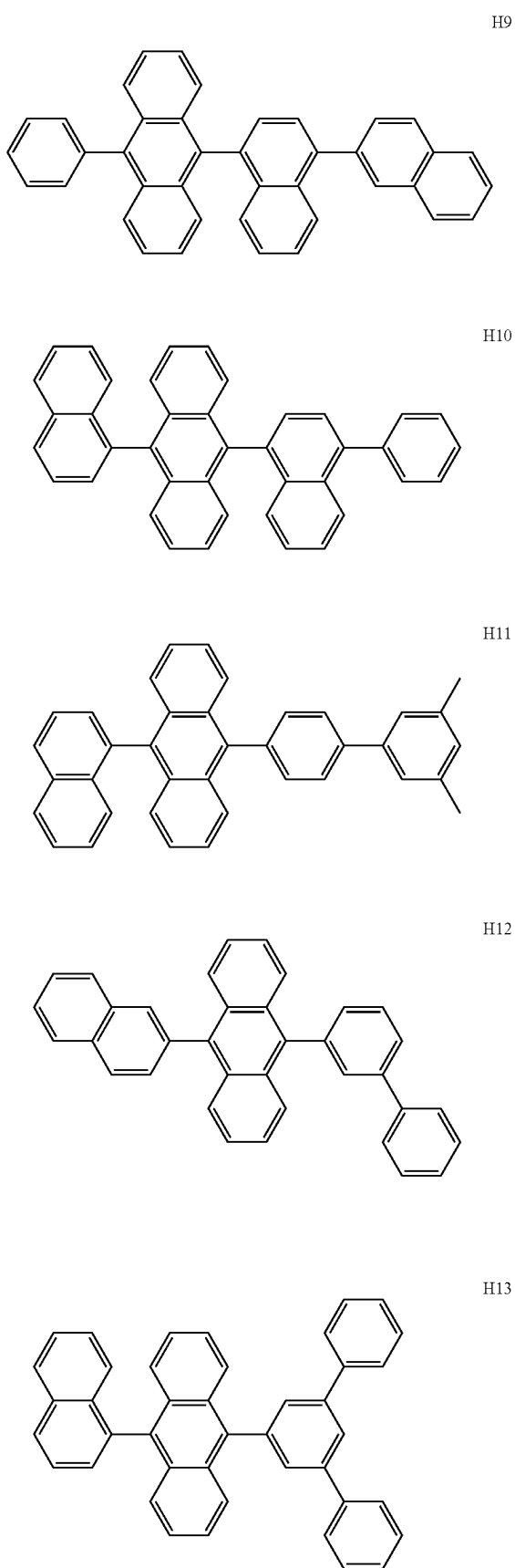
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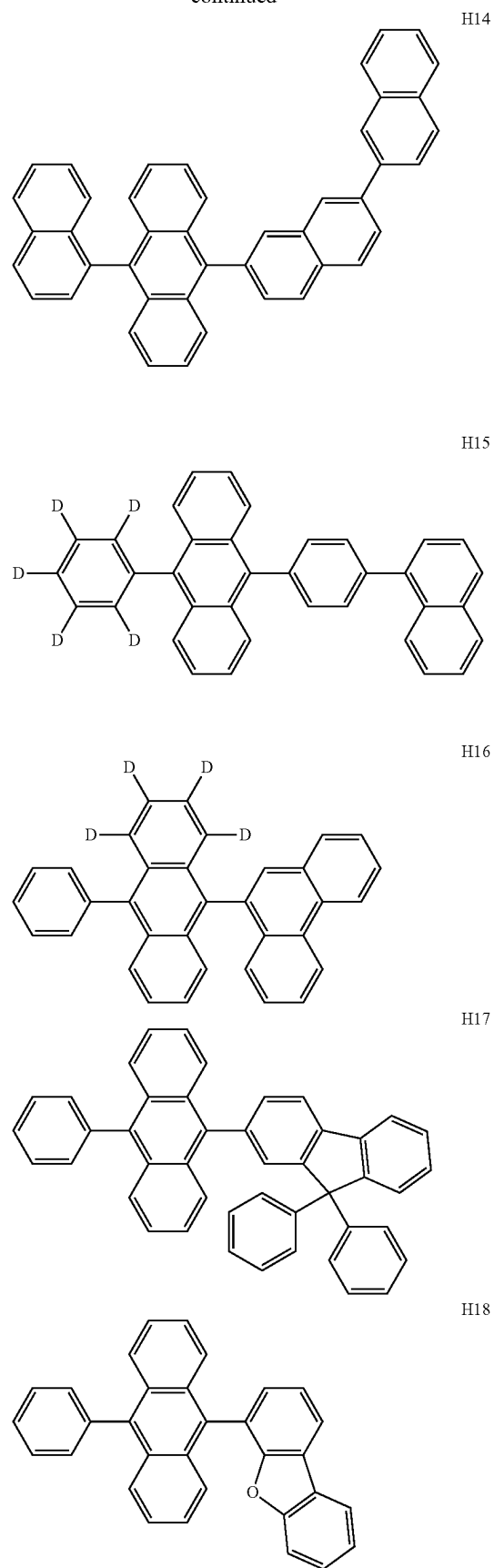


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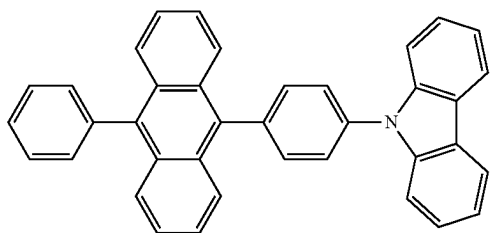
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H19

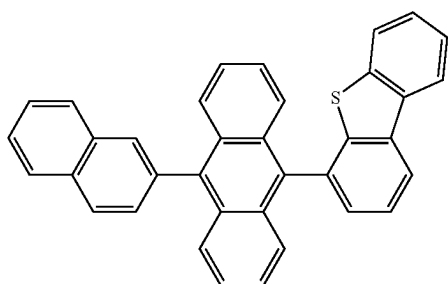


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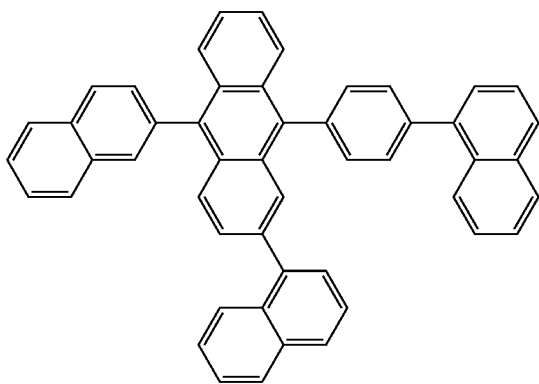
H20



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H21

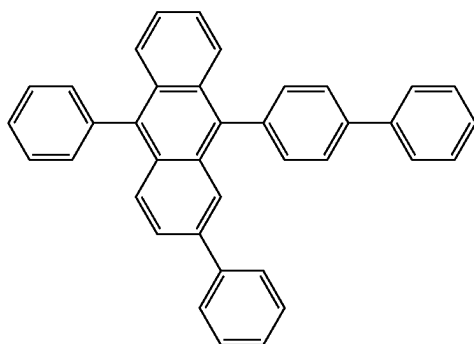


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H22



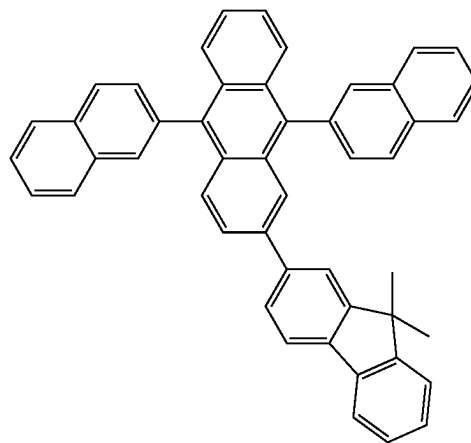
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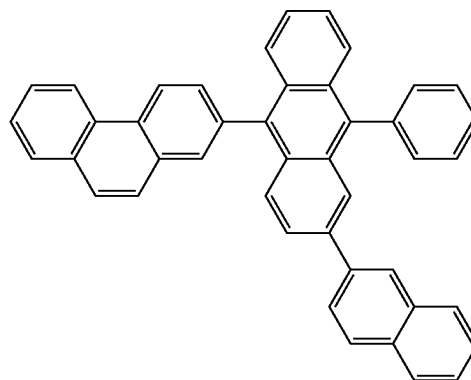
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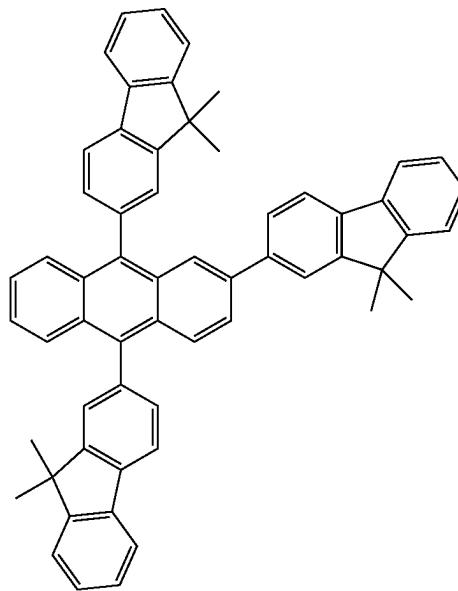
H23



H24

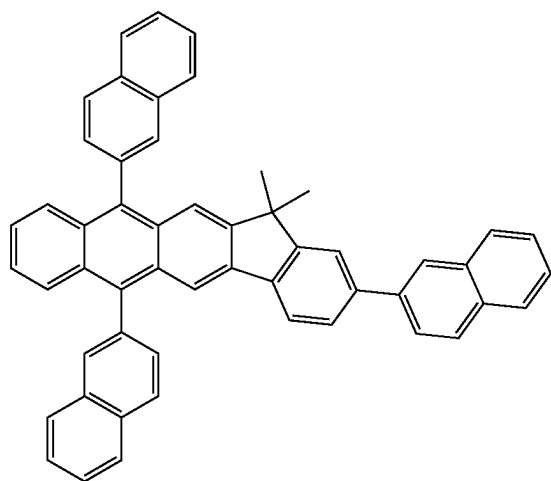
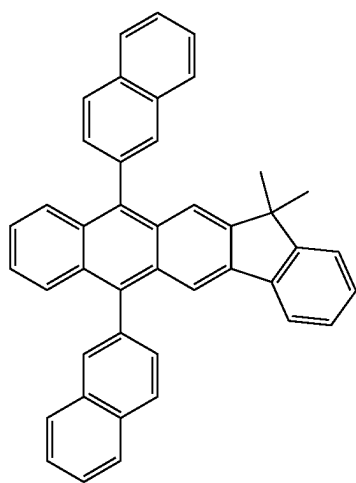
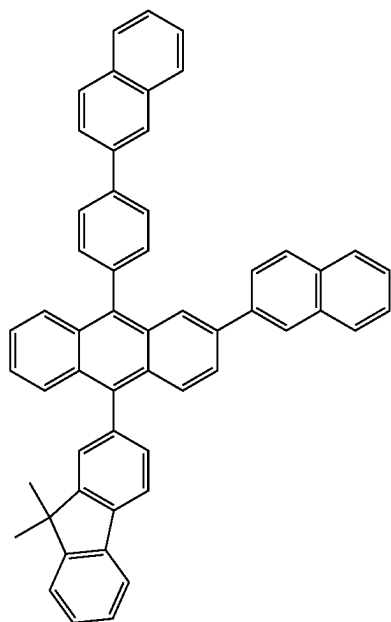


H25



75

-continued



76

-continued

H26

H29

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H27

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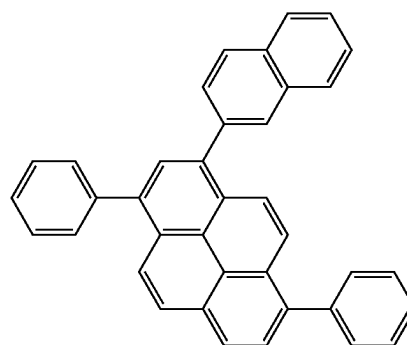
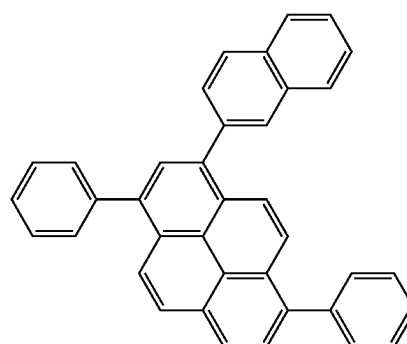
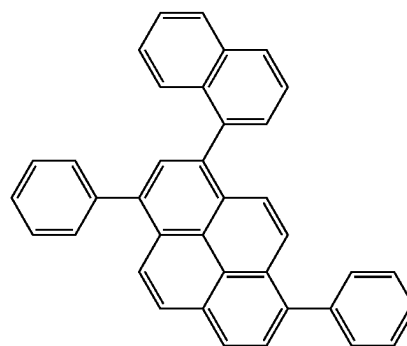
H28

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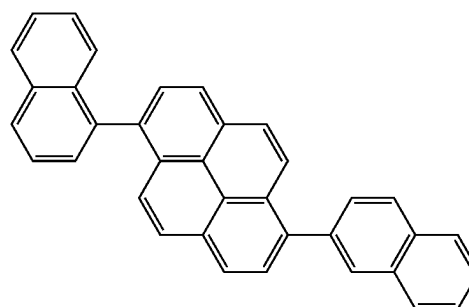
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H30

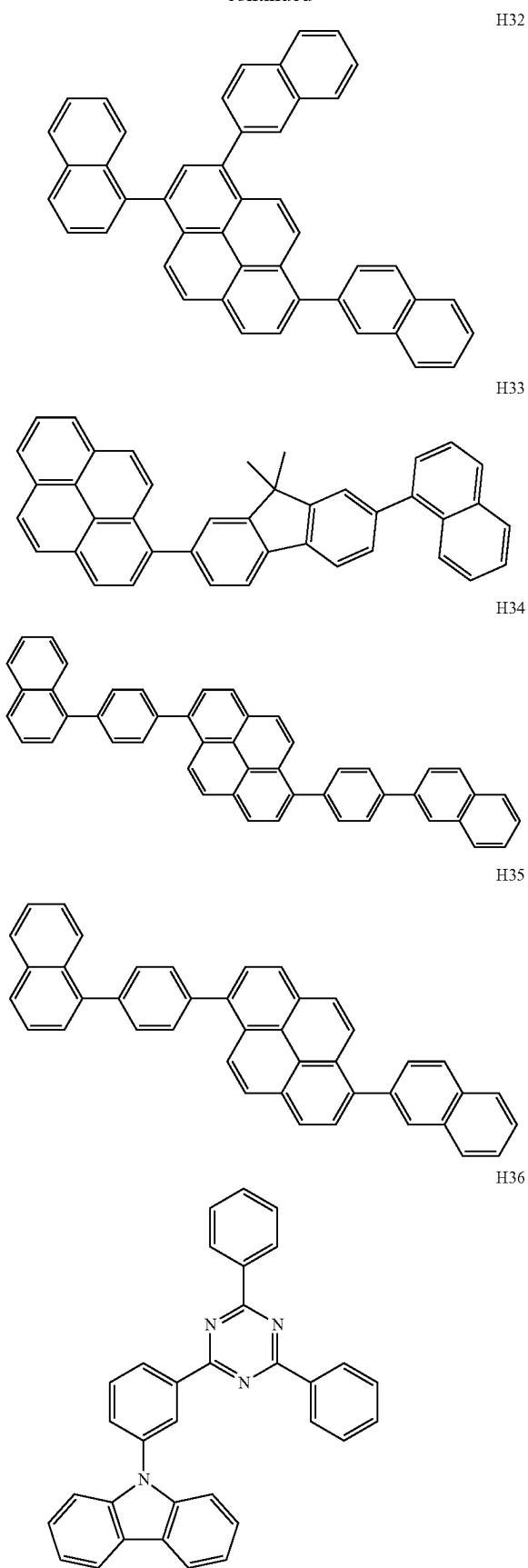
H30

H31

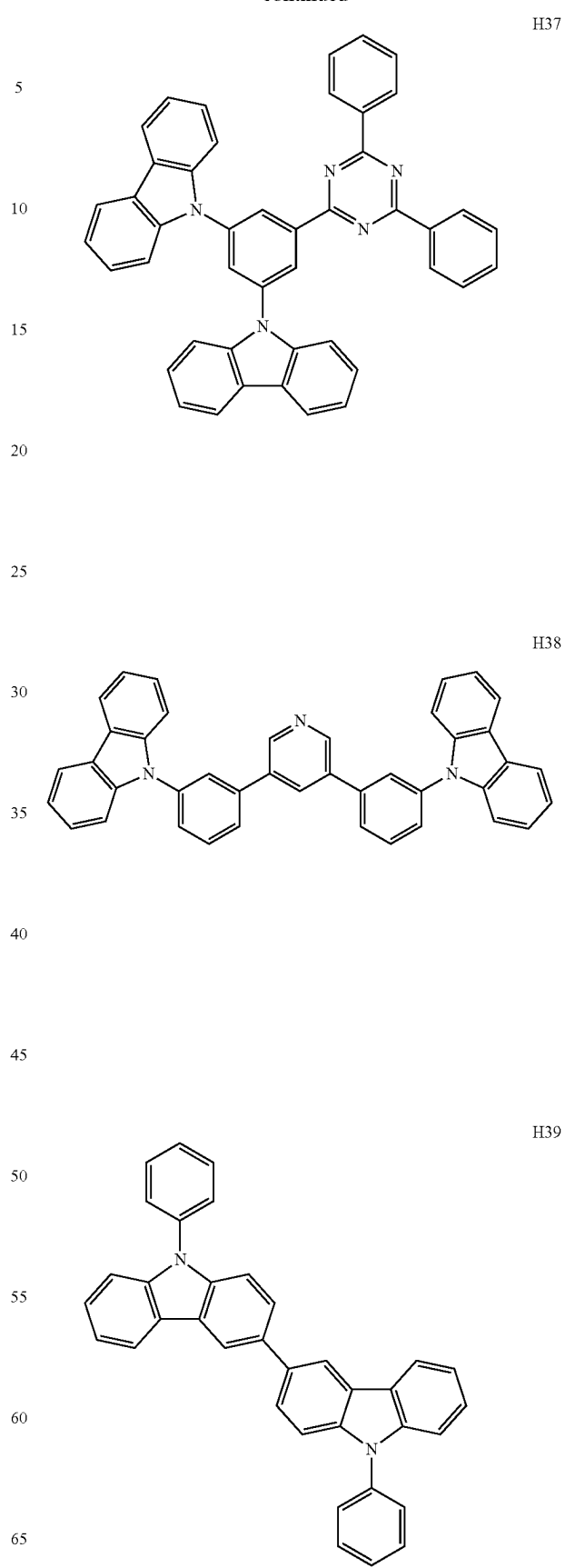


77

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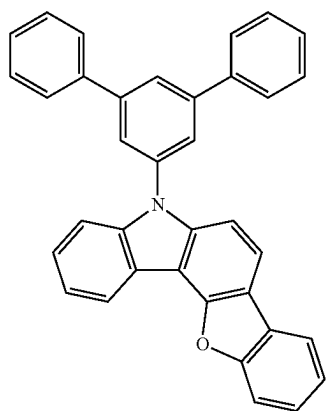
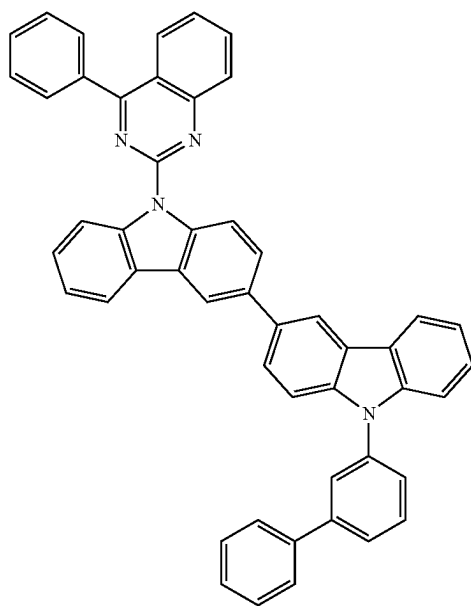
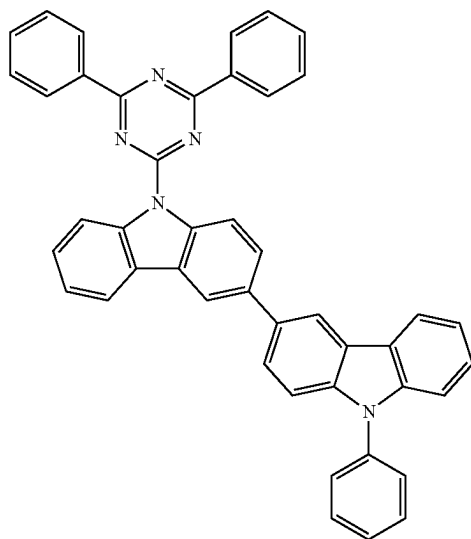
**78**

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**80**

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H40

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H41

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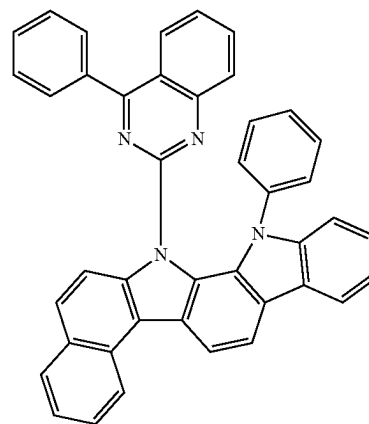
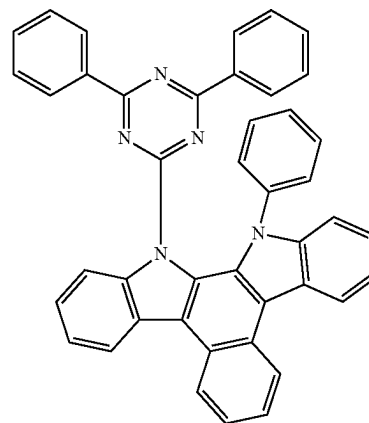
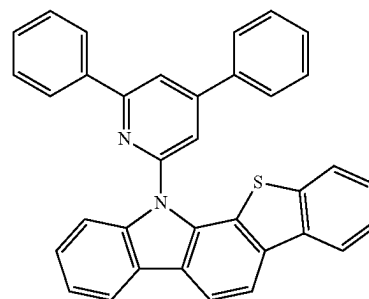
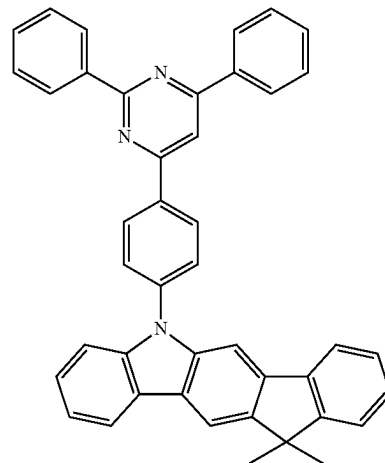
H42

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H43

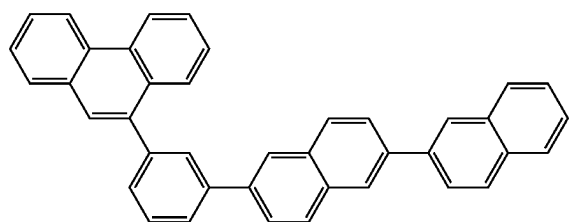
H44

H45

H46

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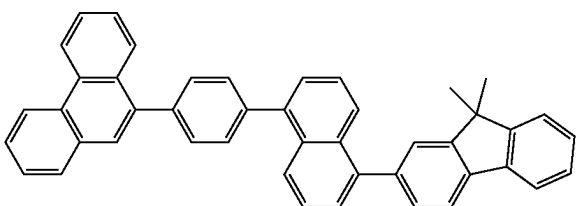


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H48

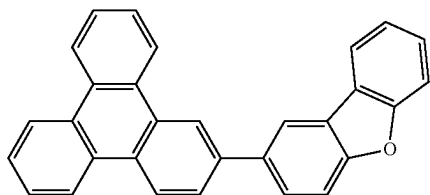
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H49

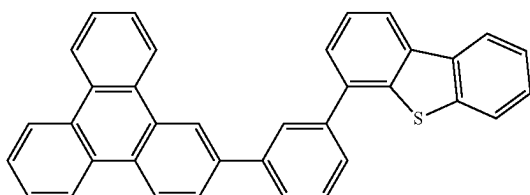
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H50

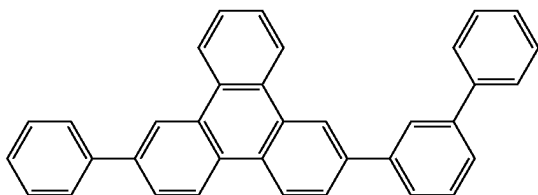
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H51

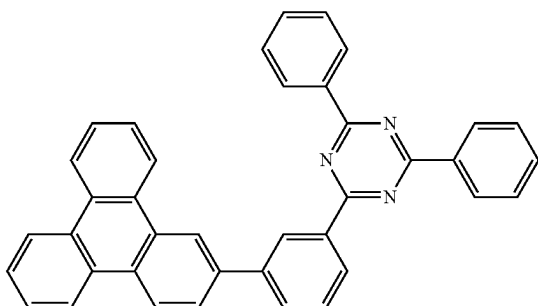
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H52

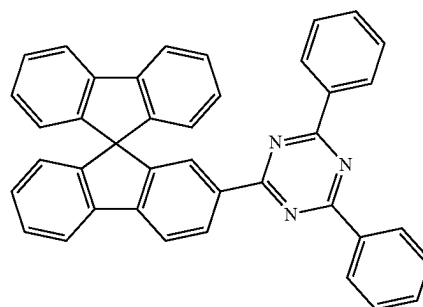
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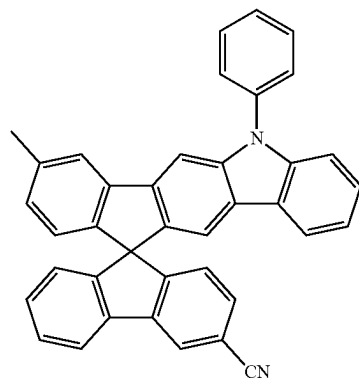
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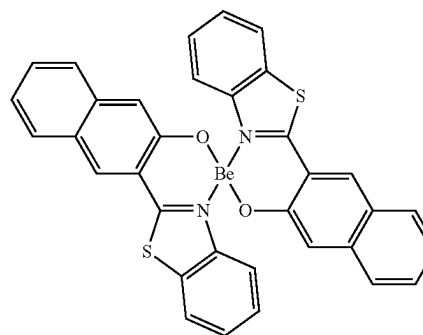
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H53



H54



H55

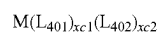
In an example embodiment, the host may include at least one selected from a silicon-containing compound (for example, BCPDS in the following examples) and a phosphine oxide-containing compound (for example, POPCPA in the following examples).

In an example embodiment, the host may include only one compound, or two or more different compounds (for example, a host used in the following examples includes BCPDS and POPCPA).

[Phosphorescent Dopant Included in Emission Layer in Organic Layer 150]

The phosphorescent dopant may include an organometallic compound represented by Formula 1:

The phosphorescent dopant may include an organometallic complex represented by Formula 401:

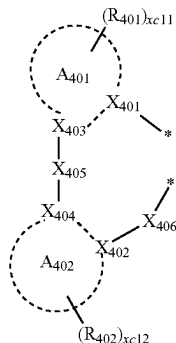


<Formula 401>

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-continued

<Formula 402>



In Formulae 401 and 402,

M may be selected from iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), and thulium (Tm),

L_{401} may be selected from ligands represented by Formula 402, and xc1 may be 1, 2, or 3, wherein, when xc1 is two or more, two or more $L_{401}(s)$ may be identical to or different from each other,

L_{402} may be an organic ligand, and xc2 may be an integer of 0 to 4, wherein, when xc2 is two or more, two or more $L_{402}(s)$ may be identical to or different from each other,

X_{401} to X_{404} may each independently be nitrogen or carbon,

X_{401} and X_{403} may be linked via a single bond or a double bond, and X_{402} and X_{404} may be linked via a single bond or a double bond,

A_{401} and A_{402} may each independently be selected from a C_5 - C_{60} carbocyclic group or a C_1 - C_{60} heterocyclic group,

X_{405} may be a single bond, $^*O^*$, $^*S^*$, $^*C(=O)^*$, $^*N(Q_{411})^*$, $^*C(Q_{411})(Q_{412})^*$, $^*C(Q_{411})=C(Q_{412})^*$, $^*C(Q_{411})=^*$, or $^*=C^*$, wherein Q_{411} and Q_{412} may each independently be hydrogen, deuterium, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group,

X_{406} may be a single bond, O, or S,

R_{401} and R_{402} may each independently be selected from hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C_1 - C_{20} alkyl group, a substituted or unsubstituted C_1 - C_{20} alkoxy group, a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_6 - C_{60} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $-Si(Q_{401})(Q_{402})(Q_{403})$, $-N(Q_{401})(Q_{402})$, $-B(Q_{401})(Q_{402})$, $-C(=O)(Q_{401})$, $-S(=O)_2(Q_{401})$, and $-P(=O)(Q_{401})(Q_{402})$, wherein Q_{401} to Q_{403} may each independently be selected from a

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C_1 - C_{10} alkyl group, a C_1 - C_{10} alkoxy group, a C_6 - C_{20} aryl group, and a C_1 - C_{20} heteroaryl group, xc11 and xc12 may each independently be an integer of 0 to 10, and

5 in Formula 402, $*$ and * each indicate a binding site to M in Formula 401.

In an example embodiment, in Formula 402, A_{401} and A_{402} may each independently be selected from a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, an indene group, a pyrrole group, a thiophene group, a furan group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a quinoxaline group, a quinazoline group, a carbazole group, a benzimidazole group, a benzofuran group, a benzothiophene group, an isobenzothiophene group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a dibenzofuran group, and a dibenzothiophene group.

In an example embodiment, in Formula 402, i) X_{401} may be nitrogen, and X_{402} may be carbon, or ii) X_{401} and X_{402} may each be nitrogen at the same time.

In an example embodiment, R_{401} and R_{402} in Formula 402 may each independently be selected from:

hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, and a C_1 - C_{20} alkoxy group;

a C_1 - C_{20} alkyl group, and a C_1 - C_{20} alkoxy group, each substituted with at least one selected from deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a phenyl group, a naphthyl group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, and a norbornenyl group; a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group;

a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, each substituted with at least one selected from deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group.

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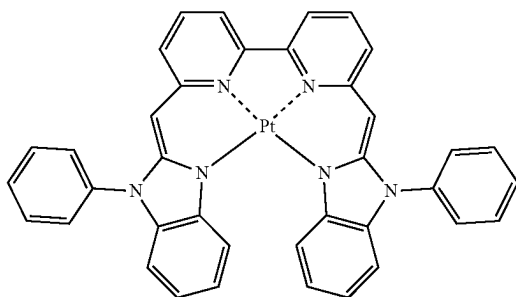
group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; and
 $-\text{Si}(\text{Q}_{401})(\text{Q}_{402})(\text{Q}_{403})$, $-\text{N}(\text{Q}_{401})(\text{Q}_{402})$, $-\text{B}(\text{Q}_{401})(\text{Q}_{402})$, $-\text{C}(=\text{O})(\text{Q}_{401})$, $-\text{S}(=\text{O})_2(\text{Q}_{401})$, and $-\text{P}(=\text{O})(\text{Q}_{401})(\text{Q}_{402})$, and

Q_{401} to Q_{403} may each independently be selected from a C_1 - C_{10} alkyl group, a C_1 - C_{10} alkoxy group, a phenyl group, a biphenyl group, and a naphthyl group.

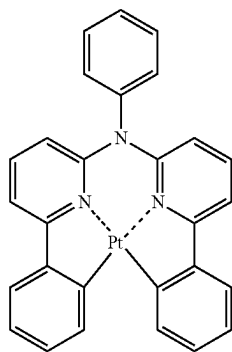
In an example embodiment, when xc1 in Formula 401 is two or more, two $\text{A}_{401}(\text{s})$ in two or more $\text{L}_{401}(\text{s})$ may be linked via X_{407} , which is a linking group, or two $\text{A}_{402}(\text{s})$ in two or more $\text{L}_{401}(\text{s})$ may be linked via X_{408} , which is a linking group (see Compounds PD1 to PD4 and PD7). X_{407} and X_{408} may each independently be a single bond, $^*\text{O}^*$, $^*\text{S}^*$, $^*\text{C}(=\text{O})^*$, $^*\text{N}(\text{Q}_{413})^*$, $^*\text{C}(\text{Q}_{413})(\text{Q}_{414})^*$, or $^*\text{C}(\text{Q}_{413})=\text{C}(\text{Q}_{414})^*$ (wherein Q_{413} and Q_{414} may each independently be hydrogen, deuterium, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group).

L_{402} in Formula 401 may be a monovalent, divalent, or trivalent organic ligand. For example, L_{402} may be selected from halogen, diketone (for example, acetylacetonate), carboxylic acid (for example, picolinate), $-\text{C}(=\text{O})$, isonitrile, $-\text{CN}$, and phosphorus (for example, phosphine, or phosphite).

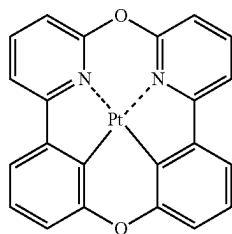
In an example embodiment, the phosphorescent dopant may be selected from, for example, Compounds PD1 to PD25:



PD1



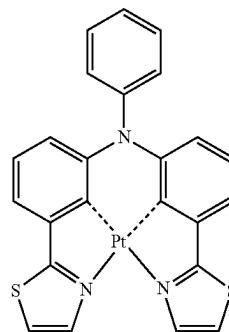
PD2



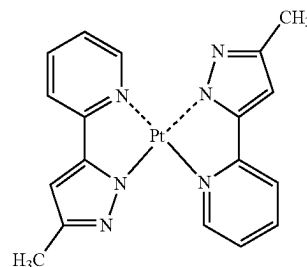
PD3

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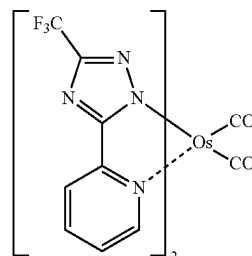
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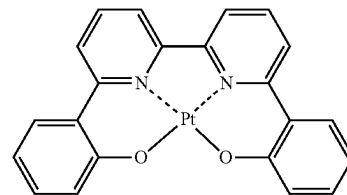
PD4



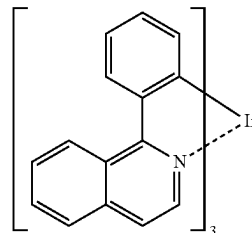
PD5



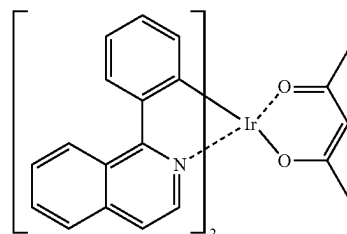
PD6



PD7



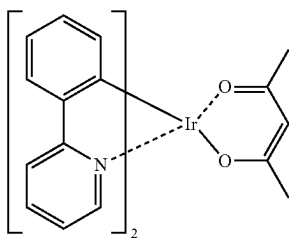
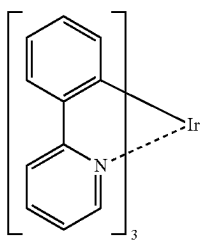
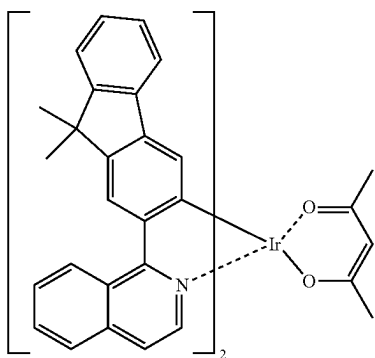
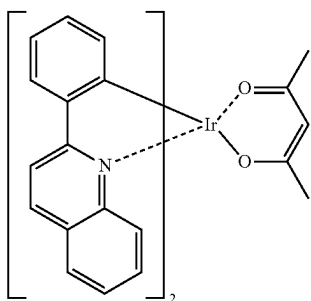
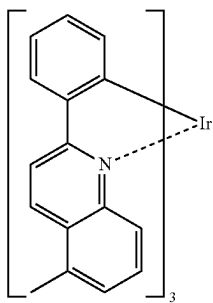
PD8



PD9

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-continued

**88**

-continued

PD10

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PD11

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PD12

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PD13

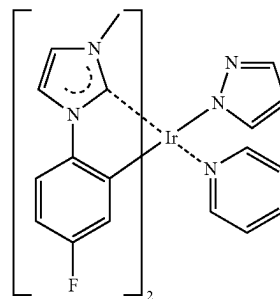
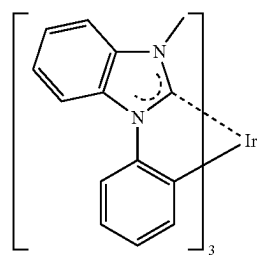
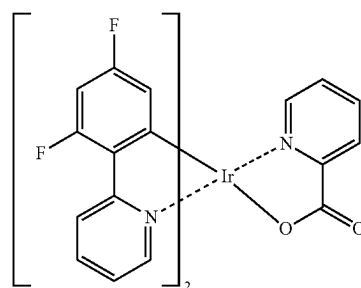
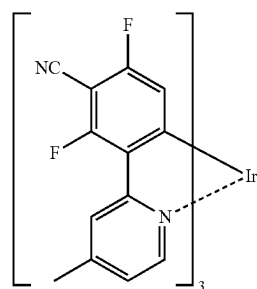
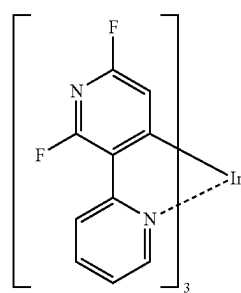
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PD14

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PD15

PD16

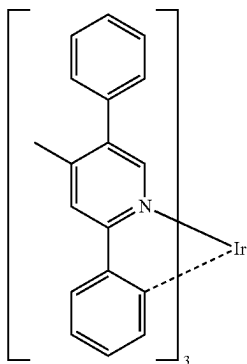
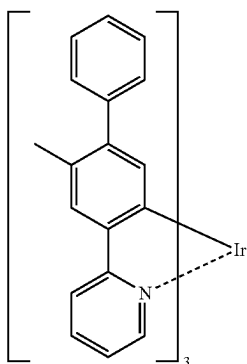
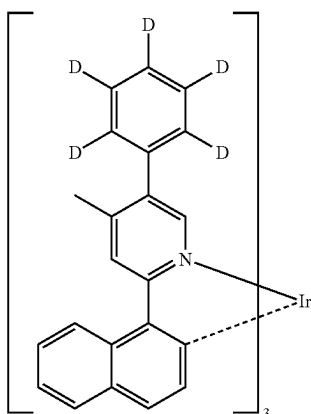
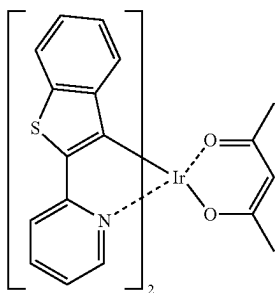
PD17

PD18

PD19

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-continued

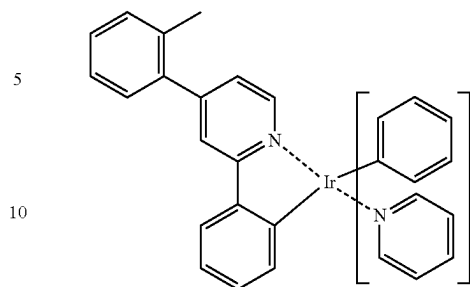


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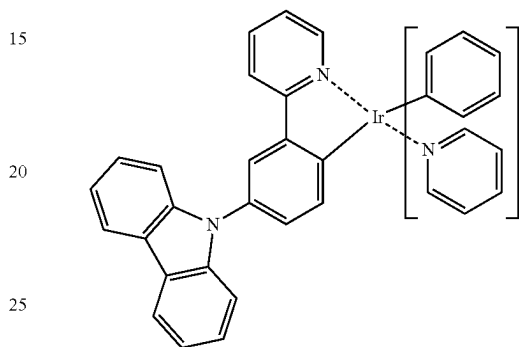
PD20

PD24



PD21

PD25



[Fluorescent Dopant in Emission Layer]

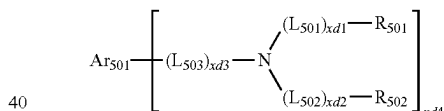
The fluorescent dopant may further include an arylamine compound or a styrylamine compound.

The fluorescent dopant may include a compound represented by Formula 501:

PD22

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<Formula 501>



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In Formula 501,

Ar₅₀₁ may be a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group,

L₅₀₁ to L₅₀₃ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkylene group, a substituted or unsubstituted C₃-C₁₀ cycloalkenylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenylene group, a substituted or unsubstituted C₆-C₆₀ arylene group, a substituted or unsubstituted C₁-C₆₀ heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

xd1 to xd3 may each independently be an integer of 0 to 3;

R₅₀₁ and R₅₀₂ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ to heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted

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C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group,

5 xd4 may be an integer of 1 to 6.

In an example embodiment, Ar₅₀₁ in Formula 501 may be selected from:

a naphthalene group, a heptalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenanthracene group, and an indenophenanthrene group; and
 a naphthalene group, a heptalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenanthracene group, and an indenophenanthrene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

In an example embodiment, L₅₀₁ to L₅₀₃ in Formula 501 may each independently be selected from:

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group; and
 a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

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biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenylyl group, a pentacenylyl group, a thiophenyl group, a furanyl group, a carbazoyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, a dibenzosilolyl group, and a pyridinyl group.

In an example embodiment, R₅₀₁ and R₅₀₂ in Formula 501 may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenylyl group, a pentacenylyl group, a thiophenyl group, a furanyl group, a carbazoyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, a dibenzosilolyl group, and a pyridinyl group; and
 a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenylyl group, a pentacenylyl group, a thiophenyl group, a furanyl group, a carbazoyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenylyl group, a pentacenylyl group, a thiophenyl group, a furanyl group, a carbazoyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, a dibenzosilolyl group, a pyridinyl group, and —Si(Q₃₁)(Q₃₂)(Q₃₃), and

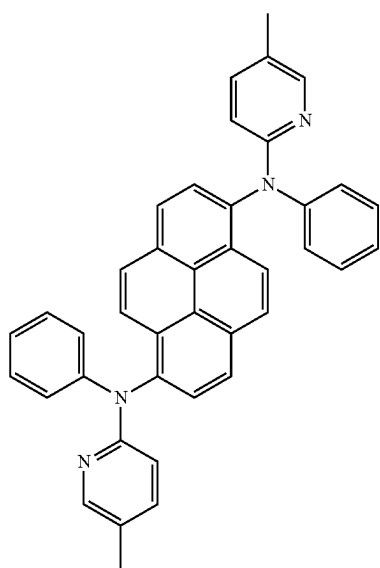
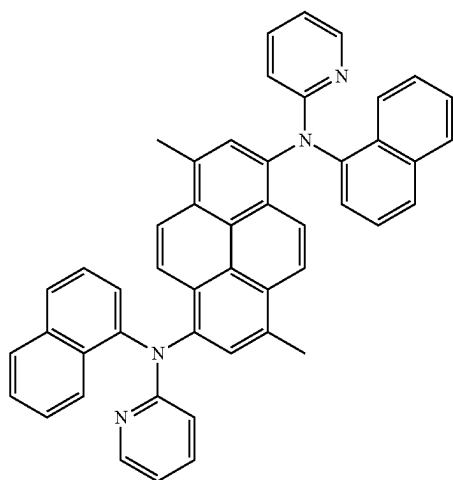
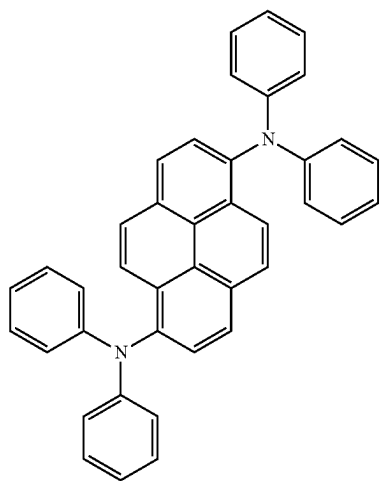
Q₃₁ to Q₃₃ may each independently be selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

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In an example embodiment, xd4 in Formula 501 may be

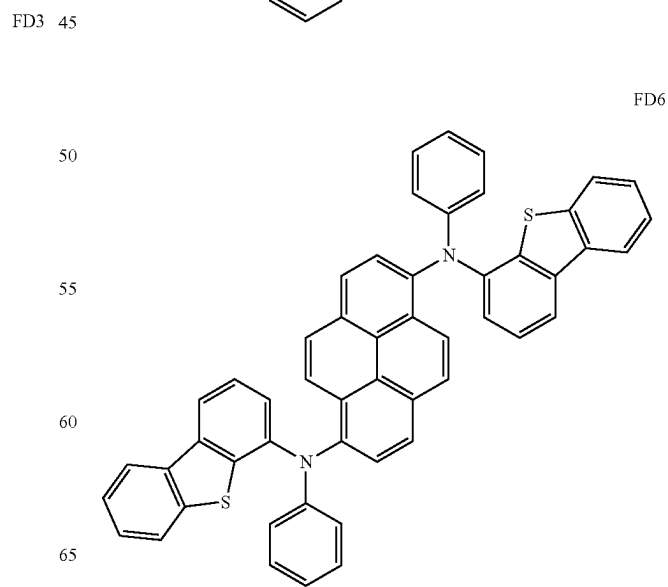
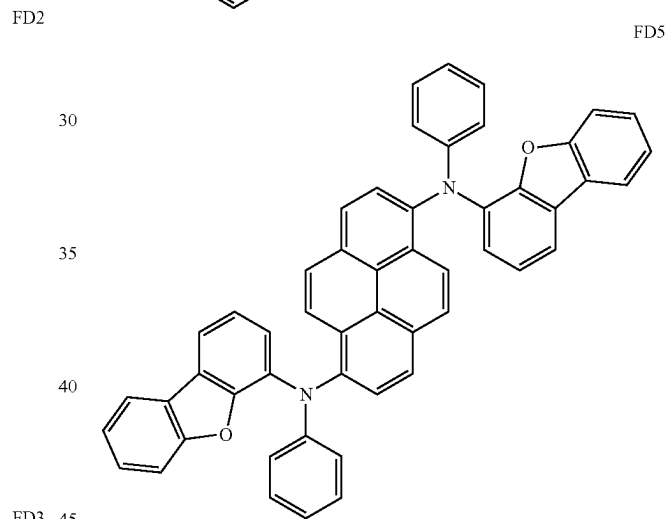
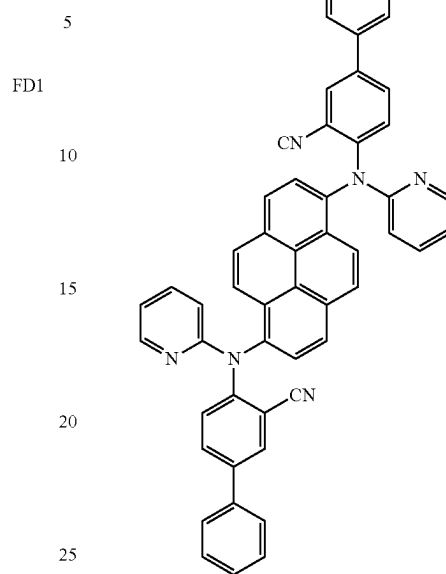
2.

For example, the fluorescent dopant may be selected from Compounds FD1 to FD22:

**94**

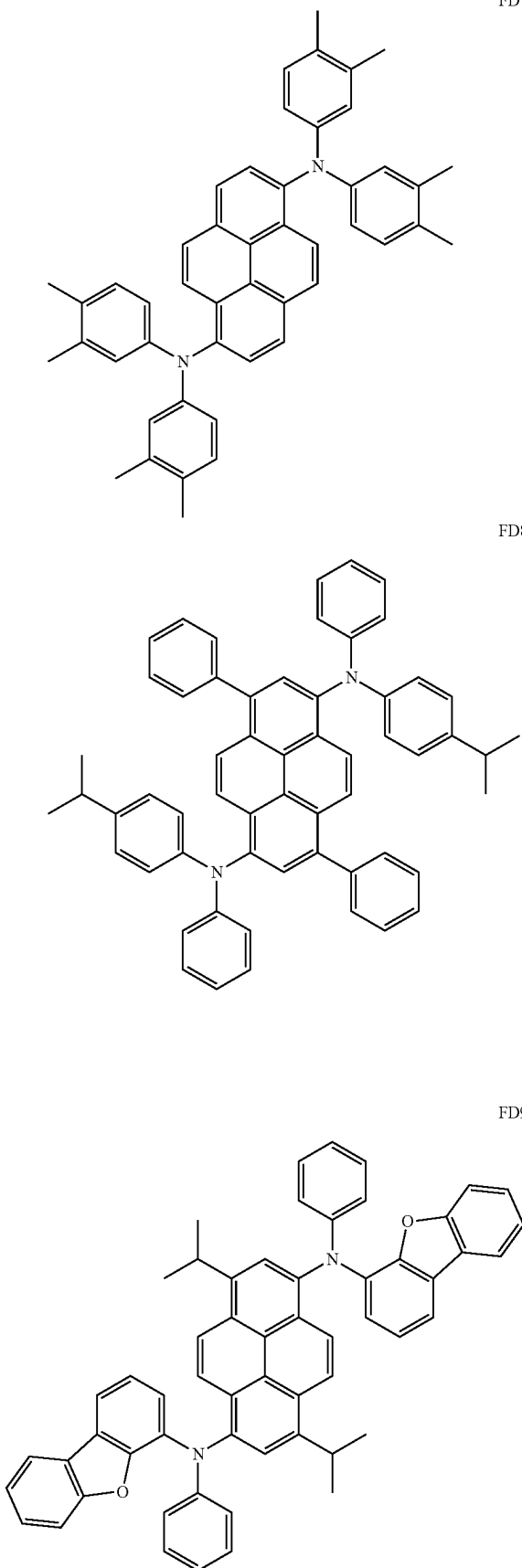
-continued

FD4



95

-continued

**96**

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FD7

FD10

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FD8

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FD12

FD13

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FD9

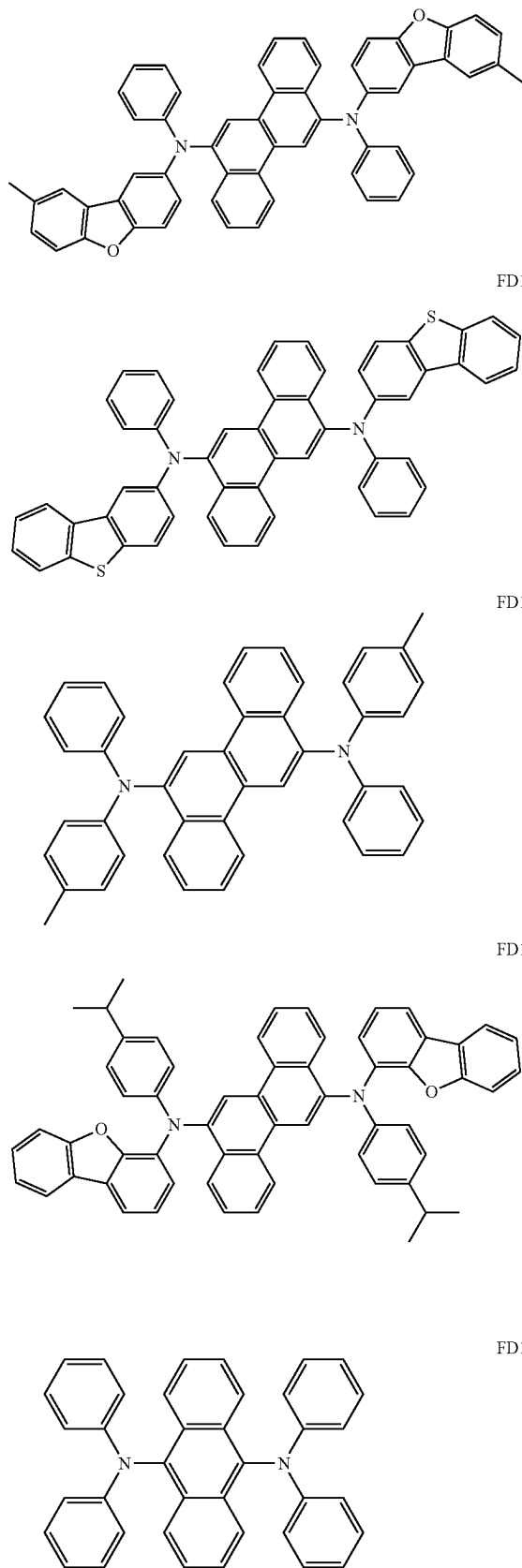
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FD14

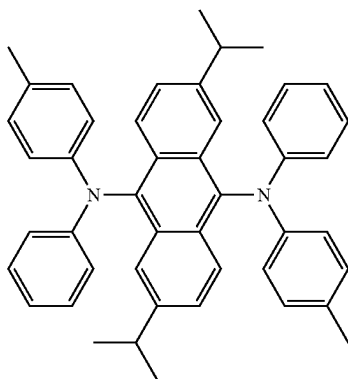
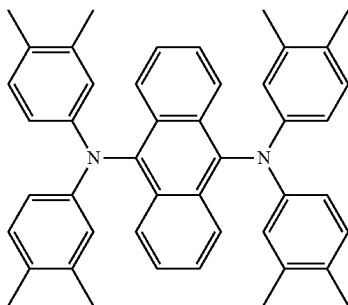
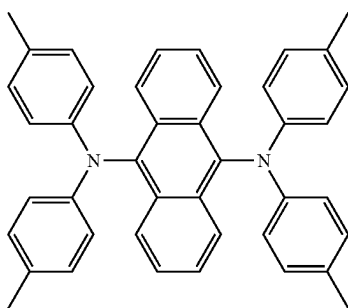
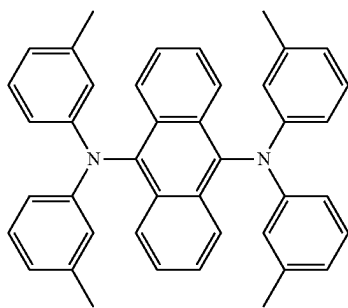
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97

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**98**

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FD15

FD19

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FD16

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FD17

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FD20

FD21

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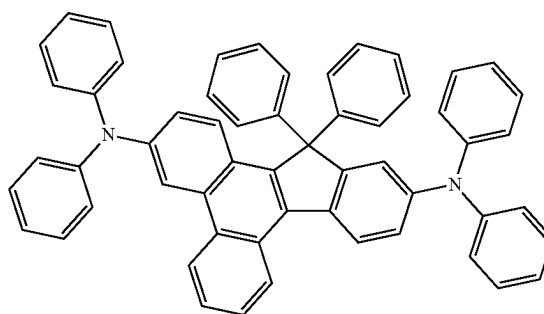
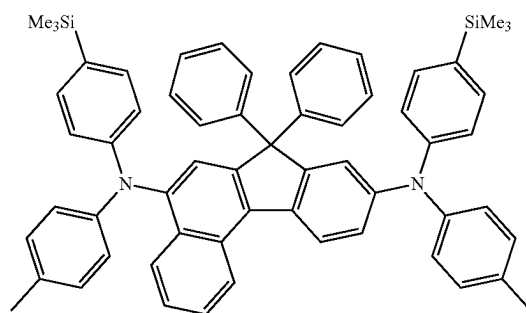
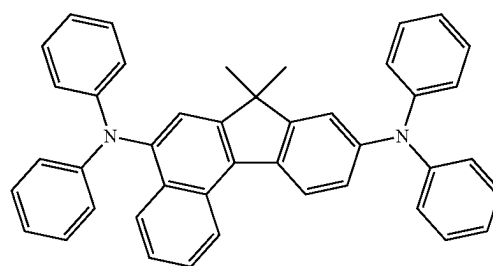
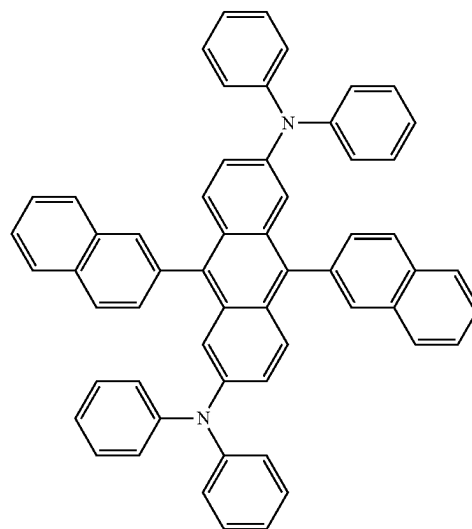
FD18

FD22

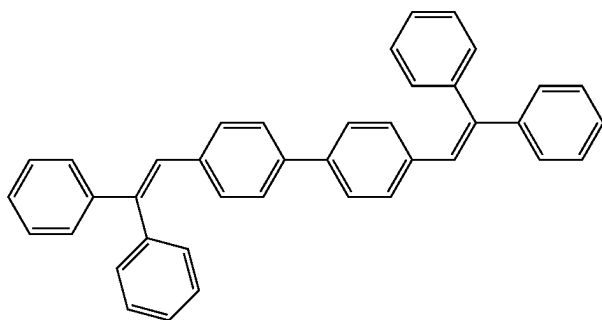
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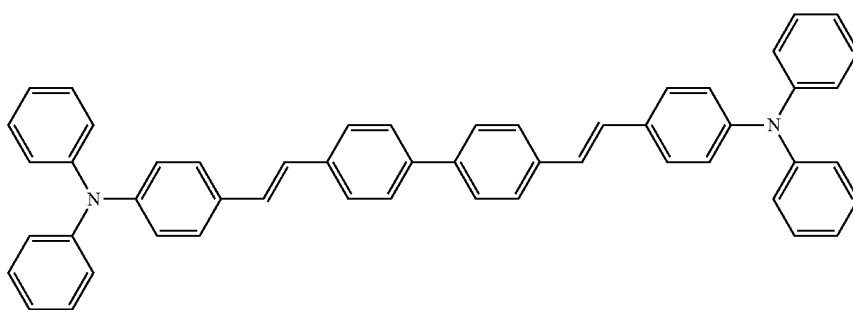
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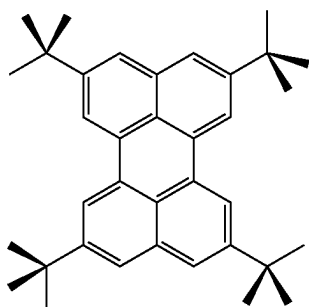
In an example embodiment, the fluorescent dopant may be selected from the following compounds.



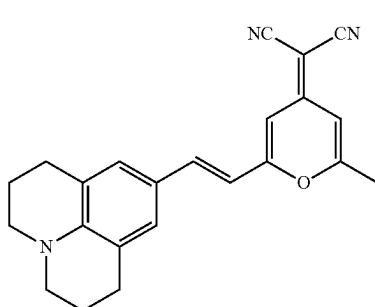
DPVBi



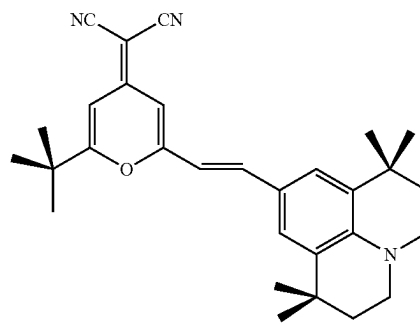
DPAVBz



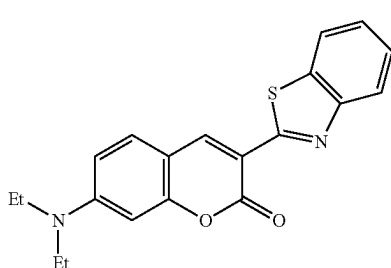
TBPz



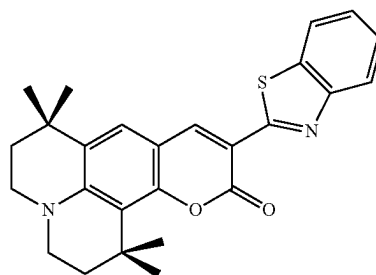
DCM



DCJTz



Coumarin 6



C545T

[Electron Transport Region in Organic Layer 150]

The electron transport region may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

The electron transport region may include at least one selected from a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, and an electron injection layer.

For example, the electron transport region may have an electron transport layer/electron injection layer structure, a hole blocking layer/electron transport layer/electron injection layer structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein for each structure, constituting layers may be sequentially stacked from an emission layer.

The electron transport region (for example, a buffer layer, a hole blocking layer, an electron control layer, or an electron transport layer in the electron transport region) may include a metal-free compound containing at least one π electron-depleted nitrogen-containing ring.

The " π electron-depleted nitrogen-containing ring" indicates a C_1 - C_{60} heterocyclic group having at least one $*-N=*$ moiety as a ring-forming moiety.

For example, the " π electron-depleted nitrogen-containing ring" may be i) a 5-membered to 7-membered heteromonocyclic group having at least one $*-N=*$ moiety, ii) a heteropolycyclic group in which two or more 5-membered to 7-membered heteromonocyclic groups each having at least one $*-N=*$ moiety are condensed with each other, or iii) a heteropolycyclic group in which at least one of 5-membered to 7-membered heteromonocyclic groups, each having at least one $*-N=*$ moiety, is condensed with at least one C_5 - C_{60} carbocyclic group.

Examples of the π electron-depleted nitrogen-containing ring include an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group.

For example, the electron transport region may include a compound represented by Formula 601:



In Formula 601,

Ar_{601} may be a substituted or unsubstituted C_5 - C_{60} carbocyclic group or a substituted or unsubstituted C_1 - C_{60} heterocyclic group,

$xe11$ may be 1, 2, or 3,

L_{601} may be selected from a substituted or unsubstituted C_3 - C_{10} cycloalkylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkylene group, a substituted or unsubstituted C_3 - C_{10} cycloalkenylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenylene group, a substituted or unsubstituted C_6 - C_{60} arylene

group, a substituted or unsubstituted C_1 - C_{60} heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group;

$xe1$ may be an integer of 0 to 5,

R_{601} may be selected from a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $-\text{Si}(Q_{601})(Q_{602})(Q_{603})$, $-\text{C}(=\text{O})(Q_{601})$, $-\text{S}(=\text{O})_2(Q_{601})$, and $-\text{P}(=\text{O})(Q_{601})(Q_{602})$, Q_{601} to Q_{603} may each independently be a C_1 - C_{10} alkyl group, a C_1 - C_{10} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group, and

$xe21$ may be an integer of 1 to 5.

In an example embodiment, at least one of $Ar_{601}(s)$ in the number of $xe11$ and $R_{601}(s)$ in the number of $xe21$ may include the π electron-depleted nitrogen-containing ring.

In an example embodiment, Ar_{601} in Formula 601 may be selected from:

a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group;

and
a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a

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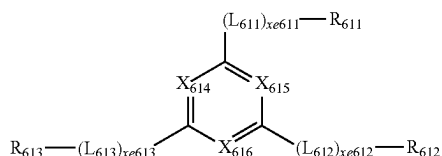
quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)(Q₃₂), and Q₃₁ to Q₃₃ may each independently be selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

When xe11 in Formula 601 is two or more, two or more Ar₆₀₁(s) may be linked via a single bond.

In an example embodiment, Ar₆₀₁ in Formula 601 may be an anthracene group.

In an example embodiment, a compound represented by Formula 601 may be represented by Formula 601-1:

<Formula 601-1>



In Formula 601-1,

X₆₁₄ may be N or C(R₆₁₄), X₆₁₅ may be N or C(R₆₁₅), and X₆₁₆ may be N or C(R₆₁₆), wherein at least one selected from X₆₁₄ to X₆₁₆ may be N,

L₆₁₁ to L₆₁₃ may each independently be the same as defined in connection with L₆₀₁,

xe611 to xe613 may each independently be the same as defined in connection with xe1,

R₆₁₁ to R₆₁₃ may each independently be the same as defined in connection with R₆₀₁, and

R₆₁₄ to R₆₁₆ may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

In an example embodiment, L₆₀₁ and L₆₁₁ to L₆₁₃ in Formulae 601 and 601-1 may each independently be selected from:

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indo-

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lylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indo-lylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indo-

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ophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group.

In an example embodiment, xe1 and xe611 to xe613 in Formulae 601 and 601-1 may each independently be 0, 1, or 2.

In an example embodiment, in Formulae 601 and 601-1, R₆₀₁ and R₆₁₁ to R₆₁₃ may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group;

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a

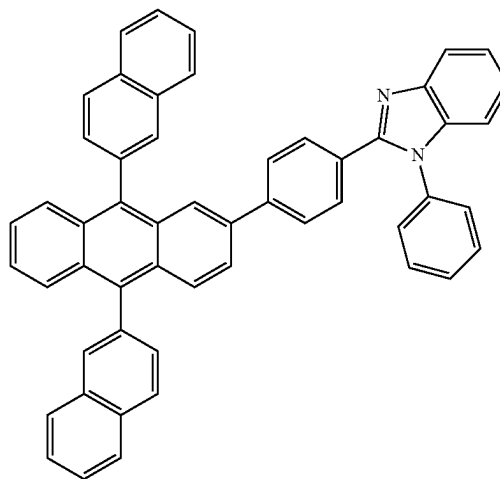
106

triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; and

—S(=O)₂(Q₆₀₁) and —P(=O)(Q₆₀₁)(Q₆₀₂), and Q₆₀₁ and Q₆₀₂ may be the same as described above.

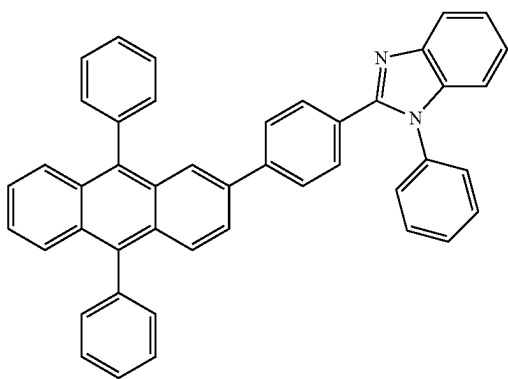
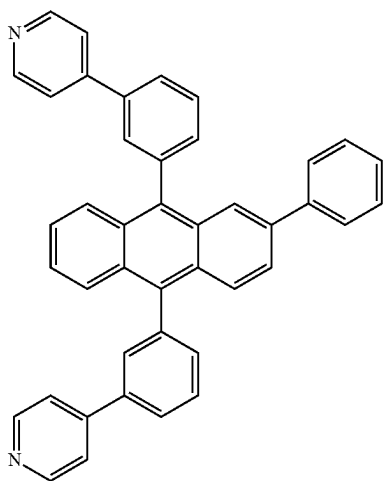
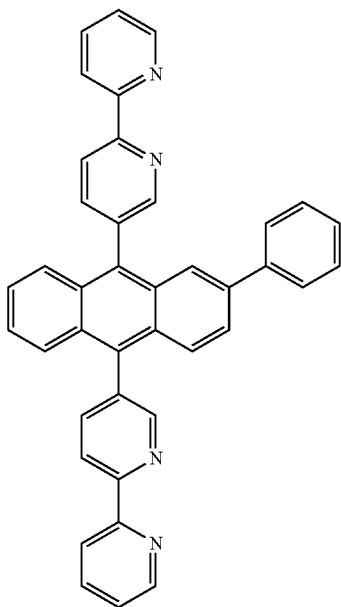
The electron transport region may include at least one compound selected from Compounds ET1 to ET36:

ET1



107

-continued

**108**

-continued

ET2

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ET3

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ET4

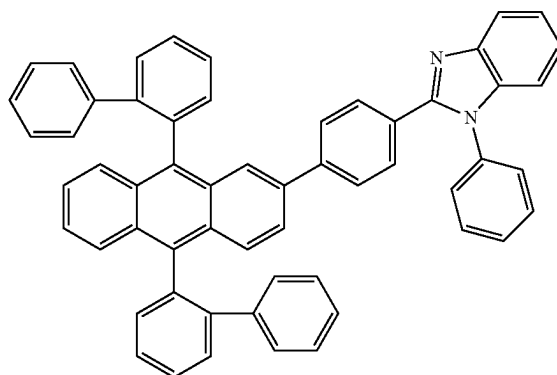
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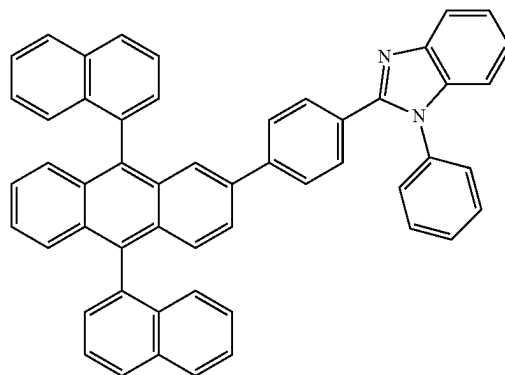
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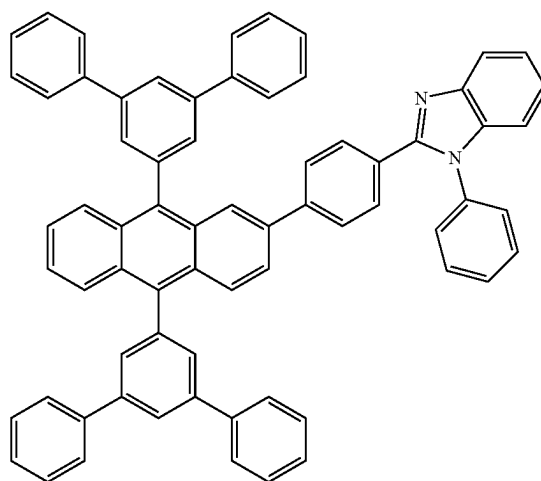
ET5



ET6

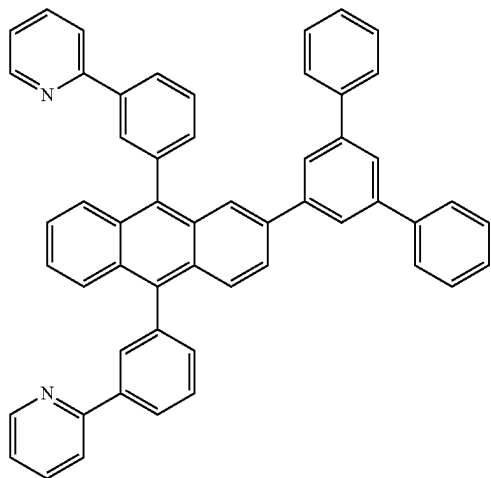


ET7



109

-continued



ET8

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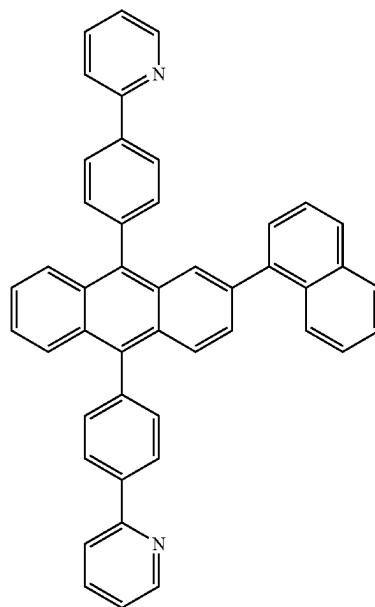
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110

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ET10

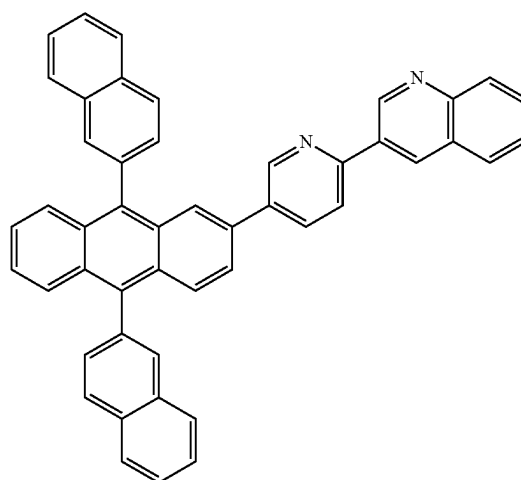
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ET9

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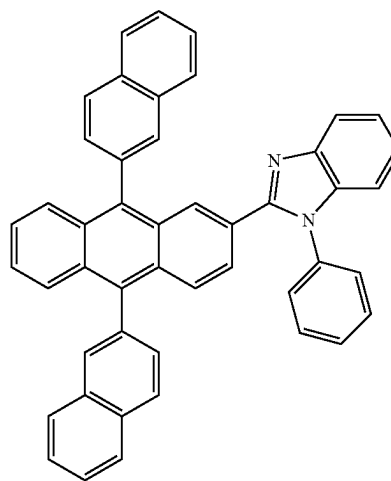
ET11

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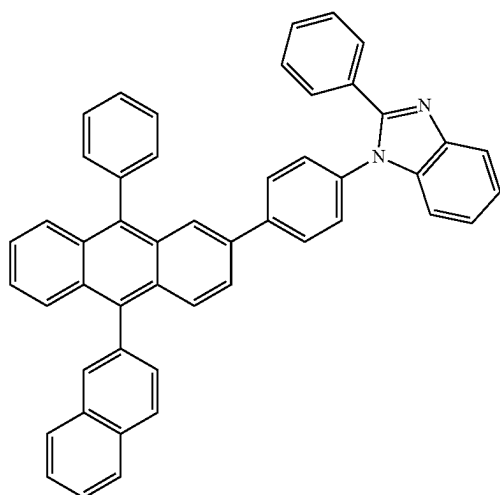


ET12

111

-continued

ET13



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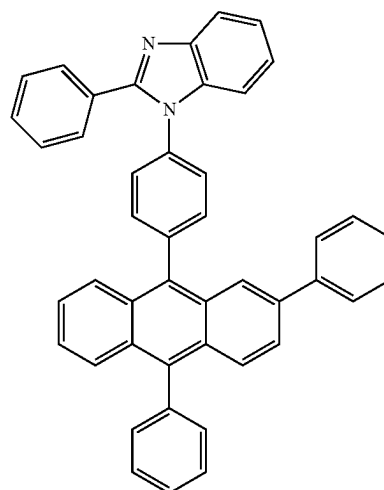
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112

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ET16



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ET14

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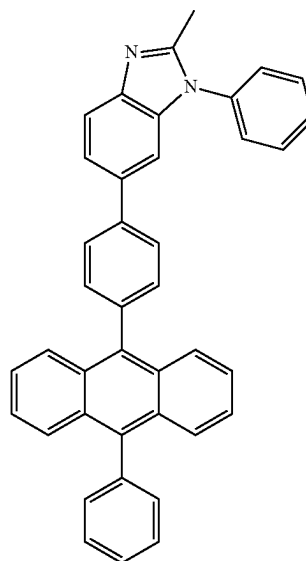
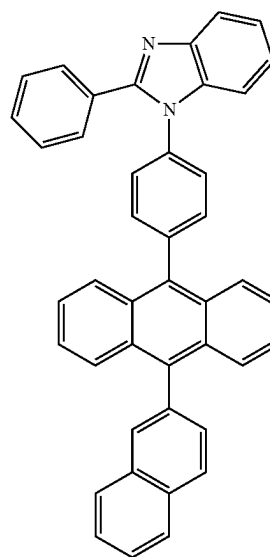
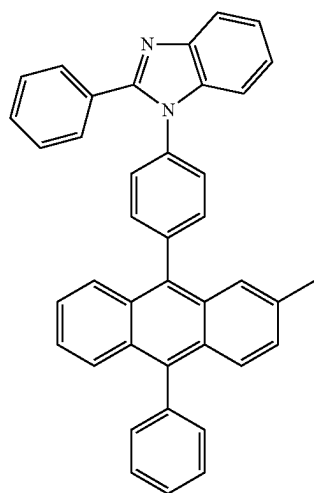
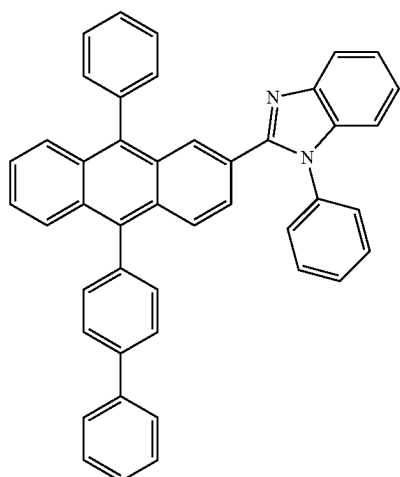
ET15

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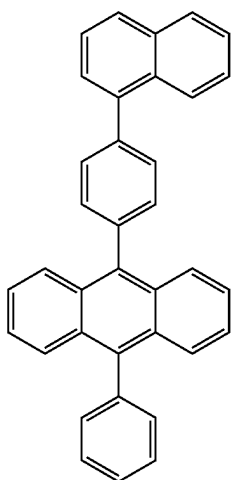
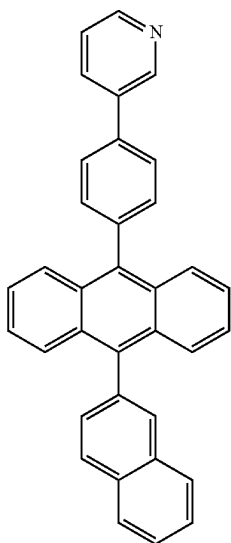
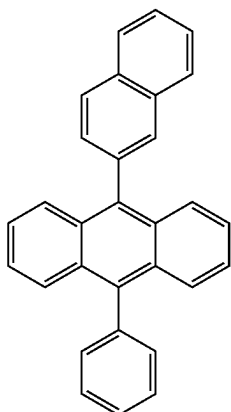


ET17

ET18

113

-continued



114

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ET19

ET22

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ET20

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ET23

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ET21

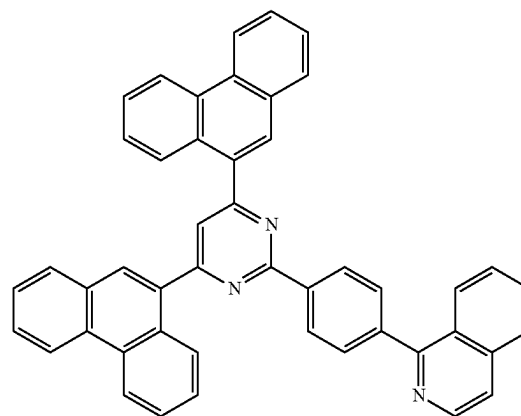
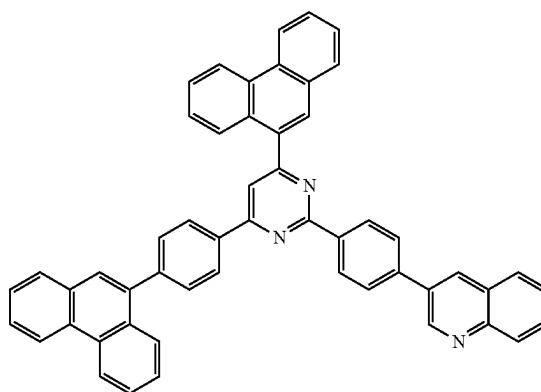
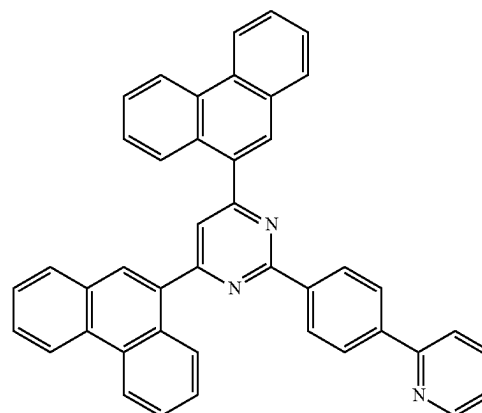
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ET24

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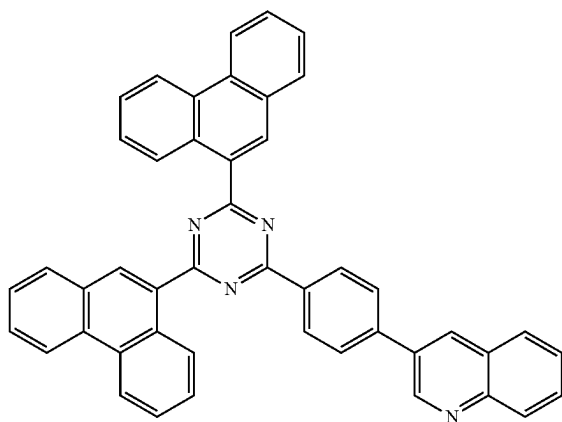
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115

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ET25



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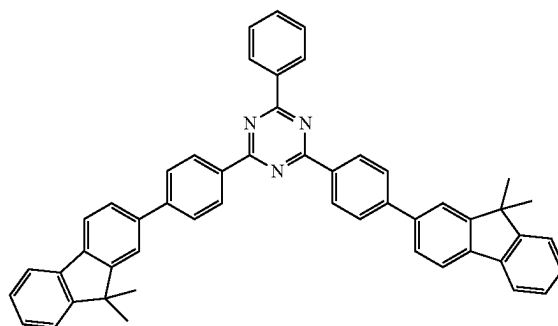
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116

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ET28



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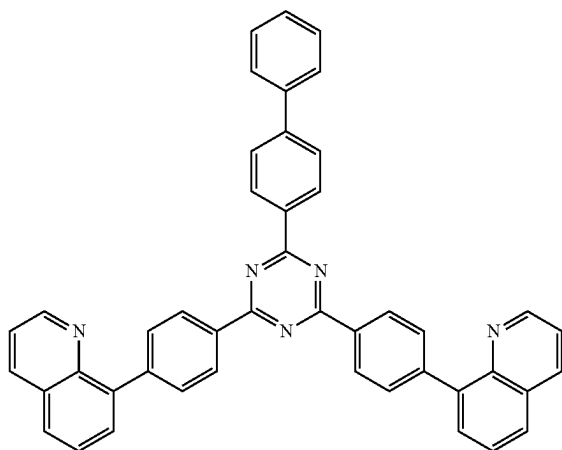
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ET29

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ET26

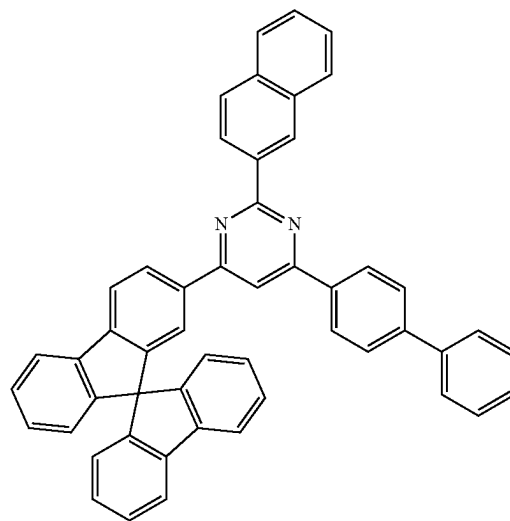


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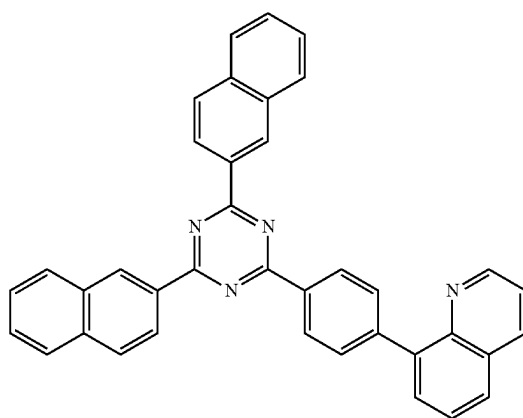
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ET30

ET27

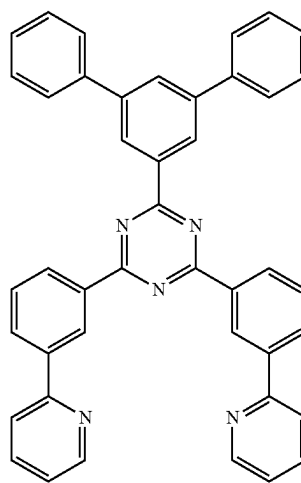
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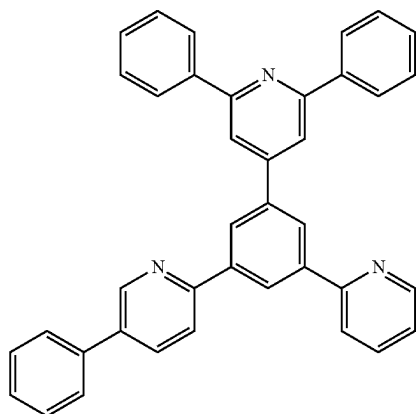
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117

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ET31

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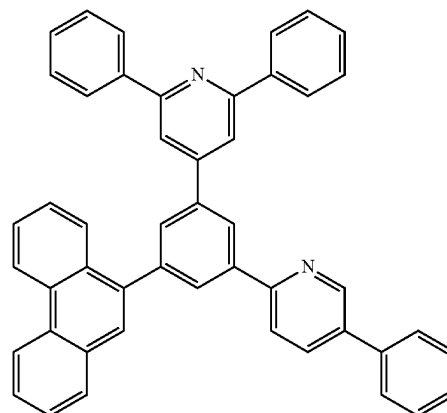
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ET34

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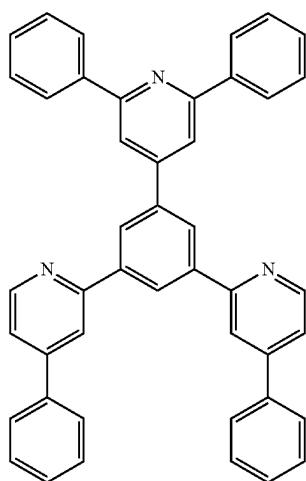
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ET32

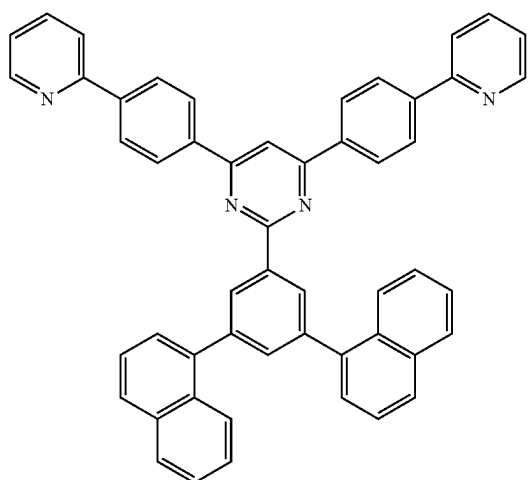


ET33

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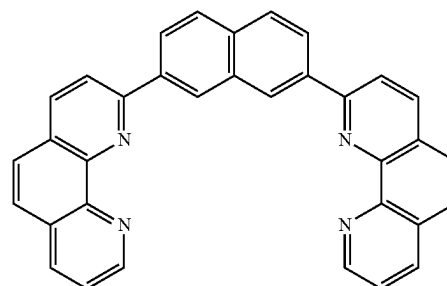
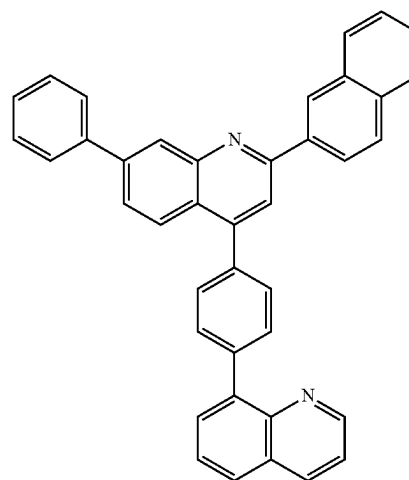
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ET35

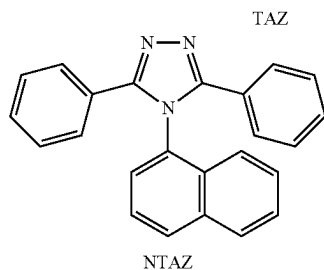
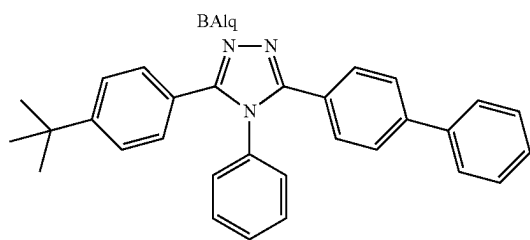
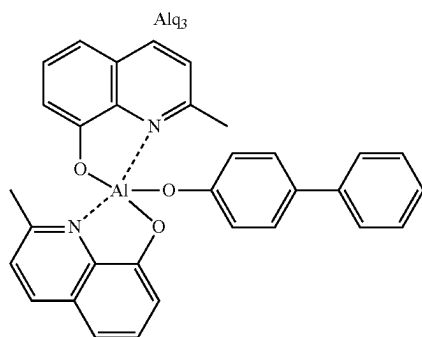
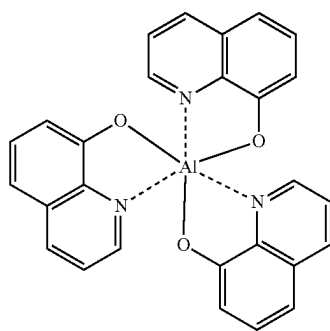
ET36



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In an example embodiment, the electron transport region may include at least one compound selected from 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Alq₃, BAlq, 3-(biphenyl-4-yl)-5-(4-tert-butylphenyl)-4-phenyl-4H-1,2,4-triazole (TAZ), and NTAZ:

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In an example embodiment, the electron transport region may include a phosphine oxide-containing compound (for example, TSPO1 used in the following examples). In an example embodiment, the phosphine oxide-containing compound may be used in a hole blocking layer in the electron transport region.

A thickness of the buffer layer, the hole blocking layer, or the electron control layer may be in a range of about 20 Å to about 1,000 Å, for example, about 30 Å to about 300 Å. When the thicknesses of the buffer layer, the hole blocking layer, and the electron control layer are within these ranges, the electron transport region may have excellent hole blocking characteristics or electron control characteristics without a substantial increase in driving voltage.

A thickness of the electron transport layer may be in a range of about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thickness of the electron transport layer is within the range described above, the

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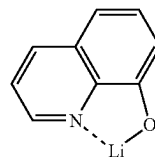
electron transport layer may have satisfactory electron transport characteristics without a substantial increase in driving voltage.

The electron transport region (for example, the electron transport layer in the electron transport region) may further include, in addition to the materials described above, a metal-containing material.

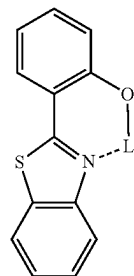
The metal-containing material may include at least one selected from alkali metal complex and alkaline earth-metal complex. The alkali metal complex may include a metal ion selected from a Li ion, a Na ion, a K ion, a Rb ion, and a Cs ion, and the alkaline earth-metal complex may include a metal ion selected from a Be ion, a Mg ion, a Ca ion, a Sr ion, and a Ba ion. A ligand coordinated with the metal ion of the alkali metal complex or the alkaline earth-metal complex may be selected from a hydroxy quinoline, a hydroxy isoquinoline, a hydroxy benzoquinoline, a hydroxy acridine, a hydroxy phenanthridine, a hydroxy phenylloxazole, a hydroxy phenylthiazole, a hydroxy diphenylloxadiazole, a hydroxy diphenylthiadiazol, a hydroxy phenylpyridine, a hydroxy phenylbenzimidazole, a hydroxy phenylbenzothiazole, a bipyridine, a phenanthroline, and a cyclopentadiene.

For example, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (lithium quinolate, LiQ) or ET-D2.

ET-D1



ET-D2



The electron transport region may include an electron injection layer that facilitates injection of electrons from the second electrode **190**. The electron injection layer may directly contact the second electrode **190**.

The electron injection layer may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

The electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or a combinations thereof.

The alkali metal may be selected from Li, Na, K, Rb, and Cs. In an example embodiment, the alkali metal may be Li, Na, or Cs. In an example embodiment, the alkali metal may be Li or Cs.

The alkaline earth metal may be selected from Mg, Ca, Sr, and Ba.

The rare earth metal may be selected from Sc, Y, Ce, Tb, Yb, and Gd.

The alkali metal compound, the alkaline earth-metal compound, and the rare earth metal compound may be selected from oxides and halides (for example, fluorides, chlorides, bromides, or iodides) of the alkali metal, the alkaline earth-metal, and the rare earth metal.

The alkali metal compound may be selected from alkali metal oxides, such as Li_2O , Cs_2O , or K_2O , and alkali metal halides, such as LiF , NaF , CsF , KF , LiI , NaI , CsI , or KI . In an example embodiment, the alkali metal compound may be selected from LiF , Li_2O , NaF , LiI , NaI , CsI , and KI .

The alkaline earth-metal compound may be selected from alkaline earth-metal oxides, such as BaO , SrO , CaO , $\text{Ba}_x\text{Sr}_{1-x}\text{O}$ ($0 < x < 1$), or $\text{Ba}_x\text{Ca}_{1-x}\text{O}$ ($0 < x < 1$). In an example embodiment, the alkaline earth-metal compound may be selected from BaO , SrO , and CaO .

The rare earth metal compound may be selected from YbF_3 , ScF_3 , ScO_3 , Y_2O_3 , Ce_2O_3 , GdF_3 , and TbF_3 . In an example embodiment, the rare earth metal compound may be selected from YbF_3 , ScF_3 , TbF_3 , YbI_3 , ScI_3 , and TbI_3 .

The alkali metal complex, the alkaline earth-metal complex, and the rare earth metal complex may include an ion of alkali metal, alkaline earth-metal, and rare earth metal as described above, and a ligand coordinated with a metal ion of the alkali metal complex, the alkaline earth-metal complex, or the rare earth metal complex may be selected from hydroxy quinoline, hydroxy isoquinoline, hydroxy benzoquinoline, hydroxy acridine, hydroxy phenanthridine, hydroxy phenyloxazole, hydroxy phenylthiazole, hydroxy diphenyloxadiazole, hydroxy diphenylthiadiazol, hydroxy phenylpyridine, hydroxy phenylbenzimidazole, hydroxy phenylbenzothiazole, bipyridine, phenanthroline, and cyclopentadiene.

The electron injection layer may include or consist of an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or a combinations thereof, as described above. In an example embodiment, the electron injection layer may further include an organic material. When the electron injection layer further includes an organic material, an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or a combinations thereof may be homogeneously or non-homogeneously dispersed in a matrix including the organic material.

A thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the range described above, the electron injection layer may have satisfactory electron injection characteristics without a substantial increase in driving voltage.

[Second Electrode 190]

The second electrode 190 may be disposed on the organic layer 150 having such a structure. The second electrode 190 may be a cathode which is an electron injection electrode, and in this regard, a material for forming the second electrode 190 may be selected from metal, an alloy, an electrically conductive compound, and a combination thereof, which have a relatively low work function.

The second electrode 190 may include at least one selected from lithium (Li), silver (Ag), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), ITO, and IZO. The second electrode 190 may be a transmissive electrode, a semi-transmissive electrode, or a reflective electrode.

The second electrode 190 may have a single-layered structure, or a multi-layered structure including two or more layers.

Layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region may be formed in a certain region by using one or more suitable methods selected from vacuum deposition, spin coating, casting, Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, and laser-induced thermal imaging.

When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by vacuum deposition, the deposition may be performed at a deposition temperature of about 100° C. to about 500° C., a vacuum degree of about 10^{-8} torr to about 10^{-3} torr, and a deposition speed of about 0.01 Å/sec to about 100 Å/sec by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by spin coating, the spin coating may be performed at a coating speed of about 2,000 rpm to about 5,000 rpm and at a heat treatment temperature of about 80° C. to about 200° C. by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

[General Definition of Substituents]

The term “ $\text{C}_1\text{-C}_{60}$ alkyl group” as used herein refers to a linear or branched aliphatic saturated hydrocarbon monovalent group having 1 to 60 carbon atoms, and examples thereof include a methyl group, an ethyl group, a propyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, a pentyl group, an isoamyl group, and a hexyl group. The term “ $\text{C}_1\text{-C}_{60}$ alkylene group” as used herein refers to a divalent group having the same structure as the $\text{C}_1\text{-C}_{60}$ alkyl group.

The term “ $\text{C}_2\text{-C}_{60}$ alkenyl group” as used herein refers to a hydrocarbon group having at least one double bond in the middle or at the terminus of the $\text{C}_2\text{-C}_{60}$ alkyl group, and examples thereof include an ethenyl group, a propenyl group, and a butenyl group. The term “ $\text{C}_2\text{-C}_{60}$ alkenylene group” as used herein refers to a divalent group having the same structure as the $\text{C}_2\text{-C}_{60}$ alkenyl group.

The term “ $\text{C}_2\text{-C}_{60}$ alkynyl group” as used herein refers to a hydrocarbon group having at least one triple bond in the middle or at the terminus of the $\text{C}_2\text{-C}_{60}$ alkyl group, and examples thereof include an ethynyl group, and a propynyl group. The term “ $\text{C}_2\text{-C}_{60}$ alkynylene group” as used herein refers to a divalent group having the same structure as the $\text{C}_2\text{-C}_{60}$ alkynyl group.

The term “ $\text{C}_1\text{-C}_{60}$ alkoxy group” as used herein refers to a monovalent group represented by —OA_{101} (wherein A_{101} is the $\text{C}_1\text{-C}_{60}$ alkyl group), and examples thereof include a methoxy group, an ethoxy group, and an isopropoxy group.

The term “ $\text{C}_3\text{-C}_{10}$ cycloalkyl group” as used herein refers to a monovalent saturated hydrocarbon monocyclic group having 3 to 10 carbon atoms, and examples thereof include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, and a cycloheptyl group. The

term “C₃-C₁₀ cycloalkylene group” as used herein refers to a divalent group having the same structure as the C₃-C₁₀ cycloalkyl group.

The term “C₁-C₁₀ heterocycloalkyl group” as used herein refers to a monovalent monocyclic group having at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom and 1 to 10 carbon atoms, and examples thereof include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuranlyl group, and a tetrahydrothiophenyl group. The term “C₁-C₁₀ heterocycloalkylene group” as used herein refers to a divalent group having the same structure as the C₁-C₁₀ heterocycloalkyl group.

The term C₃-C₁₀ cycloalkenyl group used herein refers to a monovalent monocyclic group that has 3 to 10 carbon atoms and at least one double bond in the ring thereof and no aromaticity, and examples thereof include a cyclopentenyl group, a cyclohexenyl group, and a cycloheptenyl group. The term “C₃-C₁₀ cycloalkenylene group” as used herein refers to a divalent group having the same structure as the C₃-C₁₀ cycloalkenyl group.

The term “C₁-C₁₀ heterocycloalkenyl group” as used herein refers to a monovalent monocyclic group that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, 1 to 10 carbon atoms, and at least one double bond in its ring. Non-limiting examples of the C₁-C₁₀ heterocycloalkenyl group include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuranlyl group, and a 2,3-dihydrothiophenyl group. The term “C₁-C₁₀ heterocycloalkenylene group” as used herein refers to a divalent group having the same structure as the C₁-C₁₀ heterocycloalkenyl group.

The term “C₆-C₆₀ aryl group” as used herein refers to a monovalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms, and a C₆-C₆₀ arylene group used herein refers to a divalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms. Non-limiting examples of the C₆-C₆₀ aryl group include a phenyl group, a naphthyl group, an anthracenyl group, a phenanthrenyl group, a pyrenyl group, and a chrysenyl group. When the C₆-C₆₀ aryl group and the C₆-C₆₀ arylene group each include two or more rings, the rings may be fused to each other.

The term “C₁-C₆₀ heteroaryl group” as used herein refers to a monovalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms. The term “C₁-C₆₀ heteroarylene group” as used herein refers to a divalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms. Non-limiting examples of the C₁-C₆₀ heteroaryl group include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, and an isoquinolinyl group. When the C₁-C₆₀ heteroaryl group and the C₁-C₆₀ heteroarylene group each include two or more rings, the rings may be condensed with each other.

The term “C₆-C₆₀ aryloxy group” as used herein refers to —OA₁₀₂ (wherein A₁₀₂ is the C₆-C₆₀ aryl group), and a C₆-C₆₀ arylthio group used herein indicates —SA₁₀₃ (wherein A₁₀₃ is the C₆-C₆₀ aryl group).

The term “monovalent non-aromatic condensed polycyclic group” as used herein refers to a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed with each other, only carbon atoms as ring-forming atoms, and at least one non-aromatic ring in its entire molecular structure. A detailed example of the monovalent non-aromatic condensed polycyclic group is a fluo-

renyl group. The term “divalent non-aromatic condensed polycyclic group” as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed polycyclic group.

The term “monovalent non-aromatic condensed heteropolycyclic group” as used herein refers to a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings condensed to each other, at least one heteroatom selected from N, O, Si, P, and S, other than carbon atoms, as a ring-forming atom, and at least one non-aromatic ring in its entire molecular structure. An example of the monovalent non-aromatic condensed heteropolycyclic group is a carbazolyl group. The term “divalent non-aromatic condensed heteropolycyclic group” as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed heteropolycyclic group.

The term “C₅-C₆₀ carbocyclic group” as used herein refers to a monocyclic or polycyclic group having 5 to 60 carbon atoms in which a ring-forming atom is a carbon atom only. The C₅-C₆₀ carbocyclic group may be an aromatic carbocyclic group or a non-aromatic carbocyclic group. The C₅-C₆₀ carbocyclic group may be a ring, such as benzene, a monovalent group, such as a phenyl group, or a divalent group, such as a phenylene group. In an example embodiment, depending on the number of substituents connected to the C₅-C₆₀ carbocyclic group, the C₅-C₆₀ carbocyclic group may be a trivalent group or a quadrivalent group.

The term “C₁-C₆₀ heterocyclic group” as used herein refers to a group having the same structure as the C₅-C₆₀ carbocyclic group, except that as a ring-forming atom, at least one heteroatom selected from N, O, Si, P, and S is used in addition to carbon (the number of carbon atoms may be in a range of 1 to 60).

In the specification, at least one substituent of the substituted C₅-C₆₀ carbocyclic group, the substituted C₁-C₆₀ heterocyclic group, the substituted C₃-C₁₀ cycloalkylene group, the substituted C₁-C₁₀ heterocycloalkylene group, the substituted C₃-C₁₀ cycloalkenylene group, the substituted C₁-C₁₀ heterocycloalkenylene group, the substituted C₆-C₆₀ arylene group, the substituted C₁-C₆₀ heteroarylene group, the substituted divalent non-aromatic condensed polycyclic group, the substituted divalent non-aromatic condensed heteropolycyclic group, the substituted C₁-C₆₀ alkyl group, the substituted C₂-C₆₀ alkenyl group, the substituted C₂-C₆₀ alkynyl group, the substituted C₁-C₆₀ alkoxy group, the substituted C₃-C₁₀ cycloalkyl group, the substituted C₁-C₁₀ heterocycloalkyl group, the substituted C₃-C₁₀ cycloalkenyl group, the substituted C₁-C₁₀ heterocycloalkenyl group, the substituted C₆-C₆₀ aryl group, the substituted C₆-C₆₀ aryloxy group, the substituted C₆-C₆₀ arylthio group, the substituted C₁-C₆₀ heteroaryl group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may be selected from:

deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, and a C₁-C₆₀ alkoxy group;

a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, and a C₁-C₆₀ alkoxy group. each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group,

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a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₁₁)(Q₁₂)(Q₁₃), —N(Q₁₁)(Q₁₂), —B(Q₁₁)(Q₁₂), —C(=O)(Q₁₁), —S(=O)₂(Q₁₁), and —P(=O)(Q₁₁)(Q₁₂);
 a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group;
 a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₂₁)(Q₂₂)(Q₂₃), —N(Q₂₁)(Q₂₂), —B(Q₂₁)(Q₂₂), —C(=O)(Q₂₁), —S(=O)₂(Q₂₁), and —P(=O)(Q₂₁)(Q₂₂); and —Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)(Q₃₂), and
 Q₁₁ to Q₁₃, Q₂₁ to Q₂₃, and Q₃₁ to Q₃₃ may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a C₁-C₆₀ alkyl group substituted with at least one selected from deuterium, —F, and a cyano group, a C₆-C₆₀ aryl group substituted with at least one selected from deuterium, —F, and a cyano group, a biphenyl group, and a terphenyl group.

The term “Ph” as used herein represents a phenyl group, the term “Me” as used herein represents a methyl group, the term “Et” as used herein represents an ethyl group, the term “ter-Bu” or “Bu^t” as used herein, represents a tert-butyl group, and the term “OMe” as used herein represents a methoxy group.

The term “biphenyl group” as used herein refers to a “phenyl group substituted with a phenyl group. The “biphenyl group” is a “substituted phenyl group” having a “C₆-C₆₀ aryl group” as a substituent.

The term “terphenyl group” as used herein refers to a “phenyl group substituted with a biphenyl group. The “ter-

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phenyl group” is a “phenyl group” having, as a substituent, a “C₆-C₆₀ aryl group substituted with a C₆-C₆₀ aryl group.”

* and * as used herein, unless defined otherwise, each refer to a binding site to a neighboring atom in a corresponding formula.

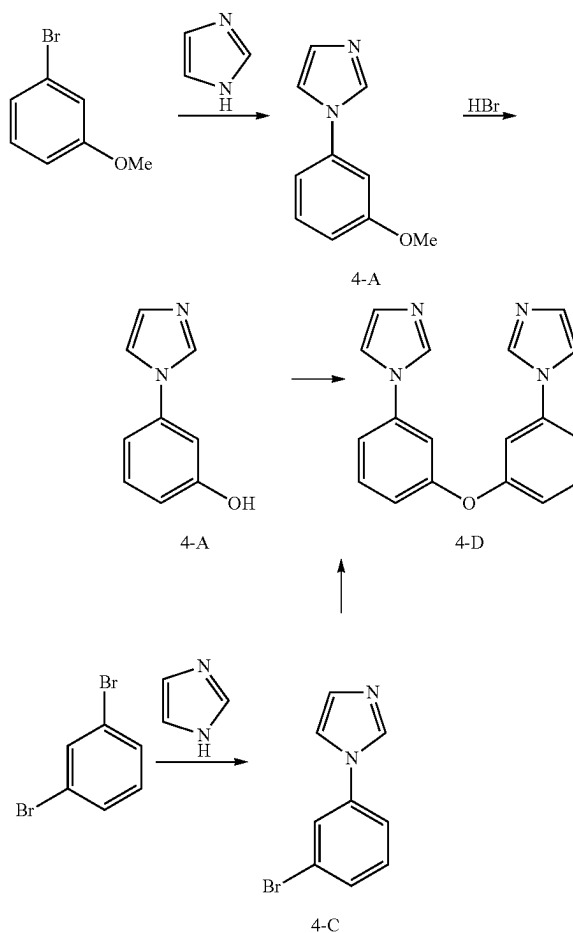
The following Examples and Comparative Examples are provided in order to highlight characteristics of one or more embodiments, but it will be understood that the Examples and Comparative Examples are not to be construed as limiting the scope of the embodiments, nor are the Comparative Examples to be construed as being outside the scope of the embodiments. Further, it will be understood that the embodiments are not limited to the particular details described in the Examples and Comparative Examples.

The expression “B was used instead of A” used in describing Synthesis Examples means that an identical number of molar equivalents of A was used in place of molar equivalents of B.

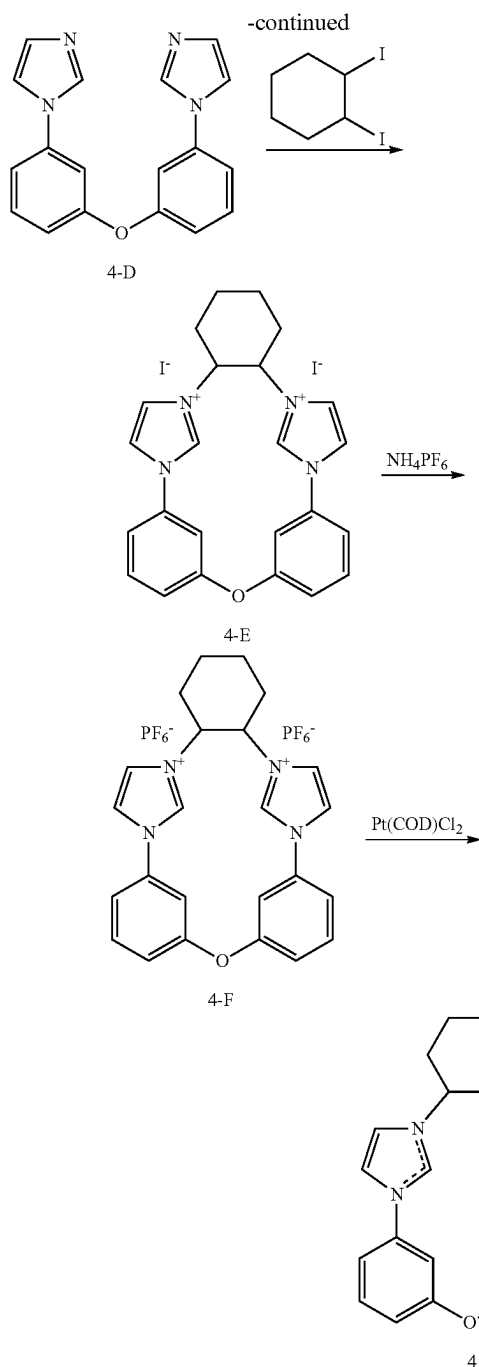
SYNTHESIS EXAMPLES

Synthesis Example 1

Synthesis of Compound 4



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1) Synthesis of Intermediate [4-A]

9.35 g (50 mmol) of bromoanisole, 3.40 g (50 mmol) of imidazole, 23 g (100 mmol) of potassium triphosphate, 1.83 g (10 mmol) of iodine copper, and 1.17 g (10 mmol) of picolinic acid were added to a reaction container and suspended in 100 mL of dimethyl sulfoxide. The reaction mixture was heated and stirred at a temperature of 160° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from

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which a solvent was removed, was separated by column chromatography to obtain 6.97 g (40 mmol) of Intermediate [4-A].

2) Synthesis of Intermediate [4-B]

6.97 g (40 mmol) of Intermediate [4-A] was suspended in an excess of bromic acid solution. The reaction mixture was heated and stirred at a temperature of 110° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature and neutralized by adding an appropriate amount of sodium hydrogen carbonate thereto. 300 mL of distilled water was added thereto and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 5.77 g (36 mmol) of Intermediate [4-B].

3) Synthesis of Intermediate [4-C]

11.80 g (50 mmol) of 1,3-dibromobenzene, 3.40 g (50 mmol) of imidazole, 23 g (100 mmol) of potassium triphosphate, 1.83 g (10 mmol) of iodine copper, and 1.17 g (10 mmol) of picolinic acid were added to a reaction container and suspended in 100 mL of dimethyl sulfoxide. The reaction mixture was heated and stirred at a temperature of 160° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 7.81 g (35 mmol) of Intermediate [4-C].

4) Synthesis of Intermediate [4-D]

5.77 g (36 mmol) of Intermediate [4-B], 7.81 g (35 mmol) of Intermediate [4-C], 16.1 g (70 mmol) of potassium triphosphate, 1.28 g (7 mmol) of iodine copper, and 0.82 g (7 mmol) of picolinic acid were added to a reaction container and suspended in 70 mL of dimethyl sulfoxide. The reaction mixture was heated and stirred at a temperature of 160° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 8.47 g (28 mmol) of Intermediate [4-D].

5) Synthesis of Intermediate [4-E]

8.47 g (28 mmol) of Intermediate [4-D] and 1.68 g (140 mmol) of 1,2-diiodocyclohexane were added to a reaction container and suspended in 100 mL of toluene. The reaction mixture was heated and stirred at a temperature of 110° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, and a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 12.5 g (19.6 mmol) of Intermediate [4-E].

6) Synthesis of Intermediate [4-F]

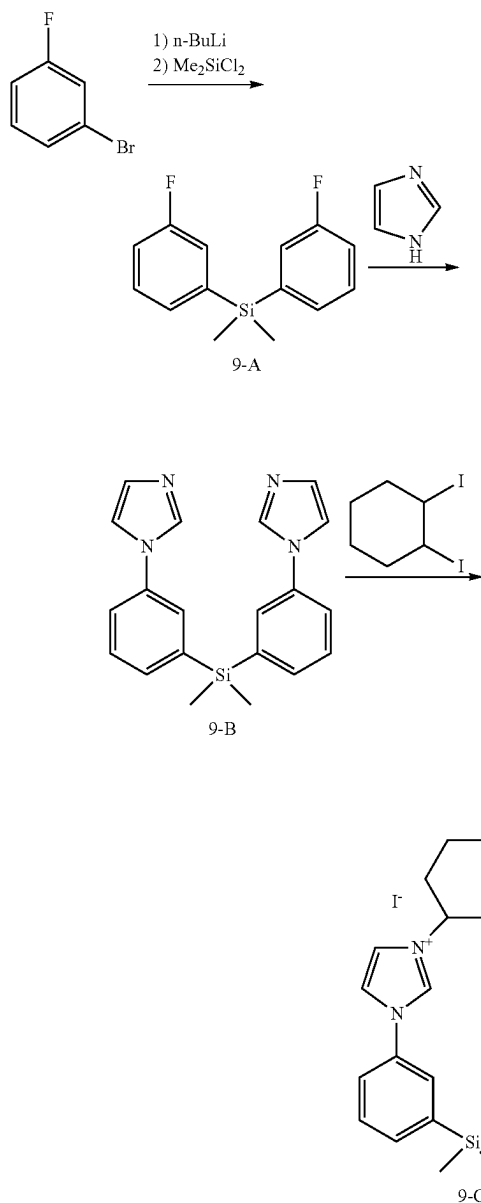
12.5 g (19.6 mmol) of Intermediate [4-E] and 9.58 g (58.8 mmol) of ammonium hexafluorophosphate were suspended in 50 mL of methyl alcohol and 50 mL of water. The reaction mixture was stirred at room temperature for 24 hours. After the reaction was completed, a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 9.25 g (13.7 mmol) of Intermediate [4-F].

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7) Synthesis of Compound 4

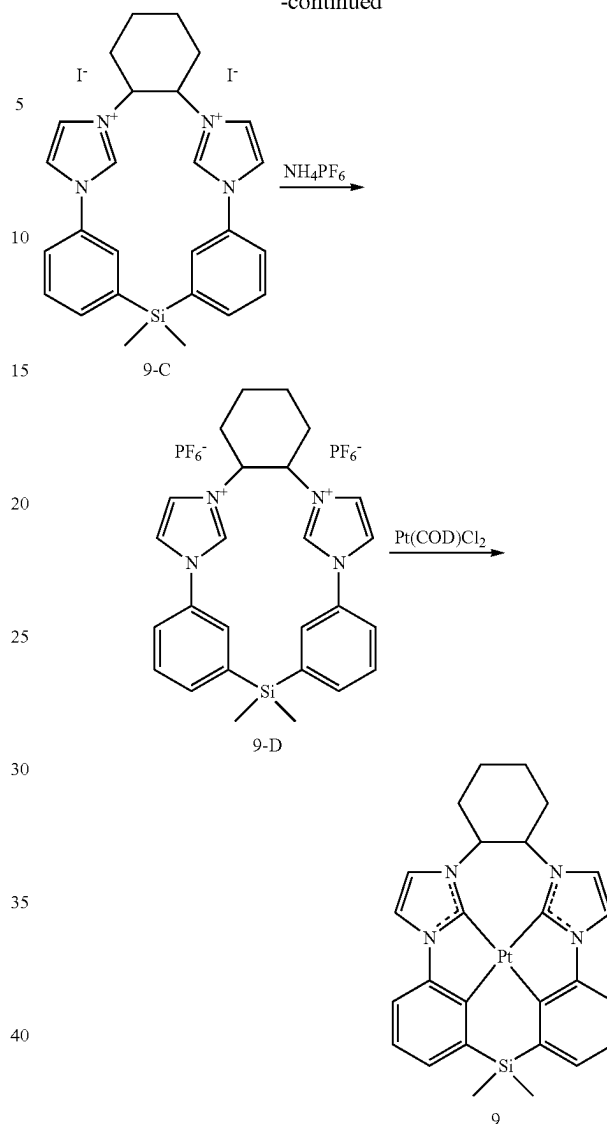
9.25 g (13.7 mmol) of Intermediate [4-F], 5.6 g (15 mmol) of dichloro(1,5-cyclooctadiene)platinum, and 3.4 g (41 mmol) of sodium acetate were suspended in 160 mL of dioxane. The reaction mixture was heated and stirred at a temperature of 110° C. for 72 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 2.37 g (4.1 mmol) of Compound 4.

Synthesis Example 2: Synthesis of Compound 9



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-continued



1) Synthesis of Intermediate [9-A]

8.75 g (50 mmol) of 1-bromo-3-fluorobenzene was suspended in 100 mL of tetrahydrofuran and stirred at a temperature of -78° C. 24 mL of $n\text{-BuLi}$ solution (2.5 M in hexane) was added to the mixed solution and stirred at the same temperature for 2 hours. After the stirring, 7.74 g (60 mmol) of dimethyldichlorosilane was added thereto and further stirred at room temperature for 24 hours. After the reaction was completed, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 9.93 g (40 mmol) of Intermediate [9-A].

2) Synthesis of Intermediate [9-B]

9.93 g (40 mmol) of Intermediate [9-A], 8.17 g (120 mmol) of imidazole, and 36.8 g (160 mmol) of potassium triphosphate were added to a reaction container and suspended in 80 mL of dimethyl sulfoxide. The reaction mixture was heated and stirred at a temperature of 160° C. for 24 hours. After the reaction was completed, the reaction

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product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 9.0 g (26 mmol) of Intermediate [9-B].

3) Synthesis of Intermediate [9-C]

9.0 g (26 mmol) of Intermediate [9-B] and 1.56 g (130 mmol) of 1,2-diiodocyclohexane were added to a reaction container and suspended in 100 mL of toluene. The reaction mixture was heated and stirred at a temperature of 110° C. for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, and a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 12.4 g (18.2 mmol) of Intermediate [9-C].

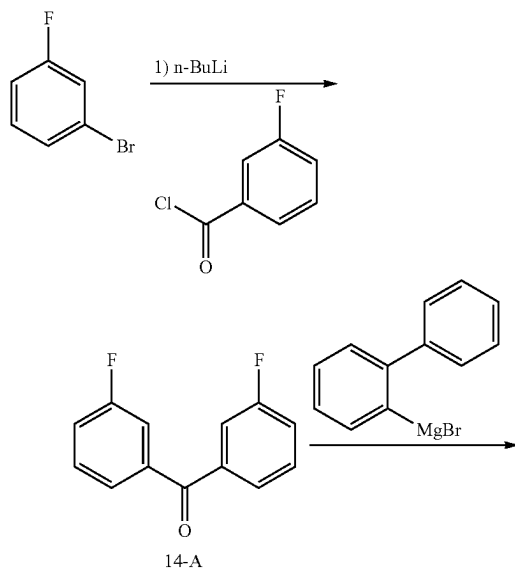
4) Synthesis of Intermediate [9-D]

12.4 g (18.2 mmol) of Intermediate [9-C] and 8.9 g (54.6 mmol) of ammonium hexafluorophosphate were added to a reaction container and suspended in 50 mL of methyl alcohol and 50 mL of water. The reaction mixture was stirred at room temperature for 24 hours. After the reaction was completed, a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 8.48 g (11.8 mmol) of Intermediate [9-D].

5) Synthesis of Compound 9

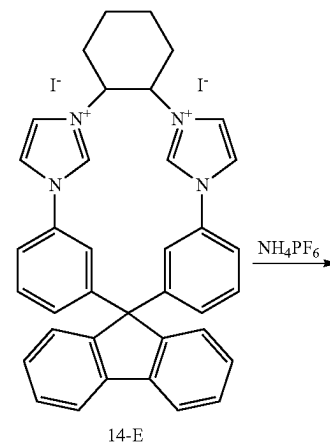
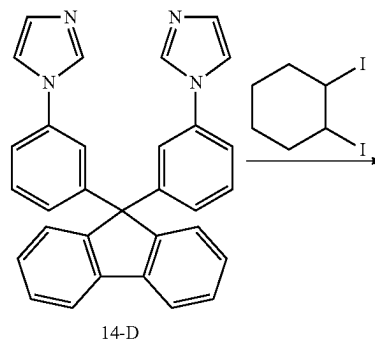
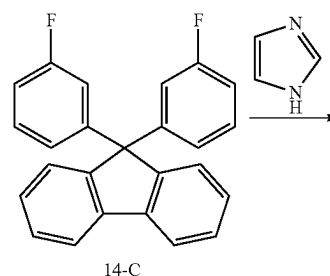
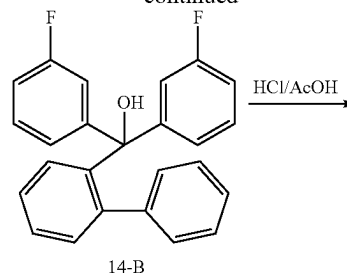
8.48 g (11.8 mmol) of Intermediate [9-D], 4.85 g (13 mmol) of dichloro(1,5-cyclooctadiene)platinum, and 2.94 g (35.4 mmol) of sodium acetate were suspended in 140 mL of dioxane. The reaction mixture was heated and stirred at a temperature of 110° C. for 72 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 1.11 g (1.8 mmol) of Compound 9.

Synthesis Example 3: Synthesis of Compound 14

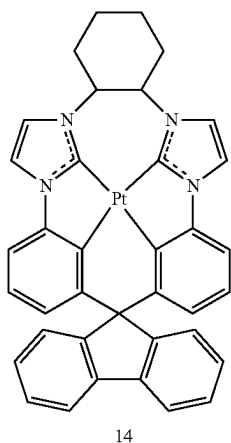
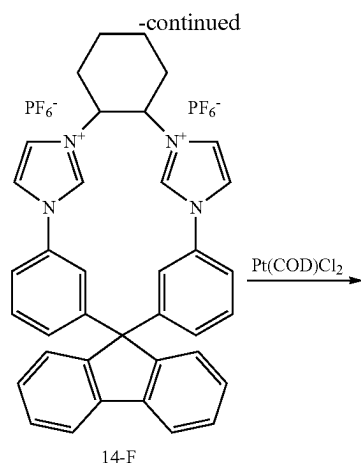


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-continued



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1) Synthesis of Intermediate [14-A]

8.75 g (50 mmol) of 1-bromo-3-fluorobenzene was suspended in 100 mL of tetrahydrofuran and stirred at a temperature of -78°C . 24 mL of *n*-BuLi solution (2.5 M in hexane) was added to the mixed solution and stirred at the same temperature for 2 hours. After the stirring, 9.51 g (60 mmol) of 3-fluorobenzoyl chloride was added thereto and further stirred at room temperature for 24 hours. After the reaction was completed, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 8.29 g (38 mmol) of Intermediate [14-A].

2) Synthesis of Intermediate [14-B]

10.7 g (45.6 mmol) of 2-bromobiphenyl, 11.1 g (45.6 mmol) of magnesium, and a catalytic amount of iodine were suspended in 100 mL of tetrahydrofuran and refluxed for 3 hours to generate a Grignard reagent. 8.29 g (38 mmol) of Intermediate [14-A] was further added to the mixture and refluxed for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, and 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 10.6 g (28.5 mmol) of Intermediate [14-B].

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3) Synthesis of Intermediate [14-C]

10.6 g (28.5 mmol) of Intermediate [14-B] was added to an excess of hydrochloric acid/acetic acid solution and refluxed for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, and 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 8.59 g (24.2 mmol) of Intermediate [14-C].

4) Synthesis of Intermediate [14-D]

8.59 g (24.2 mmol) of Intermediate [14-C] and 1.45 g (121 mmol) of imidazole were added to a reaction container and suspended in 100 mL of toluene. The reaction mixture was heated and stirred at a temperature of 110°C . for 24 hours. After the reaction was completed, the reaction product was cooled to room temperature, and a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 7.61 g (9.68 mmol) of Intermediate [14-D].

5) Synthesis of Intermediate [14-E]

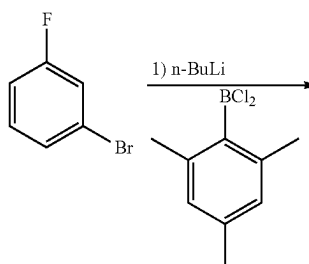
7.61 g (9.68 mmol) of Intermediate [14-D] and 4.72 g (29 mmol) of ammonium hexafluorophosphate were added to a reaction container and suspended in 20 mL of methyl alcohol and 20 mL of water. The reaction mixture was stirred at room temperature for 24 hours. After the reaction was completed, a solid generated therefrom was filtered and washed with ether. The washed solid was dried to obtain 6.37 g (7.74 mmol) of Intermediate [14-E].

6) Synthesis of Compound 14

6.37 g (7.74 mmol) of Intermediate [14-E], 3.2 g (8.51 mmol) of dichloro(1,5-cyclooctadiene)platinum, and 1.92 g (23.2 mmol) of sodium acetate were suspended in 90 mL of dioxane. The reaction mixture was heated and stirred at a temperature of 110°C . for 72 hours. After the reaction was completed, the reaction product was cooled to room temperature, 300 mL of distilled water was added thereto, and an organic layer was extracted therefrom by using ethyl acetate. The extracted organic layer was washed with a saturated aqueous sodium chloride solution and dried by using sodium sulfate. A residue, from which a solvent was removed, was separated by column chromatography to obtain 1.34 g (1.86 mmol) of Compound 14.

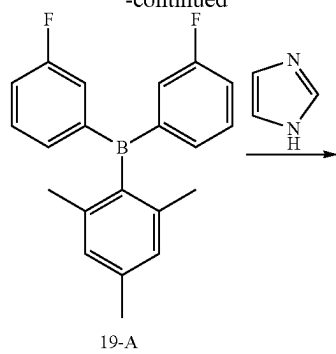
Synthesis Example 4

Synthesis of Compound 19

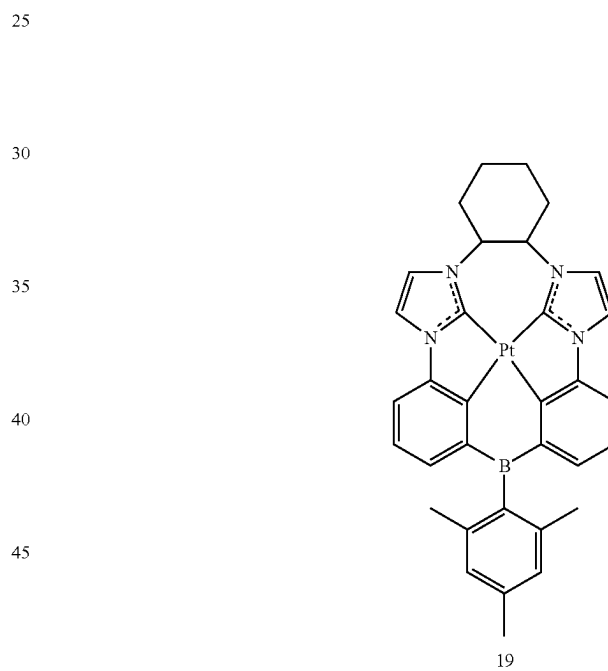
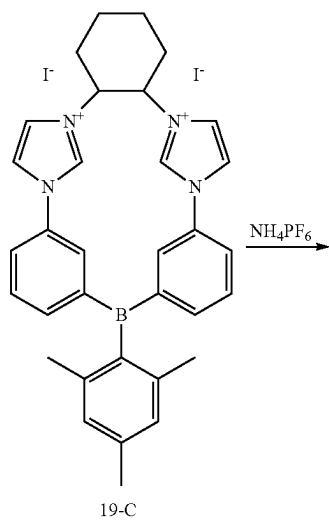
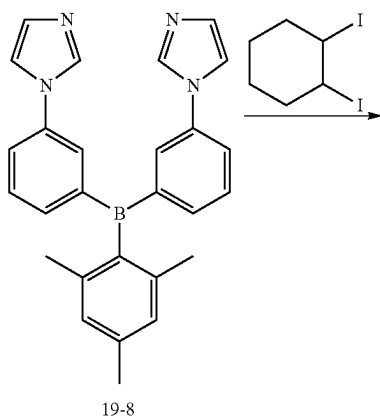
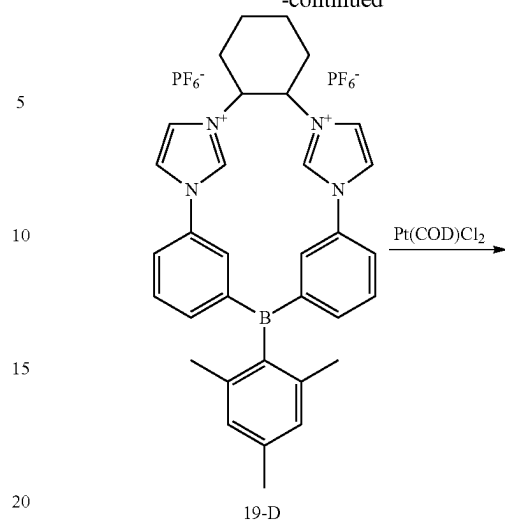


135

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**136**

-continued



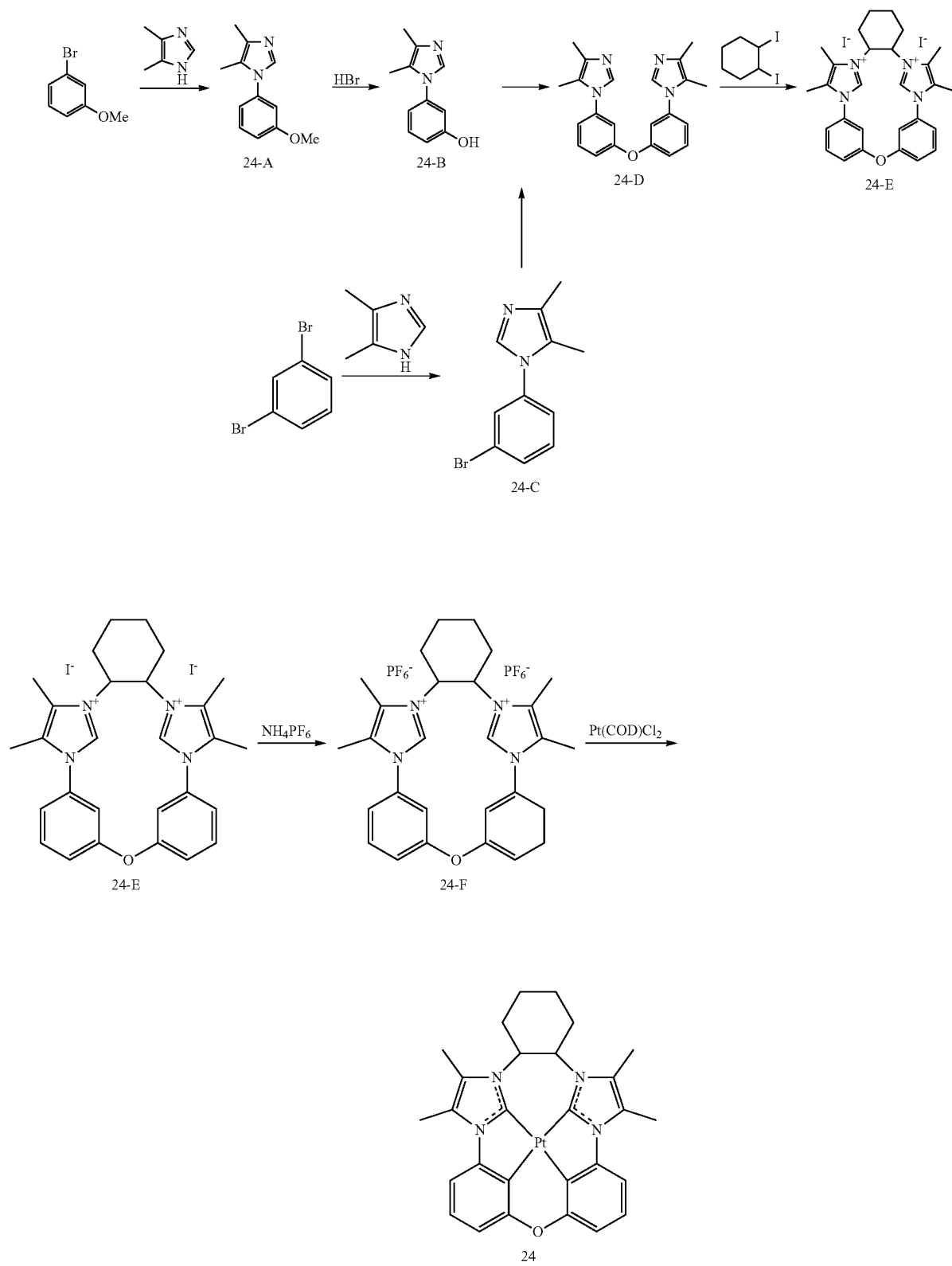
Compound 19 was obtained in the same manner as in
 Synthesis Example 2, except that dichloro(2,4,6-trimeth-
 ylphenyl)borane was used instead of dimethyldichlorosi-
 lane.

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Synthesis Example 5

Synthesis of Compound 24

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Compound 24 was obtained in the same manner as in Synthesis Example 1, except that 4,5-dimethyl-1H-imidazole was used instead of imidazole.

¹H NMR and MS/FAB of the compounds synthesized in Synthesis Examples 1 to 5 are shown in Table 1.

Synthesis methods of compounds other than Compounds shown in Table 1 may also be easily recognized by those of skill in the art by referring to the synthesis mechanisms and methods described above.

TABLE 1

Compound	¹ H NMR (CDCl ₃ , 400 MHz)	MS/FAB	
		found	calc.
4	0.82-1.81 (8H, m), 2.62 (2H, m), 6.45-6.55 (6H, m), 7.10 (2H, m), 8.55 (2H, m)	575.1290	575.1285
9	0.25 (6H, s), 1.11-1.89 (8H, m), 2.52 (2H, m), 5.83 (1H, m), 6.31 (2H, m), 6.52 (2H, m), 6.81 (2H, m), 7.22 (1H, m), 8.74 (2H, m)	617.1584	617.1574
14	1.1-1.98 (8H, m), 2.65 (2H, m), 6.57 (2H, m), 6.71-6.82 (4H, m), 7.09-7.48 (8H, m), 7.91 (2H, m), 8.75 (2H, m)	723.1953	723.1962
19	0.81-1.91 (8H, m), 2.18 (3H, s), 2.41 (6H, s), 2.42 (2H, m), 6.46 (2H, m), 6.81 (2H, m), 7.01 (2H, m), 7.24-7.65 (4H, m), 8.51 (2H, m)	689.2285	689.2290
24	0.84-1.85 (8H, m), 2.31 (12H, s) 2.63 (2H, m), 6.41-6.45 (4H, m), 7.12 (2H, m)	631.1910	631.1911

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EXAMPLES

Example 1

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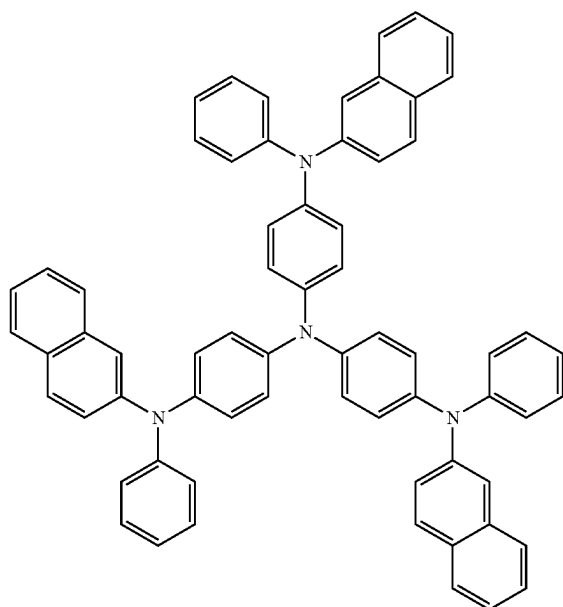
As a substrate and an ITO anode, a Corning 15 Ω/cm² (1,200 Å) ITO glass substrate was cut to a size of 50 mm×50 mm×0.7 mm, sonicated with isopropyl alcohol and pure water each for 5 minutes, and then cleaned by exposure to ultraviolet rays and ozone for 30 minutes. Then, the ITO glass substrate was provided to a vacuum deposition apparatus.

With reference to the compounds illustrated below, 2-TNATA was vacuum-deposited on the ITO anode formed on the ITO glass substrate to form a hole injection layer having a thickness of 600 Å, and NPB was vacuum-deposited on the hole injection layer to form a hole transport layer having a thickness of 300 Å.

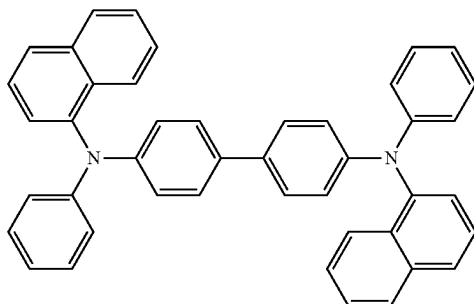
BCPDS and POPCPA (BCPDS to POPCPA weight ratio of 1:1) as a co-host and Compound 4 as a dopant were co-deposited on the hole transport layer as a co-host to dopant weight ratio of 90:10 to form an emission layer having a thickness of 300 Å.

TSPO1 was deposited on the emission layer to form a hole blocking layer having a thickness of 50 Å, Alga was deposited on the hole blocking layer to form an electron transport layer having a thickness of 300 Å, LiF was deposited on the electron transport layer to form an electron injection layer having a thickness of 10 Å, and Al was vacuum-deposited on the electron injection layer to form a cathode having a thickness of 3,000 Å, thereby completing the manufacture of an organic light-emitting device.

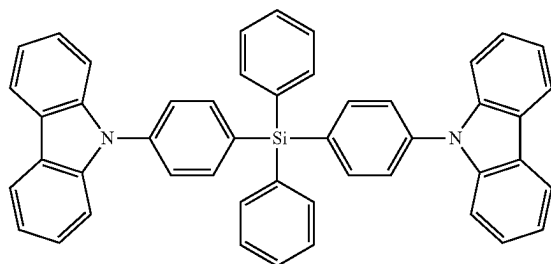
141



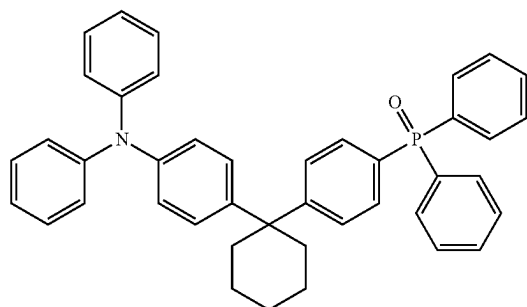
2-TNATA



NPB



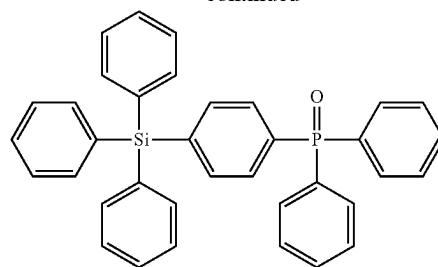
BCPDS



POPCPA

142

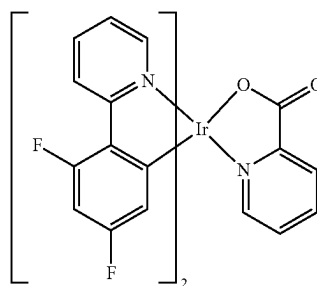
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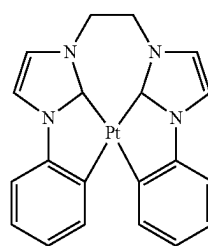
TSP01

Examples 2 to 5 and Comparative Examples 1 to 4

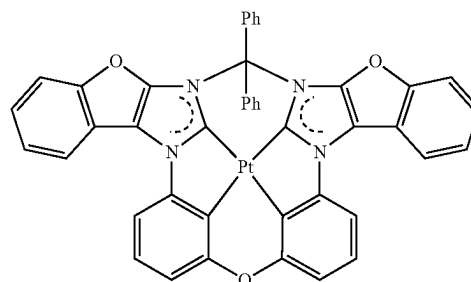
Organic light-emitting devices were manufactured in the same manner as in Example 1, except that the compounds shown in Table 2 were each used instead of Compound 1 as a dopant in forming an emission layer.



Flrpic



<Compound A>



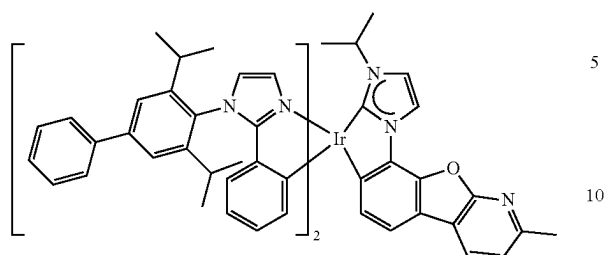
<Compound B>

(Ph indicates a phenyl group)

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-continued

<Compound C>



Evaluation Example 1

The driving voltage, current density, brightness, luminance efficiency, emission color, and wavelength of maximum emission of the organic light-emitting devices manufactured according to Examples 1 to 5 and Comparative Examples 1 to 4 were measured by using Keithley SMU 236 and a luminance meter PR650, and results thereof are shown in Table 2.

TABLE 2

	Dopant compound	Driving voltage (V)	Current density (mA/cm ²)	Brightness (cd/m ²)	Luminance efficiency (cd/A)	Emission color	wavelength of maximum emission (nm)
Example 1	Compound 4	5.30	50	4125	8.25	Blue	490
Example 2	Compound 9	5.45	50	4025	8.05	Blue	470
Example 3	Compound 14	5.55	50	4033	8.07	Blue	474
Example 4	Compound 19	5.43	50	4024	8.05	Blue	468
Example 5	Compound 24	5.58	50	4055	8.11	Blue	495
Comparative Example 1	Flpic	6.56	50	3870	7.74	Blue	478
Comparative Example 2	Compound A	7.10	50	2957	6.89	Blue	460
Comparative Example 3	Compound B	6.83	50	3800	7.34	Blue	500
Comparative Example 4	Compound C	6.23	50	3759	7.12	Blue	480

Referring to Table 2, it can be seen that the organic light-emitting devices of Examples 1 to 5 had a low driving voltage and high brightness and luminance efficiency, as compared with those of the organic light-emitting devices of Comparative Examples 1 to 4.

As described above, an organic light-emitting device including the organometallic compound may have a low driving voltage, high brightness, and high luminance efficiency.

Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with

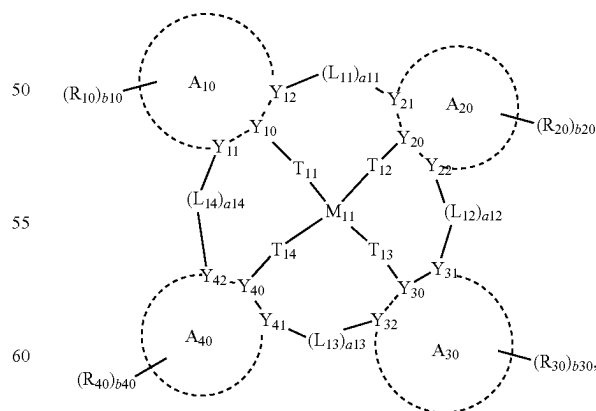
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other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the claims.

What is claimed is:

1. An organic light-emitting device comprising: a first electrode; a second electrode; and an organic layer between the first electrode and the second electrode and comprising an emission layer, wherein the organic layer comprises an organometallic compound represented by Formula 1:

<Formula 1>



wherein, in Formula 1,

M_{11} is selected from platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium

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(Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), and thulium (Tm),
 Y_{10} , Y_{20} , Y_{30} , and Y_{40} are each independently N or C,
 Y_{11} , Y_{12} , Y_{21} , Y_{22} , Y_{31} , Y_{32} , Y_{41} , and Y_{42} are each 5
independently N or C, A_{30} , and A_{40} are each independently selected from a C_5 - C_{60} carbocyclic group and a C_1 - C_{60} heterocyclic group,
 T_{11} to T_{14} are each independently selected from a single 10
bond, $*-O-*'$, and $*-S-*'$,
 L_{11} is selected from a substituted or unsubstituted C_3 - C_{10} cycloalkylene group, $*-O-*'$, $*-S-*'$, $*-C(R_1)$
 $(R_2)-*$, $*-C(R_1)=*$, $*=C(R_1)-*$, $*-C(R_1)=C$
 $(R_2)-*$, $*-C(=O)-*$, $*-C(=S)-*$, $*-C\equiv C-*$, 15
 $*-B(R_1)-*$, $*-N(R_1)-*$, $*-P(R_1)-*$, $*-Si(R_1)$
 $(R_2)-*$, $*-P(R_1)(R_2)-*$, and $*-Ge(R_1)(R_2)-*$,
a11 is selected from 1, 2, 3, 4, and 5,
at least one selected from L_{11} (s) in the number of a11 is 20
a substituted or unsubstituted C_3 - C_{10} cycloalkylene
group,
 L_{12} and L_{14} are each independently selected from a single
bond, $*-O-*'$, $*-S-*'$, $*-C(R_1)(R_2)-*$, $*-C$
 $(R_1)=*$, $*=C(R_1)-*$, $*-C(R_1)=C(R_2)-*$, $*-C$
 $(=O)-*$, $*-C(=S)-*$, $*-C\equiv C-*$, $*-B(R_1)-*$, 25
 $*-N(R_1)-*$, $*-P(R_1)-*$, $*-Si(R_1)(R_2)-*$, $*-P$
 $(R_1)(R_2)-*$, and $*-Ge(R_1)(R_2)-*$,
 L_{13} is selected from a single bond, $*-O-*'$, $*-S-*'$,
 $*-B(R_1)-*$, $*-N(R_1)-*$, $*-P(R_1)-*$, $*-P(R_1)$
 $(R_2)-*$, and $*-Ge(R_1)(R_2)-*$, wherein, if L_{11} is a 30
cyclohexylene group, L_{13} is $*-B(R_1)-*$,
a12, and a14 are each independently selected from 0, 1, 2,
3, 4, and 5,
a13 is selected from 1, 2, 3, 4, and 5,
 R_1 , R_2 , R_{10} , R_{20} , R_{30} , and R_{40} are each independently 35
selected from hydrogen, deuterium, —F, —Cl, —Br,
—I, a hydroxyl group, a cyano group, a nitro group, an
amidino group, a hydrazino group, a hydrazono group,
a substituted or unsubstituted C_1 - C_{60} alkyl group, a
substituted or unsubstituted C_2 - C_{60} alkenyl group, a 40
substituted or unsubstituted C_2 - C_{60} alkynyl group, a
substituted or unsubstituted C_1 - C_{60} alkoxy group, a
substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a
substituted or unsubstituted C_1 - C_{10} heterocycloalkyl
group, a substituted or unsubstituted C_3 - C_{10} cycloalk- 45
enyl group, a substituted or unsubstituted C_1 - C_{10} hetero-
cycloalkenyl group, a substituted or unsubstituted
 C_6 - C_{60} aryl group, a substituted or unsubstituted
 C_6 - C_{60} aryloxy group, a substituted or unsubstituted
 C_6 - C_{60} arylthio group, a substituted or unsubstituted 50
 C_1 - C_{60} heteroaryl group, a substituted or unsubstituted
 C_1 - C_{60} hetero aryloxy group, a substituted or unsub-
stituted C_1 - C_{60} hetero arylthio group, a substituted or
unsubstituted monovalent non-aromatic condensed
polycyclic group, a substituted or unsubstituted mono- 55
valent non-aromatic condensed heteropolycyclic
group, —Si(Q_1)(Q_2)(Q_3), —B(Q_1)(Q_2), —N(Q_1)(Q_2),
—P(Q_1)(Q_2), —C(=O)(Q_1), —S(=O)(Q_1),
—S(=O)₂(Q_1), —P(=O)(Q_1)(Q_2), and —P(=S)(Q_1)
(Q_2), 60
 R_1 and R_2 are optionally linked to form a substituted or
unsubstituted C_5 - C_{60} carbocyclic group or a substituted
or unsubstituted C_1 - C_{60} heterocyclic group,
 R_1 and R_{10} ; R_1 and R_{20} ; R_1 and R_{30} ; or R_1 and R_{40} are
optionally linked to form a substituted or unsubstituted 65
 C_5 - C_{60} carbocyclic group or a substituted or unsubsti-
tuted C_1 - C_{60} heterocyclic group,

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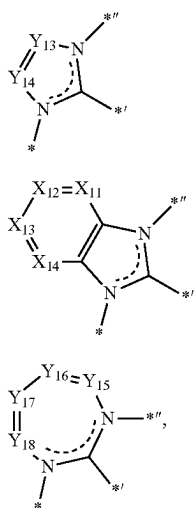
b10, b20, b30, and b40 are each independently selected
from 1, 2, 3, 4, 5, 6, 7, and 8,
* and *' each indicate a binding site to a neighboring
atom,
at least one substituent of the substituted C_5 - C_{60} carbo-
cyclic group, the substituted C_1 - C_{60} heterocyclic group,
the substituted C_3 - C_{10} cycloalkylene group, the substi-
tuted C_1 - C_{60} alkyl group, the substituted C_2 - C_{60} alkenyl
group, the substituted C_2 - C_{60} alkynyl group, the substi-
tuted C_1 - C_{60} alkoxy group, the substituted
 C_3 - C_{10} cycloalkyl group, the substituted C_1 - C_{10} hetero-
cycloalkyl group, the substituted C_3 - C_{10} cycloalk-
enyl group, the substituted C_1 - C_{10} heterocycloalkenyl
group, the substituted C_6 - C_{60} aryl group, the substi-
tuted C_6 - C_{60} aryloxy group, the substituted C_6 - C_{60}
arylthio group, the substituted C_1 - C_{60} heteroaryl group,
the substituted monovalent non-aromatic condensed
polycyclic group, and the substituted monovalent non-
aromatic condensed heteropolycyclic group is selected
from:
deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a
cyano group, a nitro group, an amidino group, a
hydrazino group, a hydrazono group, a C_1 - C_{60} alkyl
group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group,
and a C_1 - C_{60} alkoxy group;
a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60}
alkynyl group, and a C_1 - C_{60} alkoxy group, each substi-
tuted with at least one selected from deuterium, —F,
—Cl, Br, —I, a hydroxyl group, a cyano group, a nitro
group, an amidino group, a hydrazino group, a hydrazono
group, a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} hetero-
cycloalkyl group, a C_3 - C_{10} cycloalkenyl group, a
 C_1 - C_{10} heterocycloalkenyl group, a C_6 - C_{60} aryl group,
a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a
 C_1 - C_{60} heteroaryl group, a monovalent non-aromatic
condensed polycyclic group, a monovalent non-aromatic
condensed heteropolycyclic group, —Si(Q_{11})
(Q_{12})(Q_{13}), —N(Q_{11})(Q_{12}), —B(Q_{11})(Q_{12}), —C(=O)
(Q_{11}), —S(=O)₂(Q_{11}), and —P(=O)(Q_{11})(Q_{12});
a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl
group, a C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} hetero-
cycloalkenyl group, a C_6 - C_{60} aryl group, a C_6 - C_{60}
aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} hetero-
aryl group, a monovalent non-aromatic condensed
polycyclic group, and a monovalent non-aromatic condensed
heteropolycyclic group;
a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl
group, a C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} hetero-
cycloalkenyl group, a C_6 - C_{60} aryl group, a C_6 - C_{60}
aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} hetero-
aryl group, a monovalent non-aromatic condensed
polycyclic group, and a monovalent non-aromatic condensed
heteropolycyclic group, each substituted with at
least one selected from deuterium, —F, —Cl, —Br,
—I, a hydroxyl group, a cyano group, a nitro group, an
amidino group, a hydrazino group, a hydrazono group,
a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60}
alkynyl group, a C_1 - C_{60} alkoxy group, a C_3 - C_{10}
cycloalkyl group, a C_1 - C_{10} heterocycloalkyl group, a
 C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} heterocycloalk-
enyl group, a C_6 - C_{60} aryl group, a C_6 - C_{60} aryloxy
group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryl
group, a monovalent non-aromatic condensed polycyclic
group, a monovalent non-aromatic condensed hetero-
polycyclic group, —Si(Q_{21})(Q_{22})(Q_{23}), —N(Q_{21})
(Q_{22}), —B(Q_{21})(Q_{22}), —C(=O)(Q_{21}), —S(=O)₂
(Q_{21}), and —P(=O)(Q_{21})(Q_{22}); and

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—Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂),
—C(=O)(Q₃₁), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)
(Q₃₂), and

Q₁ to Q₃, Q₁₁ to Q₁₃, Q₂₁ to Q₂₃, and Q₃₁ to Q₃₃ are each
independently selected from hydrogen, deuterium, —F,
—Cl, —Br, —I, a hydroxyl group, a cyano group, a
nitro group, an amidino group, a hydrazino group, a
hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alk-
enyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy
group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocy-
cloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀
heterocycloalkenyl group, a C₆-C₆₀ aryl group, a
C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a
C₁-C₆₀ heteroaryl group, a C₁-C₆₀ hetero aryloxy
group, a C₁-C₆₀ hetero arylthio group, a monovalent
non-aromatic condensed polycyclic group, a monova-
lent non-aromatic condensed heteropolycyclic group, a
C₁-C₆₀ alkyl group substituted with at least one
selected from deuterium, —F, and a cyano group, a
C₆-C₆₀ aryl group substituted with at least one selected
from deuterium, —F, and a cyano group, a biphenyl
group, and a terphenyl group, and

A₁₀ and A₂₀ are each independently selected from groups
represented by Formulae 10-1 to 10-3:



wherein, in Formulae 10-1 to 10-3,

Y₁₃ to Y₁₈ and X₁₁ to X₁₄ are each independently N or
C(R₁₀), and

*, *, and *'' each indicate a binding site to a neighboring
atom.

2. The organic light-emitting device of claim 1, wherein
the first electrode is an anode,

the second electrode is a cathode,

the organic layer further comprises a hole transport region
between the first electrode and the emission layer and
an electron transport region between the emission layer
and the second electrode,

the hole transport region comprises at least one layer
selected from a hole injection layer, a hole transport
layer, a buffer layer, an emission auxiliary layer, and an
electron blocking layer, and

the electron transport region comprises at least one layer
selected from a hole blocking layer, an electron trans-
port layer, and an electron injection layer.

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3. The organic light-emitting device of claim 2, wherein
the emission layer comprises the organometallic com-
pound.

4. The organic light-emitting device of claim 1, wherein
the emission layer emits blue light having a maximum
emission wavelength of about 440 nm to about 500 nm.

5. The organic light-emitting device of claim 3, wherein
the emission layer comprises a host and a dopant, and
the dopant comprises the organometallic compound.

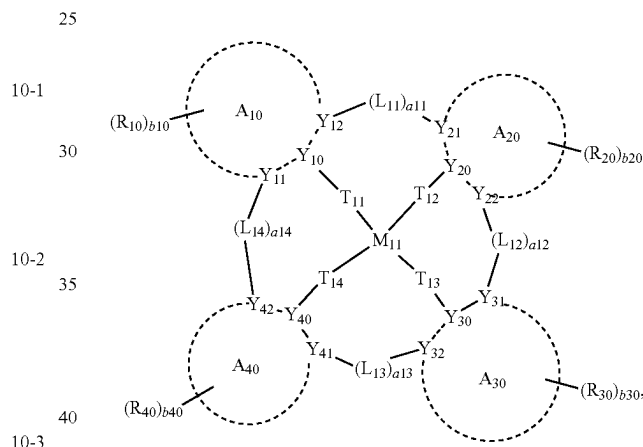
6. The organic light-emitting device of claim 3, wherein
the emission layer further comprises a host, and
the host comprises a silyl-containing compound, a phos-
phine oxide-containing compound, or a combination
thereof.

7. The organic light-emitting device of claim 2, wherein
the electron transport region comprises a hole blocking
layer,

the hole blocking layer directly contacts the emission
layer, and

the hole blocking layer comprises a phosphine oxide-
containing compound or a silyl-containing compound.

8. An organometallic compound represented by Formula
1:



wherein M₁₁ is selected from platinum (Pt), palladium
(Pd), copper (Cu), silver (Ag), gold (Au), rhodium
(Rh), iridium (Ir), ruthenium (Ru), osmium (Os), tita-
nium (Ti), zirconium (Zr), hafnium (Hf), europium
(Eu), terbium (Tb), and thulium (Tm),

Y₁₀, Y₂₀, Y₃₀, and Y₄₀ are each independently N or C,
Y₁₁, Y₁₂, Y₂₁, Y₂₂, Y₃₁, Y₃₂, Y₄₁, and Y₄₂ are each
independently N or C,

A₃₀ and A₄₀ are each independently selected from a
C₅-C₆₀ carbocyclic group and a C₁-C₆₀ heterocyclic
group,

T₁₁ to T₁₄ are each independently selected from a single
bond, *—O—*, and *—S—*,

L₁₁ is selected from a substituted or unsubstituted C₃-C₁₀
cycloalkylene group, *—O—*, *—S—*, *—C(R₁)
(R₂)—*, *—C(R₁)=*, *—C(R₁)—*, *—C(R₁)=C
(R₂)—*, *—C(=O)—*, *—C(=S)—*, *—C≡C—*,
—B(R₁)—, *—N(R₁)—*, *—P(R₁)—*, *—Si(R₁)
(R₂)—*, *—P(R₁)(R₂)—*, and *—Ge(R₁)(R₂)—*,

a11 is selected from 1, 2, 3, 4, and 5,

at least one selected from L₁₁ (s) in the number of a11 is
a substituted or unsubstituted C₃-C₁₀ cycloalkylene
group,

L₁₂ and L₁₄ are each independently selected from a single
bond, *—O—*, *—S—*, *—C(R₁)(R₂)—*, *—N

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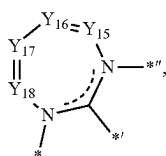
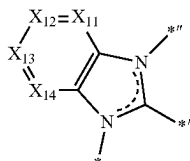
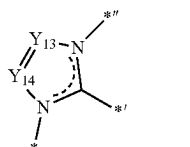
$(R_1)-*$, $*-P(R_1)-*$, $*-Si(R_1)(R_2)-*$, $*-P(R_1)(R_2)-*$, and $*-Ge(R_1)(R_2)-*$,
 L_{13} is selected from a single bond, $*-O-*$, $*-S-*$,
 $*-B(R_1)-*$, $*-N(R_1)-*$, $*-P(R_1)-*$, $*-P(R_1)(R_2)-*$, and $*-Ge(R_1)(R_2)-*$, wherein, if L_{11} is a
 cyclohexylene group, L_{13} is $*-B(R_1)-*$,
 a_{12} , and a_{14} are each independently selected from 0, 1, 2,
 3, 4, and 5,
 a_{13} is selected from 1, 2, 3, 4, and 5,
 R_1 , R_2 , R_{10} , R_{20} , R_{30} , and R_{40} are each independently
 selected from hydrogen, deuterium, $-F$, $-Cl$, $-Br$,
 $-I$, a hydroxyl group, a cyano group, a nitro group, an
 amidino group, a hydrazino group, a hydrazono group,
 a substituted or unsubstituted C_1-C_{60} alkyl group, a
 substituted or unsubstituted C_2-C_{60} alkenyl group, a
 substituted or unsubstituted C_2-C_{60} alkynyl group, a
 substituted or unsubstituted C_1-C_{60} alkoxy group, a
 substituted or unsubstituted C_3-C_{10} cycloalkyl group, a
 substituted or unsubstituted C_1-C_{10} heterocycloalkyl
 group, a substituted or unsubstituted C_3-C_{10} cycloalk-
 enyl group, a substituted or unsubstituted C_1-C_{10} hetero-
 cycloalkenyl group, a substituted or unsubstituted
 C_6-C_{60} aryl group, a substituted or unsubstituted
 C_6-C_{60} aryloxy group, a substituted or unsubstituted
 C_6-C_{60} arylthio group, a substituted or unsubstituted
 C_1-C_{60} heteroaryl group, a substituted or unsubstituted
 C_1-C_{60} hetero aryloxy group, a substituted or unsub-
 stituted C_1-C_{60} hetero arylthio group, a substituted or
 unsubstituted monovalent non-aromatic condensed
 polycyclic group, a substituted or unsubstituted mono-
 valent non-aromatic condensed heteropolycyclic
 group, $-Si(Q_1)(Q_2)(Q_3)$, $-B(Q_1)(Q_2)$, $-N(Q_1)(Q_2)$,
 $-P(Q_1)(Q_2)$, $-C(=O)(Q_1)$, $-S(=O)(Q_1)$,
 $-S(=O)_2(Q_1)$, $-P(=O)(Q_1)(Q_2)$, and $-P(=S)(Q_1)(Q_2)$,
 R_1 and R_2 are optionally linked to form a substituted or
 unsubstituted C_5-C_{60} carbocyclic group or a substituted
 or unsubstituted C_1-C_{60} heterocyclic group,
 R_1 and R_{10} ; R_1 and R_{20} ; R_1 and R_{30} ; or R_1 and R_{40} are
 optionally linked to form a substituted or unsubstituted
 C_5-C_{60} carbocyclic group or a substituted or unsubsti-
 tuted C_1-C_{60} heterocyclic group,
 b_{10} , b_{20} , b_{30} , and b_{40} are each independently selected
 from 1, 2, 3, 4, 5, 6, 7, and 8,
 $*$ and $*$ each indicate a binding site to a neighboring
 atom,
 at least one substituent of the substituted C_5-C_{60} carbo-
 cyclic group, the substituted C_1-C_{60} heterocyclic group,
 the substituted C_3-C_{10} cycloalkylene group, the substi-
 tuted C_1-C_{60} alkyl group, the substituted C_2-C_{60} alkenyl
 group, the substituted C_2-C_{60} alkynyl group, the substi-
 tuted C_1-C_{60} alkoxy group, the substituted
 C_3-C_{10} cycloalkyl group, the substituted C_1-C_{10} hetero-
 cycloalkyl group, the substituted C_3-C_{10} cycloalk-
 enyl group, the substituted C_1-C_{10} heterocycloalkenyl
 group, the substituted C_6-C_{60} aryl group, the substi-
 tuted C_6-C_{60} aryloxy group, the substituted C_6-C_{60}
 arylthio group, the substituted C_1-C_{60} heteroaryl group,
 the substituted monovalent non-aromatic condensed
 polycyclic group, and the substituted monovalent non-
 aromatic condensed heteropolycyclic group is selected
 from:
 deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a
 cyano group, a nitro group, an amidino group, a
 hydrazino group, a hydrazono group, a C_1-C_{60} alkyl
 group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group,
 and a C_1-C_{60} alkoxy group;

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a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60}
 alkynyl group, and a C_1-C_{60} alkoxy group, each sub-
 stituted with at least one selected from deuterium, $-F$,
 $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro
 group, an amidino group, a hydrazino group, a hydrazo-
 no group, a C_3-C_{10} cycloalkyl group, a C_1-C_{10} hetero-
 cycloalkyl group, a C_3-C_{10} cycloalkenyl group, a
 C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group,
 a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a
 C_1-C_{60} heteroaryl group, a monovalent non-aromatic
 condensed polycyclic group, a monovalent non-aro-
 matic condensed heteropolycyclic group, $-Si(Q_{11})(Q_{12})(Q_{13})$,
 $-N(Q_{11})(Q_{12})$, $-B(Q_{11})(Q_{12})$, $-C(=O)(Q_{11})$,
 $-S(=O)_2(Q_{11})$, and $-P(=O)(Q_{11})(Q_{12})$;
 a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl
 group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} hetero-
 cycloalkenyl group, a C_6-C_{60} aryl group, a C_6-C_{60}
 aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} hetero-
 aryl group, a monovalent non-aromatic condensed
 polycyclic group, and a monovalent non-aromatic con-
 densed heteropolycyclic group;
 a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl
 group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} hetero-
 cycloalkenyl group, a C_6-C_{60} aryl group, a C_6-C_{60}
 aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} hetero-
 aryl group, a monovalent non-aromatic condensed
 polycyclic group, and a monovalent non-aromatic con-
 densed heteropolycyclic group, each substituted with at
 least one selected from deuterium, $-F$, $-Cl$, $-Br$,
 $-I$, a hydroxyl group, a cyano group, a nitro group, an
 amidino group, a hydrazino group, a hydrazono group,
 a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60}
 alkynyl group, a C_1-C_{60} alkoxy group, a C_3-C_{10}
 cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a
 C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalk-
 enyl group, a C_6-C_{60} aryl group, a C_6-C_{60} aryloxy
 group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl
 group, a monovalent non-aromatic condensed polycy-
 clic group, a monovalent non-aromatic condensed het-
 eropolycyclic group, $-Si(Q_{21})(Q_{22})(Q_{23})$, $-N(Q_{21})(Q_{22})$,
 $-B(Q_{21})(Q_{22})$, $-C(=O)(Q_{21})$, $-S(=O)_2(Q_{21})$,
 and $-P(=O)(Q_{21})(Q_{22})$; and
 $-Si(Q_{31})(Q_{32})(Q_{33})$, $-N(Q_{31})(Q_{32})$, $-B(Q_{31})(Q_{32})$,
 $-C(=O)(Q_{31})$, $-S(=O)_2(Q_{31})$, and $-P(=O)(Q_{31})(Q_{32})$, and
 Q_1 to Q_3 , Q_{11} to Q_{13} , Q_{21} to Q_{23} , and Q_{31} to Q_{33} are each
 independently selected from hydrogen, deuterium, $-F$,
 $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a
 nitro group, an amidino group, a hydrazino group, a
 hydrazono group, a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl
 group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy
 group, a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocyclo-
 alkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10}
 heterocycloalkenyl group, a C_6-C_{60} aryl group, a
 C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a
 C_1-C_{60} heteroaryl group, a C_1-C_{60} hetero aryloxy
 group, a C_1-C_{60} hetero arylthio group, a monovalent
 non-aromatic condensed polycyclic group, a monova-
 lent non-aromatic condensed heteropolycyclic group, a
 C_1-C_{60} alkyl group substituted with at least one
 selected from deuterium, $-F$, and a cyano group, a
 C_6-C_{60} aryl group substituted with at least one
 selected from deuterium, $-F$, and a cyano group, a
 biphenyl group, and a terphenyl group, and

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A₁₀ and A₂₀ are each independently selected from groups represented by Formulae 10-1 to 10-3:



wherein, in Formulae 10-1 to 10-3,

Y₁₃ to Y₁₈ and X₁₁ to X₁₄ are each independently N or C(R₁₀), and

*, *' and *'' each indicate a binding site to a neighboring atom.

9. The organometallic compound of claim 8, wherein

M₁₁ is selected from Pt, Pd, Cu, Ag, and Au.

10. The organometallic compound of claim 8, wherein

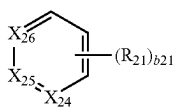
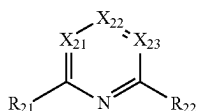
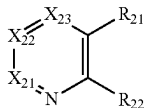
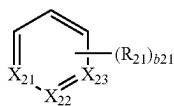
at least one selected from Y₁₀ and Y₂₀ is C,

(i) when Y₁₀ is C, a bond between Y₁₀ and T₁₁ or a bond between Y₁₀ and M₁₁ is a coordinate bond, and

(ii) when Y₂₀ is C, a bond between Y₂₀ and T₁₂ or a bond between Y₂₀ and M₁₁ is a coordinate bond.

11. The organometallic compound of claim 8, wherein

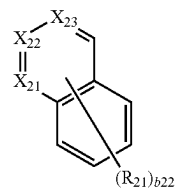
A₃₀ and A₄₀ are each independently selected from groups represented by Formulae 2-1 to 2-43:



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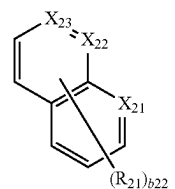
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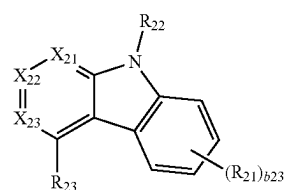
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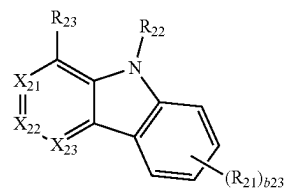
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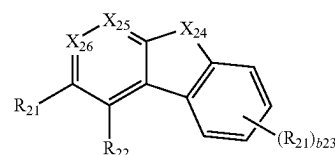
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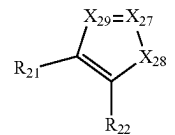
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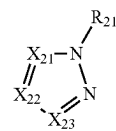
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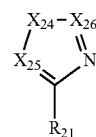
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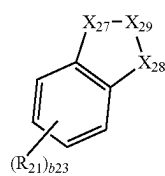
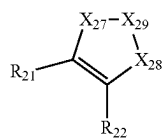
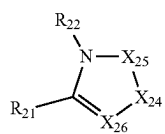
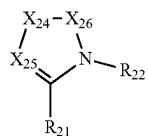
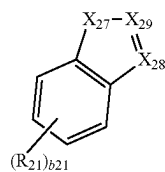
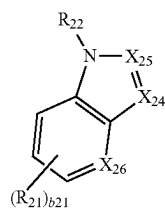
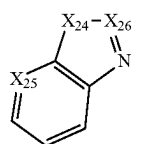
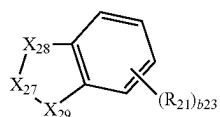
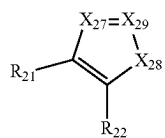
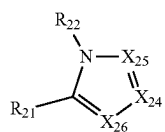
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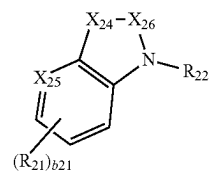
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**154**

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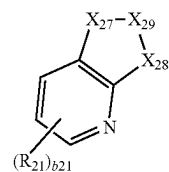
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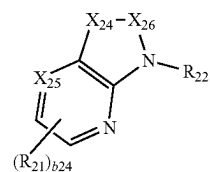
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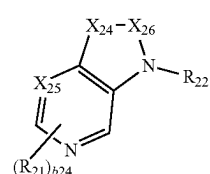
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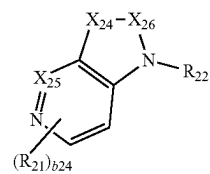
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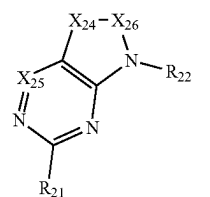
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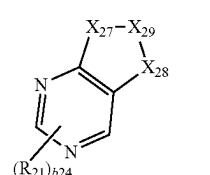
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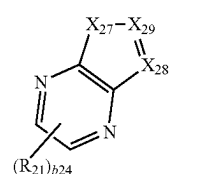
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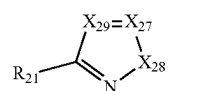
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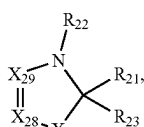
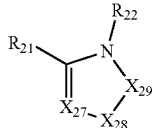
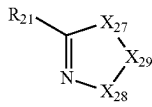
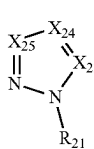
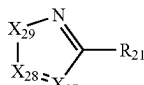
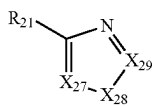
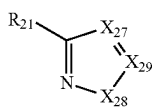
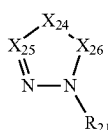
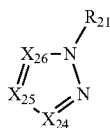
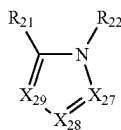
2-30

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-continued



wherein, in Formulae 2-1 to 2-43,

X₂₁ to X₂₃ are each independently selected from C(R₂₄) and C—*, and at least two selected from X₂₁ to X₂₃ are each independently C—*,

X₂₄ is N—*, and X₂₅ and X₂₆ are each independently selected from C(R₂₄) and C—*, wherein at least one selected from X₂₅ and X₂₆ is C—*,

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X₂₇ and X₂₈ are each independently selected from N, N(R₂₅), and N—*, and X₂₉ is selected from C(R₂₄) and C—*, wherein i) at least one selected from X₂₇ and X₂₈ is N—*, and X₂₉ is C—*, or ii) X₂₇ and X₂₈ are each independently N—*, and X₂₉ is C(R₂₄),

R₂₁ to R₂₅ are each independently the same as described in connection with R₁₀ in claim 8,

b₂₁ is selected from 1, 2, and 3,

b₂₂ is selected from 1, 2, 3, 4, and 5,

b₂₃ is selected from 1, 2, 3, and 4,

b₂₄ is selected from 1 and 2, and

* indicates a binding site to a neighboring atom.

12. The organometallic compound of claim 8, wherein each of T₁₁ to T₁₄ is a single bond;

T₁₁ is selected from O and S, and T₁₂ to T₁₄ are each independently a single bond;

T₁₂ is selected from O and S, and T₁₁, T₁₃, and T₁₄ are each independently a single bond;

T₁₃ is selected from O and S, and T₁₁, T₁₂, and T₁₄ are each independently a single bond; or

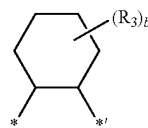
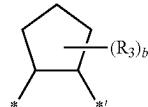
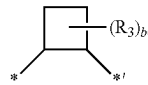
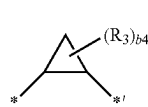
T₁₄ is selected from O and S, and T₁₁, T₁₂, and T₁₃ are each independently a single bond.

13. The organometallic compound of claim 8, wherein at least one selected from L₁₁ (s) in the number of all is selected from:

a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, and a cyclodecanylene group; and

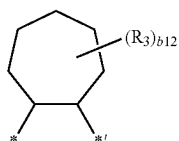
a cyclopropylene group, a cyclobutylene group, a cyclopentylene group, a cyclohexylene group, a cycloheptylene group, a cyclooctylene group, a cyclononylene group, and a cyclodecanylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkenyl group, a C₁-C₂₀ alkynyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triptylenyl group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, a triazinyl group, a dibenzofuranyl group, and a dibenzothiophenyl group.

14. The organometallic compound of claim 8, wherein at least one selected from L₁₁ (s) in the number of all is selected from groups represented by Formulae 3-1 to 3-5:



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-continued



wherein, in Formulae 3-1 to 3-5,

R_3 is selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkenyl group, a C_1 - C_{20} alkynyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, a triazinyl group, a dibenzofuranyl group, and a dibenzothiophenyl group,

b_4 is an integer from 1 to 4,

b_6 is an integer from 1 to 6,

b_8 is an integer from 1 to 8,

b_{10} is an integer from 1 to 10,

b_{12} is an integer from 1 to 12, and

* and *' each indicate a binding site to a neighboring atom.

15. The organometallic compound of claim 8, wherein L_{12} and L_{14} are each independently a single bond, and L_{13} is selected from *—O—*', *—S—*', *—N(R_1)—*', and *—B(R_1)—*.

16. The organometallic compound of claim 8, wherein R_1 , R_2 , R_{10} , R_{20} , R_{30} , and R_{40} are each independently selected from:

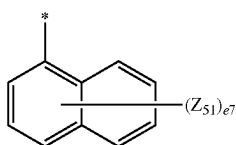
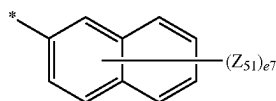
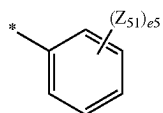
hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C_1 - C_{20} alkyl group, and a C_1 - C_{20} alkoxy group;

a C_1 - C_{20} alkyl group and a C_1 - C_{20} alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, and a biphenyl group;

a group represented by one selected from Formulae 5-1 to 5-25; and

a group represented by one selected from Formulae 6-1 to 6-55, and

R_1 and R_2 are optionally linked to form a cyclopentane group, a cyclohexane group, a benzene group, a naphthylene group, or a fluorene group:

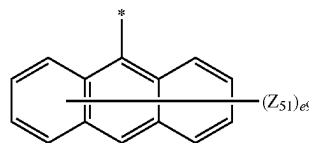


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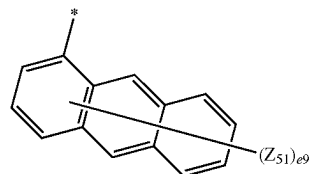
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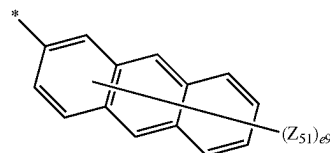


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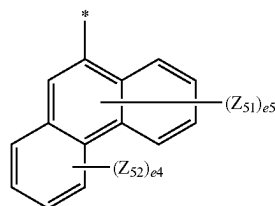


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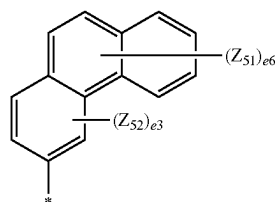
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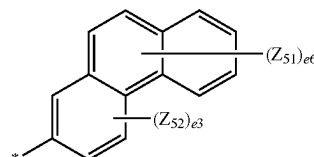
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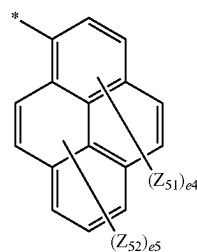
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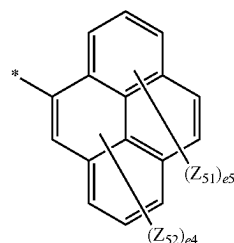
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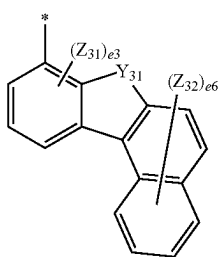
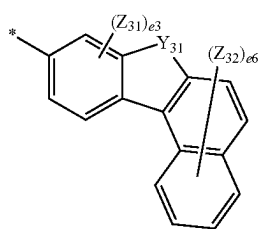
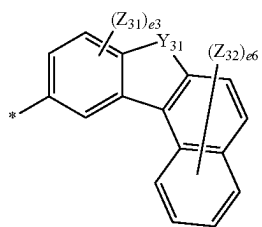
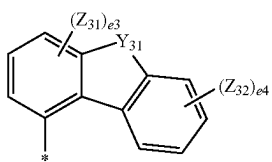
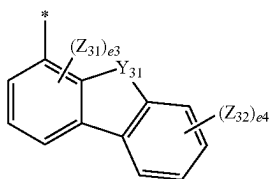
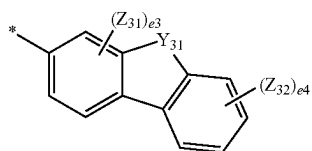
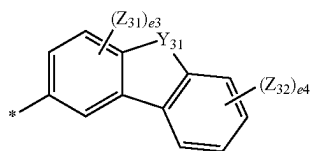
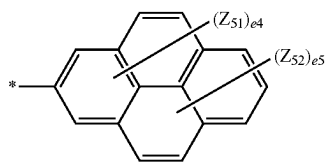
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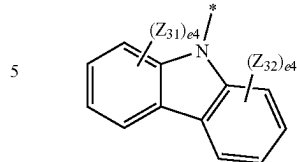
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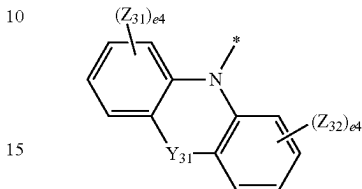
**160**

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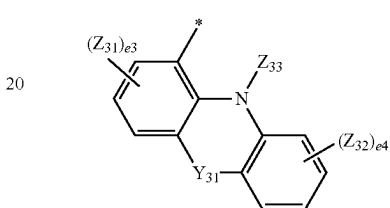
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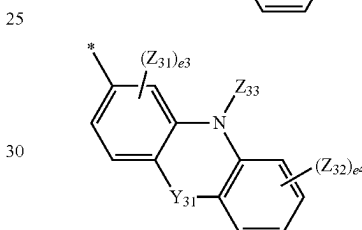
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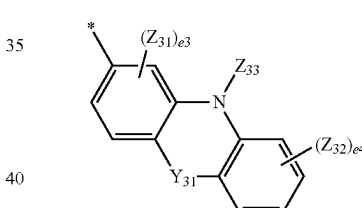
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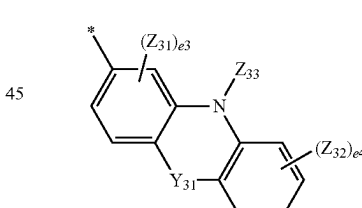
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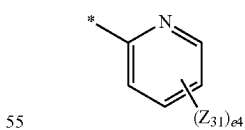
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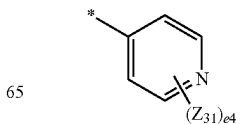
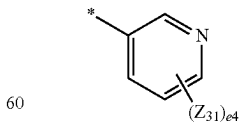
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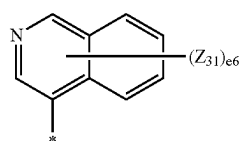
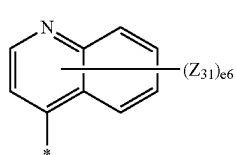
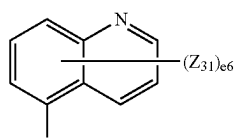
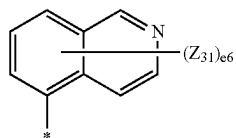
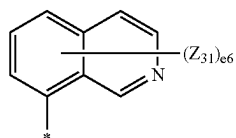
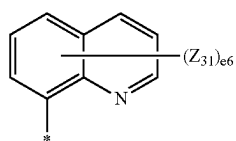
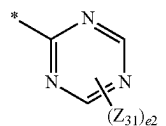
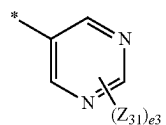
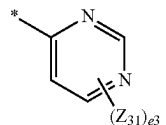
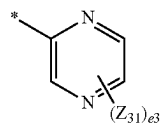
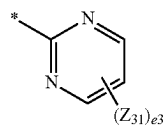
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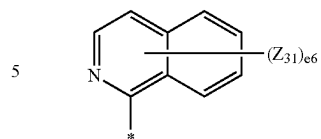
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**162**

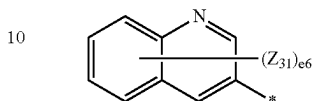
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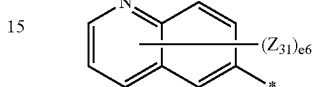
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6-5



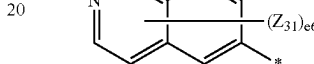
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6-6



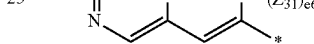
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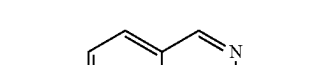
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6-9



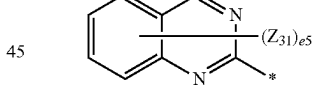
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6-10



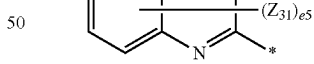
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6-11



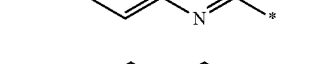
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6-12



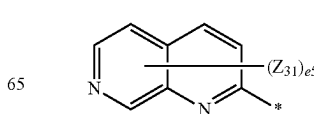
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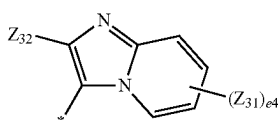
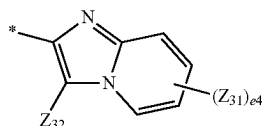
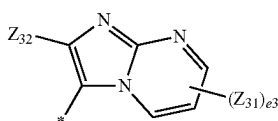
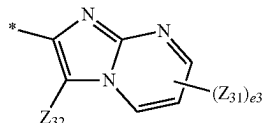
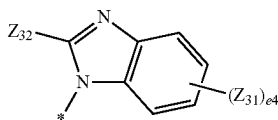
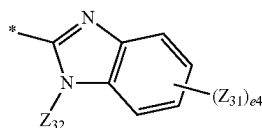
6-25

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6-27

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wherein, in Formulae 5-1 to 5-25 and 6-1 to 6-55, Y_{31} is O, S, C(Z_{34})(Z_{35}), N(Z_{34}), or Si(Z_{34})(Z_{35}), Z_{31} to Z_{35} are each independently selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkenyl group, a C_1 - C_{20} alkynyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triperylene group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, and a triazinyl group,

e_2 is 1 or 2,

e_3 is an integer from 1 to 3,

e_4 is an integer from 1 to 4,

e_5 is an integer from 1 to 5,

e_6 is an integer from 1 to 6,

e_7 is an integer from 1 to 7,

e_9 is an integer from 1 to 9, and

* indicates a binding site to a neighboring atom.

17. The organometallic compound of claim **8**, wherein R_1 , R_2 , R_{10} , R_{20} , R_{30} , and R_{40} are each independently selected from:

hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C_1 - C_{20} alkyl group, and a C_1 - C_{20} alkoxy group;

a C_1 - C_{20} alkyl group and a C_1 - C_{20} alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, and a biphenyl group;

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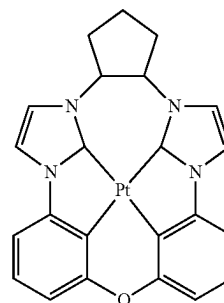
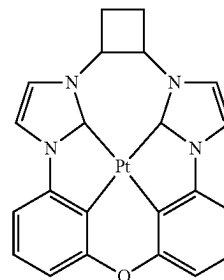
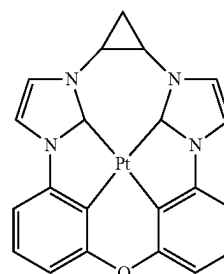
a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, and a dibenzocarbazolyl group; and

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazolyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, and a dibenzocarbazolyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a cyano group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, and a biphenyl group, and

R_1 and R_2 are optionally linked to form a cyclopentane group, a cyclohexane group, or a fluorene group.

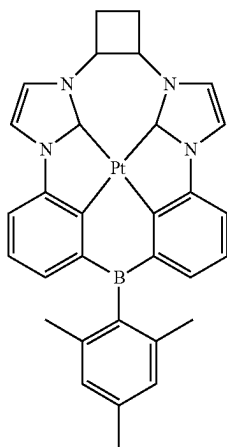
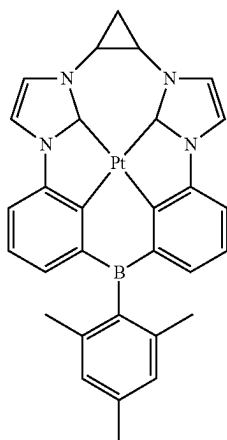
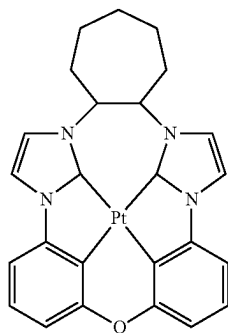
18. An organometallic compound, wherein

the organometallic compound is selected from Compounds 1 to 3 and 5 to 20:

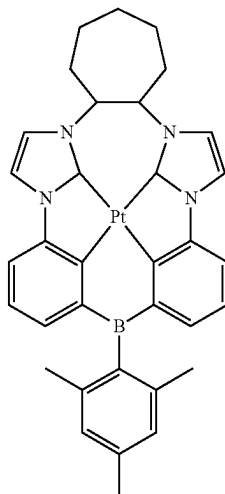
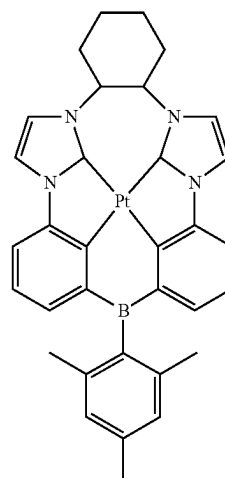
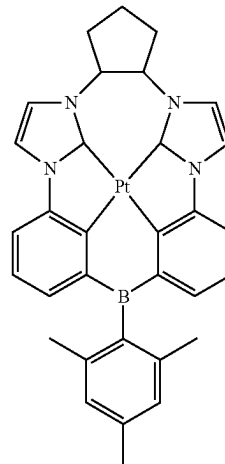


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